Recovery Boundaries and Quantization Limits for Low-Rank Neural Adaptations under Restricted Oracles*

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Abstract. Low-rank adapters concentrate task-specific behavior in a small parameter subspace that prediction APIs returning scores expose. We give an oracle-by-oracle recovery theory over a public base model: score-bearing outputs recover observable heads exactly, and known row space cuts later per-class observations from $h + 1$ to $r + 1$. Matching obstructions: hard labels erase scale, index-only APIs hide never-returned classes, and endpoint logits identify hidden adapters only modulo the final-head nullspace.

Our main result is negative, and about defenses. Quantizing scores to significant digits is provably scale-invariant, and we prove a computable indistinguishability floor binding every algorithm at its chosen query set. But that quantifier is the whole story: an attacker placing each cell boundary at the midpoint of its uncertainty drives its own floor to zero, and defeats both families to within 1.22×. What bounds recovery is the attacker's *request* resolution: d-digit requests against p-digit scores admit no algorithm at any budget beating $\frac{1}{8}10^{-(p+d+1)}$; enumeration puts the floor at $\approx \frac{1}{4} \cdot 10^{-(p+d)}$. It is scalar (the matrix case is open) and needs a server that canonicalizes requests onto a decimal grid; under binary64 parsing we found no useful floor. On real transformer and vision models the units are worth 1.40× at a bounded endpoint and 178–182× at a module interface, while at a LayerNorm endpoint *absolute* rounding defeats recovery outright; we give the test for which regime applies. Scale-invariant units are not themselves a security property.

Keywords: Cryptanalytic model extraction · Low-rank adaptation · Prediction-API oracles · Quantization defenses · Indistinguishability bounds · Machine learning for security

1 Introduction

Cryptanalytic model extraction treats a deployed neural network as a keyed primitive and asks whether chosen queries reveal its hidden parameters. This re-

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search line began with raw-output extraction at CRYPTO 2020 [6], was strengthened to polynomial-time ReLU extraction at EUROCRYPT 2024 [4], and recently moved to hard-label extraction at ASIACRYPT 2024 [9] and EUROCRYPT 2025 [5].

This paper studies a complementary structural secret: a low-rank adaptation applied to a known base model. This setting is motivated by parameter-efficient fine-tuning and LoRA-style deployments, where the base model may be public while the task-specific update is proprietary. Existing LoRA extraction work demonstrates functional extraction from synthetic data [53], but does not give a cryptanalytic parameter-recovery model with formal oracle guarantees.

Research question. Which part of a known-base low-rank adapter is observable under a given API, and when do API restrictions collapse parameter recovery to a quotient, a decision boundary, or no identifiable parameter object at all?

Thesis. Score-bearing oracles make observable adapters exactly recoverable; hard labels, index-only outputs and rank-deficient endpoints reduce recovery to a quotient, a boundary arrangement, or nothing identifiable; and *numerical* degradation of the returned scores sits between the two, weak against an adaptive attacker for a reason that is fixable. That reason is the *units* of the perturbation rather than its size, and how much fixing it buys is decided by three measurable properties of the deployment. Section 5 makes this precise; Section S19 measures where it binds.

1.1 Positioning

Much of the positive analysis solves linear systems, so the framing needs stating. We use "cryptanalytic" in the sense of Carlini et al. [6]: the model is a keyed primitive, the adapter the key, the API the oracle, and the goal exact key recovery rather than approximate imitation. Three features separate this from generic system identification. *Exactness*: the target is the parameter up to a characterized symmetry group, so positive results report machine-precision errors and negative results are indistinguishability statements. *A known public component*: as in a known-plaintext setting the adversary knows the base computation and recovers only the secret difference. *Quotient-first accounting*: LoRA factors carry a GL_r gauge, probabilities an additive shift, hard labels a positive scaling, rank-deficient endpoints a kernel, and the transcript equivalence class is primary (Section 3).

The incentive is also quantitative. Where the interface is observable the cost is one linear solve per adapted module: the trained rank-4 update of Section 8.3, a 768×768 matrix, is recovered at $8.37 \cdot 10^{-15}$ from $h + 1 = 769$ forward queries, against 3600 forward-and-backward passes to train the 147,456-parameter adapter — and the real gap is that training needs 1800 *labeled in-domain* examples and the attacker needs none. The advantage belongs to the interface, not the attack: the same target costs $2.15 \cdot 10^4$ calls behind a forcible-row endpoint, $5.17 \cdot 10^5$ behind a scoreless one, and is not recoverable behind a rank-deficient classifier (Theorem 9).

Our observable-head recovery result (Section 4) is closest to the equation-solving attacks of Tramèr et al. [48], who recovered multiclass parameters from confidence-bearing queries and studied score rounding as a countermeasure. Two differences drive everything below. The secret here is a rank-r update over a *public* base, so once the row space is identified each further class row costs $r+1$ rather than $h+1$ observations (Proposition 1) — an acceleration that Theorem 3 shows cannot be certified earlier. They also evaluate rounding against a *fixed* attacker; we let the attacker amplify (Section 5) and choose where the thresholds fall relative to its own uncertainty (Section 5.2), which is what decides whether the defense holds. Section S5 gives the full comparison.

1.2 Contributions

Our load-bearing new objects are Theorems 6 and 7 — a computable bound at the attacker's chosen query set, and an exact floor over every query set — with Proposition 2 measuring the distance between them and the boundary search of Section 5.2 closing it: scale-invariant units are not themselves a security property. The rest — Theorems 1–5 and 8–10 — buy the *model* that makes "what is recoverable" well posed per interface; Table 16 (Section S23) maps each to its oracle, object, and experiment.

Observable recovery. We formalize known-base low-rank adapter extraction as a chosen-query oracle problem and prove exact recovery for affine and public-feature heads, score-bearing top-k rows, full-rank endpoint inversions, and observable final-head adapters over public transformer features; once the row space is identified, low rank cuts later per-class observations from $h+1$ to $r+1$, and rank-revealing selection on the *public* head cuts forced-row probing on a `distilgpt2` endpoint from 50,257 rows to 769 per prompt at no oracle cost.

Quotient-corrected impossibility. We prove that row-space acceleration cannot be certified before r independent rows are observed, that hard labels erase scale, that partial index-only APIs can hide never-returned classes, and that endpoint logits identify hidden adapters only through the downstream quotient, after quotienting out the r^2-dimensional LoRA gauge.

A defense that survives two named levers — and the one that defeats it. The stable-recovery bound separates a defender-controlled perturbation from an attacker-controlled design, giving scaling laws (Corollary 1) we confirm to within the predicted constants. An absolute grid is scale-dependent and a relative one is not, so quantizing to significant digits is provably immune to both levers (Theorem 5), leaving the attacker 178–182× worse off at no greater service cost against a *trained* LoRA target, and against an attacker whose *estimator* is built for that defense. Theorem 6 bounds every algorithm's error *at the query set it chooses*: it vanishes as $1/A$ under an absolute grid and is invariant under a relative one.

The negative result we owe our own recommendation. Because that floor is quantified inside the attacker's choice of $\mathcal{Q}$, it certifies a transcript rather than guaranteeing a defense. An attacker placing a cell boundary at the midpoint of its uncertainty interval halves that interval per response (Proposition 2), drives

its own floor to zero, and, using only the fact that cells *have* boundaries, defeats both quantizer families identically: the 178× advantage becomes at most 1.22× (Table 2), and the same attacker recovers the deployed bias rather than being given it. Scale-invariance of the units is thus not by itself a security property. What governs recovery instead is the attacker's request resolution; a norm cap reaches the stronger attacker that searches boundary and scale jointly, but only against an absolute grid, making the cap and not the units the control doing the work (Section S11). We delimit the remaining gain rather than assert it: amplification needs a feature map that is not scale-invariant, a reachable amplified query, and a design that keeps rank — LayerNorm removes the first, pixel bounds the second, ReLU inactivity the third. Finally, "absolute versus relative" and "deterministic versus stochastic" are independent, inducing a 2×2 taxonomy in which each cell is breached by exactly the lever it is exposed to, and relative–deterministic is the only cell immune to *those two* — though not, by Section 5.2, to boundary seeking. With experiments in which scoreless APIs remove the recovery equations outright, this separates defenses that *remove* the linear signal from those that merely *perturb* it.

2 Related Work

Three strands bear on this paper: cryptanalytic neural extraction, prediction-API model stealing, and adapter-specific attacks and verification. Section S5 surveys each with its citations; here we keep the two comparisons our claims rest on — differential privacy, and recovery under quantized measurements — since those are where a reader is entitled to suspect we are reinventing or overclaiming.

Defenses, output perturbation, and differential privacy. Prior defenses detect unnatural queries [29], perturb outputs under utility constraints [42, 47], watermark adaptively [54], exit early [35], train for extraction resistance [31], add logit noise [50], or censor top-k scores [38]; these target functional extraction, distribution recovery, or architecture hiding, and the observational ones face known provable limits [30]; Section 5 adds an adaptive-attacker evaluation. Our absolute–stochastic cell is output perturbation in the sense of [12], and averaging repeats to breach it is the classical composition argument [13]; we claim no novelty there; Corollary 1(2) is a prediction we verify. Three things do not follow. Amplification breaches *deterministic* perturbation, where no budget is spent and composition says nothing. DP calibrates noise to *sensitivity*, an absolute quantity, so its instantiations — and prediction-API defenses built the same way [42] — land in the row whose certificate Theorem 6 shows vanishes with scale, and where Corollary 1's attacker realizes the loss. And our target is exact parameter recovery under a known public base, so the operative quantity is that floor, not an ε-δ statement over neighboring datasets.

Recovery from quantized measurements. What a quantized answer leaks is older than its appearance in extraction, and two literatures already ask our question.

Quantized system identification bounds what is recoverable from binary-valued sensors and shows the input design governs the accuracy attainable [52]; one-bit compressed sensing bounds reconstruction from sign measurements with *fixed* thresholds [27], and — the part that matters here — choosing each threshold from the estimate built out of previous answers replaces that precision floor with error decaying *exponentially* in the number of measurements [3, 14]. Section 5.2 applies that mechanism to significant-digit output quantization, and is why we do not claim the relative grid is attacker-independent. The transfer is partial; the difference is the subject of Section 5.3: those results let the adversary name the threshold, whereas ours may name only a query *scale* and must aim it at boundaries the quantizer has already fixed. Granting the stronger model removes the floor entirely, by five to nine orders at a matched budget (Table 10), so the exponential-decay guarantee is about freely placed thresholds and not about quantized APIs. Rounded confidences were already an obstacle to equation solving [48], but as a fixed loss of precision rather than as thresholds an adversary places.

3 Preliminaries

This section fixes notation, defines every architectural term and oracle used later, and names the quantities reported in the experiments.

3.1 Model and secret

Let $W_0 \in \mathbb{R}^{m\times d}$ be a public base classifier and let $\Delta = UV^\top$, with $U \in \mathbb{R}^{m\times r}$ and $V \in \mathbb{R}^{d\times r}$, be an unknown rank-$r$ adapter. The deployed affine model is $f(x) = (W_0 + \Delta)x + b$. The adversary knows W_0, m, d, and an upper bound on r, and may make adaptive chosen queries to an oracle derived from f.

Definition 1 (Architectural vocabulary). *An* affine head *is a map $x \mapsto Wx + b$; it is* adapted *when $W = W_0 + \Delta$ with W_0 public and Δ an unknown rank-r update. A* public-feature head *is $x \mapsto (H_0 + \Delta)\phi(x) + b$ with the feature map $\phi : \mathbb{R}^p \to \mathbb{R}^h$ and base head $H_0 \in \mathbb{R}^{m\times h}$ public and only Δ, b secret; ϕ may be nonlinear, being public preprocessing composed with a frozen public network. For a network with final hidden representation $h(x) \in \mathbb{R}^h$, the* endpoint head *is the last linear map $C \in \mathbb{R}^{m\times h}$ with bias β, and the* endpoint logits *are $y(x) = Ch(x)+\beta$. An adapter is* observable *when the adapted component is the endpoint head itself, and* hidden *when it lies strictly upstream of C. An endpoint interface is* forcible-row *if the adversary may nominate an output coordinate and make the API report on it, e.g. via a per-row logit bias large enough to push that row into the returned set; a* forced-row call *counts as one call per nominated coordinate per query. Two heads are* functionally equivalent *on a domain if they induce the same top-1 label throughout it — weaker than parameter equality, being invariant under positive rescaling of all logits and any shared additive constant.*

3.2 The oracle family

Table 1 defines the oracle family studied in this paper. Each is a deterministic (or, where noted, randomized) map from a query to an observation, and each abstracts a deployed prediction interface. Query cost is counted in oracle calls throughout; where an interface needs several calls per query, that is stated explicitly. We write $f(x)$ for the deployed score vector and $y(x) = Ch(x) + \beta$ for endpoint logits with public C, β.

Table 1. The oracle family. Definition 2.

Oracle	Response to a query x
$\mathcal{O}_{\text{raw}}$	The full score vector $f(x) \in \mathbb{R}^m$.
$\mathcal{O}^{\text{score}}_{\text{top-k}}$	Indices of the k largest coordinates of $f(x)$ *with* their scores; coordinates outside the top k return nothing.
$\mathcal{O}^{\text{idx}}_{\text{top-k}}$	The same ordered indices but *no* scores. For $k = m$ this is a complete ranking oracle; for $k = 1$ it is $\mathcal{O}_{\text{label}}$.
$\mathcal{O}_{\text{label}}$	Only $\arg\max_c f_c(x)$.
$\mathcal{O}_{\text{logprob}}$	Normalized $\log \text{softmax}(f(x))$ on a subset of coordinates with known temperature; equivalently, logits up to one unknown additive shift $\alpha(x)$ per query.
$\mathcal{O}_{\text{bias}}$	Accepts a per-coordinate bias $\beta' \in \mathbb{R}^m$ and responds according to $f(x) + \beta'$, with $\|\beta'\|_\infty$ bounded by an API-imposed range. Scored variant returns log-probabilities; *scoreless* variant returns only the winning index.
$\mathcal{O}_{\text{module}}$	The raw output of one designated internal module. Models a debugging or embedding endpoint; strictly stronger than $\mathcal{O}_{\text{end}}$.
$\mathcal{O}_{\text{end}}$	Only the endpoint logits $y(x) = Ch(x) + \beta$, with C, β public. No hidden states, module outputs, or intermediate scores.

Definition 2 (Oracle family). *The oracles* $\mathcal{O}_{\text{raw}}$, $\mathcal{O}^{\text{score}}_{\text{top-k}}$, $\mathcal{O}^{\text{idx}}_{\text{top-k}}$, $\mathcal{O}_{\text{label}}$, $\mathcal{O}_{\text{logprob}}$, $\mathcal{O}_{\text{bias}}$, $\mathcal{O}_{\text{module}}$, *and* $\mathcal{O}_{\text{end}}$ *are as specified in Table 1.*

3.3 Recovery goals and reported quantities

Three goals are in play: *parameter recovery* of $W_0 + \Delta$, b or Δ numerically, *subspace recovery* of the row or column space of Δ up to the factorization ambiguity $UQ, VQ^{-\top}$, and *functional recovery* of a classifier with the oracle's labels but possibly not its logit scale. Throughout, *recovery* means the maximal oracle-identifiable object the transcript determines rather than the adapter parameters — which is why the paper is framed as recovery boundaries, and why we never claim recovery of arbitrary hidden transformer LoRA factors from endpoint logits. Recovery error is relative Frobenius error on the adapter, bias error a separate absolute norm, agreement the held-out top-1 match rate, and leakage the *observable quotient* $\text{rank}\, J_Q / \dim$ of the non-gauge update space; Definition 4 in

Section S4 states all of them, with the fidelity metrics Sections S19 and S10 report alongside label agreement, which is blind to everything but the argmax.

Definition 3 (Transcript equivalence class). *Fix a public base model, a class $\mathcal{A}$ of rank-r adapter effects, a query set Q, and a restricted oracle with transcript map $T_Q : \mathcal{A} \to \mathcal{Y}_Q$. Write $\Delta_1 \sim_Q \Delta_2$ iff $T_Q(\Delta_1) = T_Q(\Delta_2)$. Since the transcript is a deterministic function of Δ, an algorithm given only $T_Q(\Delta)$ receives identical input on every member of $[\Delta] \in \mathcal{A}/\sim_Q$, so that class is the maximal guaranteed recoverable object. We state this as a definition rather than a theorem: it is immediate, and its content is the* accounting *it imposes.*

The class absorbs LoRA factor gauge, additive softmax shifts for probability outputs, positive logit-scale symmetries for label-only or ranking oracles, unreturned rows for partial top-k, and hidden-adapter directions in the kernel of the downstream endpoint map; every later result either collapses it to a point or exhibits it as nontrivial. Proofs are inline where short and otherwise in Section S1; all S-numbered sections are supporting material, following the references.

4 Recovery from Score-Bearing Oracles

Theorem 1 (Observable affine-head recovery). *Let an adapted head have the form $f(x) = (H_0 + \Delta)\phi(x) + b$, where H_0 and $\phi : \mathbb{R}^p \to \mathbb{R}^h$ are public. If $\mathcal{O}_{\text{raw}}$ is available on $h+1$ queries whose augmented features $(\phi(x_i), 1)$ have rank $h+1$, then $H_0 + \Delta$, b, and Δ are exactly recoverable over exact arithmetic. Under $\mathcal{O}^{\text{score}}_{\text{top-k}}$, the same conclusion holds for any class row returned on $h+1$ such independent feature vectors. The affine case $f(x) = Wx + b$ is the instance $\phi = \text{id}$, $h = d$; the endpoint-head transformer case is the instance $\phi = \Phi$ for a public frozen transformer feature map.*

Proof. Each returned score for class c gives one linear equation $\langle h_c, \phi(x_i)\rangle + b_c = y_{i,c}$ in the $h+1$ unknowns of the class row and bias. Stacking $h+1$ independent augmented feature vectors gives a nonsingular linear system; solving it recovers the row. Repeating over all raw-logit rows, or all top-k rows satisfying the independence condition, recovers the corresponding adapted head rows. Subtracting the public base head gives Δ.

Reachability of the rank hypothesis. The hypothesis constrains the *reachable* feature set $\phi(\mathcal{X})$, not $\mathbb{R}^h$, and is this theorem's main practical caveat. It holds whenever $\operatorname{span}\phi(\mathcal{X}) = \mathbb{R}^h$, immediate for $\phi = \text{id}$ on an unconstrained domain but not otherwise: a ReLU map zeroes roughly half the coordinates of any $\phi(x)$ on digits, and for text the reachable set is the image of a discrete token space. We therefore verify the achieved augmented rank rather than assume it, reporting it in every table (769 of 769 in Table 19); natural-input pools attain it easily, the condition being spanning rather than hitting a prescribed vector. Reachability returns in Section 5 as the binding constraint on the attacker's design.

Proposition 1 (Low-rank scored top-k recovery). *Let $\Delta \in \mathbb{R}^{m\times h}$ have rank r. Suppose scored top-k observations fully recover r linearly independent adapter rows, identifying the row space of Δ. Then any additional class row c is recoverable from $r+1$ scored observations whose projected augmented vectors have rank $r+1$, using a basis $B \in \mathbb{R}^{r\times h}$ for the recovered row space.*

Proof. Deferred to Section S1. Knowing the row space leaves only the $r+1$ unknowns α_c, b_c in $\delta_c = \alpha_c B$.

Theorem 2 (Rank-adaptive recovery from an upper bound). *Let $\Delta \in \mathbb{R}^{m\times h}$ have unknown rank $\rho \le R$. Suppose a score-bearing transcript fully recovers adapter rows whose matrix M has singular values $\sigma_1 \ge \cdots \ge \sigma_\rho > \tau \ge \sigma_{\rho+1}$ for a declared threshold τ. Then thresholding the spectrum of M identifies ρ and the row space of Δ, and each additional returned row is recoverable from $\rho+1$ scored observations with projected augmented rank $\rho+1$. Under noisy scores this recovers the τ-numerical rank.*

Theorem 3 (Row-space identification lower bound). *Consider scored top-k transcripts for a rank-r adapter. Suppose the transcript fully identifies adapter rows with span dimension $s < r < h$, no returned score equations constrain the remaining $r-s$ row-space directions, and every query separates its returned classes from its unreturned ones by a strictly positive score margin. Then the row space is not identifiable: two rank-r adapters can induce the same transcript — the same returned indices* and *scores — but have different row spaces. Consequently the $r+1$ recovery step of Proposition 1 cannot be certified until the transcript has forced r independent row directions.*

Proof. Deferred to Section S1; the construction completes the unidentified row space in two different ways that the transcript cannot separate.

4.1 Algorithms and query cost

Both attacks used in the experiments are stated in full in the supporting material, with query accounting that charges probes yielding no equation. Algorithm 1 costs exactly $|\mathcal{P}|$ oracle calls, each yielding k scalar equations over the returned classes; a class is solved iff it is returned on $h+1$ queries of rank $h+1$ (or $\hat{\rho}+1$ after the row space is known), and a class never returned is provably unidentified by Theorem 8. Algorithm 2 costs $|\mathcal{Q}| \cdot |S|$ calls, multiplied by the binary-search depth $\lceil \log_2(2B/\epsilon) \rceil$ in the scoreless variant, which is charged in full. Choosing S by rank-revealing QR rather than probing all m rows is what cuts the `distilgpt2` cost from $1.41 \cdot 10^6$ to $2.15 \cdot 10^4$ forced-row calls (Table 27): $|S| = 769$ rather than $m = 50{,}257$.

5 Adaptive Attackers and Score-Degradation Defenses

A natural defense is to return degraded scores: round them to a grid, or add noise. We bound the resulting recovery error, observe that the bound factors

into a defender-controlled and an attacker-controlled term, and measure what an attacker who optimizes its own term achieves.

Theorem 4 (Stable recovery under bounded score perturbations). *Let $X \in \mathbb{R}^{n\times(h+1)}$ be the augmented query-feature matrix used for observable-head recovery, with full column rank, and let $Y = X\Theta$ be the exact score matrix. If the oracle returns $\widetilde{Y} = Y + E$, least-squares recovery gives*

$$\|\widetilde{\Theta} - \Theta\|_F \le \|X^+\|_2 \, \|E\|_F.$$

For an evaluation feature vector $\bar{z} = (\phi(z), 1)$, every recovered logit differs from the exact one by at most $\|\bar{z}\|_2\|X^+\|_2\|E\|_F$, so the top label is unchanged whenever the exact top-two logit margin at z exceeds twice this quantity. If scores are rounded to a grid of width τ, then $\|E\|_F \le \tau\sqrt{nm}/2$.

Proof. Least squares gives $\widetilde{\Theta} = X^+\widetilde{Y} = \Theta + X^+E$, and the Frobenius bound follows by submultiplicativity. The logit perturbation at $\bar{z}$ is $\bar{z}^\top(\widetilde{\Theta} - \Theta)$, so Cauchy–Schwarz gives the per-class bound. Rounding each of the nm returned scores to grid width τ introduces entrywise error at most $\tau/2$.

The bound is a product, and the defender controls only one factor. The defender controls $\|E\|_F$ through the grid width or noise level. The attacker controls $\|X^+\|_2$ through the choice of query design, and can also shrink the *effective* $\|E\|_F$ by averaging. A defense evaluated against a fixed query distribution therefore measures neither factor at its adversarial value. This is the gap the next corollary quantifies.

Corollary 1 (Adaptive scaling laws). *Consider an attacker who may (i) scale queries by a factor A, (ii) repeat each design point R times and average, and (iii) project the recovered update onto rank $\le R_{\max}$.*

1. *If the feature map is positively homogeneous on the queried region, scaling by A scales the non-constant columns of X by A while a fixed absolute grid leaves $\|E\|_F$ unchanged. The linear-head component of the error then decays like $1/A$. The intercept component does not, because the constant column of the augmented design is invariant under scaling.*
2. *If the perturbation is independent additive noise of standard deviation σ, averaging R repetitions of the same design point yields effective noise $\sigma/\sqrt{R}$, so the error decays like $1/\sqrt{R}$. If the perturbation is deterministic rounding, repetition returns identical values and yields no improvement.*
3. *Projection onto the declared rank bound removes the component of the estimation error orthogonal to the rank-$R_{\max}$ manifold, contributing a constant factor rather than a rate.*

Two consequences are worth stating because they are what the experiments confirm. First, item 2 predicts a sharp *qualitative* separation between rounding and noise: the same lever that defeats one is useless against the other. Second, item 1 predicts that amplification helps the linear part but not the intercept, so

an attacker who wants both must spend a separate unamplified probe budget on the intercept once the head is known. Our implementation does exactly that, and Section 8.4 reports the effect.

5.1 A defense that neither lever defeats

Corollary 1 localizes the weakness precisely: an absolute grid fixes $\|E\|_F$ while the attacker inflates the design, so the ratio that governs recovery collapses. This suggests the repair. If the perturbation is made *proportional* to the score rather than absolute, amplification inflates signal and perturbation together and the ratio is invariant. Relative quantization — reporting a fixed number of significant digits, rather than a fixed number of decimal places — does exactly this.

Theorem 5 (Scale-invariance of relative quantization). *Let the oracle return $Q_\varepsilon(y)$ with $|Q_\varepsilon(y) - y| \le \varepsilon|y|$ entrywise, as achieved by reporting p significant digits with $\varepsilon = 5 \cdot 10^{-p}$. Suppose the public feature map φ is* asymptotically positively homogeneous *on the queried region: $Z(A)/A \to Z_\infty$ as $A \to \infty$ with Z_∞ of full column rank, so $Z(A) = A\big(Z_\infty + o(1)\big)$. Then:*

1. Amplification is useless. *The least-squares error of the linear head does not decay with A: it is* exactly invariant *where A is a power of the quantization base, φ is exactly homogeneous, and $b = 0$; in general it stays within one quantization step of that value, plus the $o(1)$ homogeneity defect. We claim a band, not a limit, because that is what the proof delivers: off that subsequence the error oscillates inside it rather than settling, by a measured $4.3\times$ across one decade (Section S9). Under an absolute grid of width τ the same computation gives error decaying like $1/A$; that contrast, not convergence, is what the argument needs.*
2. Averaging is useless. *Q_ε is a deterministic function of the query, so repeated queries return identical values: no variance reduction. Neither lever moves the least-squares error, which stays $O(\varepsilon)$ for that estimator, improving only if $X^\top E = 0$ — grid-aligned scores are sufficient for that cancellation, not necessary; what remains available to* it *is the conditioning of the* reachable *design.*

Remark 1 (What item 2 does not say). Items 1–2 concern the two levers of Corollary 1 and the estimator using them, not every attacker: neither is a lower bound. An attacker choosing *where the cell boundaries fall* relative to its own uncertainty reaches far below ε at the same oracle, as Section 5.2 constructs and measures.

Proof (sketch; full proof in Section S1). Relative quantization commutes with scaling by a power of the base, $Q_\varepsilon(cY) = cQ_\varepsilon(Y)$, so the perturbation carries the same factor A that the pseudoinverse removes: $Z(A) = A(Z_\infty + o(1))$ gives $Z(A)^+ = A^{-1}(Z_\infty^+ + o(1))$ and $E(A) = A(E_\infty + o(1))$, leaving $Z(A)^+E(A) = Z_\infty^+E_\infty + o(1)$. Off the base-power subsequence the two sides of that commutation differ by up to one quantization step, itself proportional to the score and

so carrying the same A — bounded, not vanishing, which is why the statement claims a band and not a limit. An absolute grid caps $\|E\|_F \le \tau\sqrt{nm}/2$ irrespective of A, so the same computation decays like $1/A$. Part (2) is immediate from determinism.

Remark 2 (The hypothesis is the weak one, and we verify it). Exact homogeneity is *false* for the maps we attack — a standardizer subtracts a mean, and $\|\varphi(Ax)\|/(A\|\varphi(x)\|)$ settles at 1.73 on `digits`, not 1 — so the theorem rests on the asymptotic hypothesis, which we verify: the defect $\|Z(A)/A - Z(4A)/4A\|_F/\|Z(4A)/4A\|_F$ decays like $1/A$, 0.56 to 0.0005 over $A = 1$ to 1024 (Table 25). Its limit cancels in $(cZ)^+(cE)$; only its existence matters.

Theorem 5 says two named levers fail — a statement about two attacks, and a defense recommendation deserves more. We now show the protection is a property of the *quantizer*, holding against every attacker.

Theorem 6 (Per-transcript indistinguishability floor). *Let the oracle return $Q(y)$ for a deterministic quantizer Q whose cells partition score space, on any query set $\mathcal{Q}$ of any size, chosen adaptively or not. For a returned score y_{jc} write $\gamma^{\uparrow}_{jc}, \gamma^{\downarrow}_{jc} > 0$ for its distances to its cell's two boundaries. Let $USV^\top$ be a singular value decomposition of Δ, of rank r, let Ξ be* one-sided, *$\Xi = UM^\top$ or $\Xi = NV^\top$, and let $T(\Xi)$ be the largest $t \ge 0$ keeping every displacement $t\, z_j^\top \Xi_c$ short of the boundary it moves* toward*; scores it does not move constrain nothing, and $T(\Xi) = \infty$ if none is moved. One sign suffices because the family is closed under $\Xi \mapsto -\Xi$. Then for every $0 \le t < T(\Xi)$ the matrix $\Delta + t\Xi$ is again exactly of rank at most r and induces an* identical *transcript. Consequently every recovery algorithm — any query budget, any design, adaptive or not, and even given the true rank — has minimax relative error at least $F(\mathcal{Q})/(1+F(\mathcal{Q}))$, for $F(\mathcal{Q}) = \frac{1}{2}\sup_{\Xi\neq 0} T(\Xi)\|\Xi\|_F/\|\Delta\|_F$, the* absolute *floor normalized by $\|\Delta\|_F$, where $\frac{1}{2}$ is tight and $F = \infty$ reads as a floor of* 1*; Section S1 derives both, and tables report F. Under amplification $z_j \mapsto Az_j$: (i) for an absolute grid of width τ, both slacks are at most τ irrespective of A, so $F(A\mathcal{Q}) = O(1/A)$ — the certificate becomes vacuous, and it is the attacker of Corollary 1 that converts this into an actual loss; (ii) for p significant digits the* cell width *at y is proportional to $|y|$ up to one decade, so where A is a power of the base the mantissas, and hence both slack ratios, are carried to themselves* when the score is exactly homogeneous and $b_c = 0$*: there $F(A\mathcal{Q}) = F(\mathcal{Q})$ exactly. Under the asymptotic homogeneity of Theorem 5, or a nonzero b_c, they converge instead to Z_∞'s, so $F(10^k\mathcal{Q}) \to F_\infty$, the floor* at the limit *— positive unless a limiting score lands on a boundary, and in general* not *$F(\mathcal{Q})$. Either way F does not decay, against $F(A\mathcal{Q}) \to 0$ in case (i), which is the contrast the defense needs. Off the subsequence we claim no positive lower bound: a continuously varying A carries scores across their cells, and a uniform bound would need a margin hypothesis we neither assume nor need. Section S9 measures all three.*

Proof (sketch; full proof in Section S1). Rank: $\Delta + tUM^\top = U(SV^\top + tM^\top)$ has inner dimension r for every t, so the alternative stays in the class exactly, not to

first order. *Indistinguishability:* $t\Xi$ moves score (j, c) by exactly $t\, z_j^\top \Xi_c$; holding it short of the boundary it heads for leaves every returned cell unchanged, so no algorithm separates the two adapters and any estimator errs by $t\|\Xi\|_F/2$ on one — attained by the midpoint, hence tight. Only returned *cells* enter, so adaptivity is irrelevant. *Scaling:* an absolute grid caps both slacks at τ while the denominator grows like A, giving $O(1/A)$; a relative grid scales cell widths with $|y|$, so at A a power of the base the mantissas, and every slack ratio, are unchanged and A cancels — exactly under exact homogeneity, and otherwise in the limit, at Z_∞'s slacks. That subsequence is unbounded, so F does not decay; off it the mantissas still move within a decade, which constrains cell *width* and not distance to an edge — hence no lower bound there.

Read the quantifier carefully, because it is the difference between a certificate and a guarantee. F is evaluated *at the query set the attacker chose*: the bound holds for every algorithm, but each faces the bound computed at its own $\mathcal{Q}$, and nothing above lower-bounds $\inf_{\mathcal{Q}} F$. Part (ii) makes the floor stable along the *amplification* family $A\mathcal{Q}$, not every family, and Section 5.2 exhibits an attacker that drives T to zero deliberately, collapsing its own floor.

$F(\mathcal{Q})$ is also computable, and we evaluate it over 10 seeds on `digits`: it is sound in all 120 configurations, and its slope in A matches the theorem on both families (Section S8, Table 21).

5.2 The lever the floor does not cover: adaptive boundary seeking

That a relative grid keeps its floor *against these levers* rests on Theorem 5 and the measurements above. It does not survive this subsection. Amplification leaves the cell slacks where it finds them, which is why Theorem 6 can hold the relative floor up along $A\mathcal{Q}$. An attacker that treats the slacks as its own variables is not so constrained, and the construction is elementary.

Proposition 2 (Boundary seeking collapses the floor). *Let the oracle return $Q(y)$ with Q any deterministic quantizer whose cells are intervals partitioning score space, and let the attacker query $x = s\, e_i$ for scales s namable to d significant digits. Fix a class c, write $\theta = W_{ci}$, so the served score is affine, $y(s) = s\theta + b_c$; neither θ nor the bias b_c is known, while Q's cell boundaries are published. Fix two, $\beta_1 \neq \beta_2$, and set $\kappa = (|s_1| + |s_2|)/|s_1 - s_2|$. As y is monotone in s, the response at scale s reveals the sign of $s - s_k$ for $s_k = (\beta_k - b_c)/\theta$: one bit per query, halving a bracket on s_k. Two such searches identify the secret outright, as $\theta = (\beta_1 - \beta_2)/(s_1 - s_2)$ and $b_c = \beta_1 - \theta s_1$, stopping only when s can no longer be named finely enough: to d digits it is realized as $s(1 + \eta)$ with $|\eta| \leq 5 \cdot 10^{-d}$, so the bracket stalls at width $|s_k \eta|$ — at relative accuracy $\Theta(10^{-d})$ on each s_k and $\Theta(\kappa 10^{-d})$ on θ, independent of the oracle's ε. The bias is an* output *of the search, not an input, and nothing unpublished is used.*

Three consequences matter, none contradicting Theorem 6. The attacker brackets a boundary from both sides, so some score is tight in each direction and $T(\Xi) \to 0$; F at *its* query set is near zero: the theorem is satisfied and

Table 2. Adaptive boundary seeking against both quantizer families, 10 seeds, median relative error of Δ. Amplification runs at a matched budget of 4808 queries; boundary seeking is Proposition 2's two-boundary search, given neither the head nor the bias, costing 1899–8543. It erases the separation amplification reproduces — the columns agree to within 1.22× — its error set by request precision.

attacker	request precision	absolute (3 dec.)	relative (4 s.d.)
amplify + WLS, A=1	unbounded	$2.38 \cdot 10^{-5}$	$2.28 \cdot 10^{-6}$
amplify + WLS, A=64	unbounded	$3.83 \cdot 10^{-7}$	$2.08 \cdot 10^{-6}$
amplify + WLS, A=1024	unbounded	$2.35 \cdot 10^{-8}$	$2.47 \cdot 10^{-6}$
boundary seeking	4 significant digits	$1.88 \cdot 10^{-4}$	$1.94 \cdot 10^{-4}$
boundary seeking	8 significant digits	$1.81 \cdot 10^{-8}$	$1.75 \cdot 10^{-8}$
boundary seeking	12 significant digits	$1.76 \cdot 10^{-12}$	$1.88 \cdot 10^{-12}$
boundary seeking	unbounded	$5.21 \cdot 10^{-16}$	$4.26 \cdot 10^{-16}$

vacuous at once, which is what a bound quantified inside the attacker's choice of $\mathcal{Q}$ permits. Second, the halving is *oracle-agnostic* — it uses only that cells have boundaries — so both families fall together and the relative grid's advantage here is nil. Third, the scale factor is bounded, $s \to (\beta_k - b_c)/\theta$, so capping $\|x\|$ does not impede *this* search: it consumes fine input *resolution*, not large *norm*.

Table 2 runs both attackers over 10 seeds on a rank-2 adapter in $\mathbb{R}^{10\times 8}$. Against amplification the measurements reproduce the paper's central claim: the relative grid holds at 2.1–$2.5 \cdot 10^{-6}$ across $A = 1$ to 1024 while the absolute grid decays $2.4 \cdot 10^{-5} \to 2.4 \cdot 10^{-8}$. Against boundary seeking the two agree to within 1.03–1.22× everywhere, governed by the attacker's input resolution alone — down to $4.3 \cdot 10^{-16}$ at unbounded request precision, ten orders below the floor the same oracle enforces against amplification, with the bias recovered alongside the head.

That near-identity is real but it is not a floor: the attacker exhausts its bisection allowance in every one of the 80 coordinate–class pairs, and an attacker that reports its budget reports nothing about the defense. An attacker that also searches the *scale*, and so stops only when no scale it can name separates its bracket, halts on its own after 25–38 steps — and there the two families do separate. The lever it separates them by is the defender's query-norm cap: raising the cap from 10^2 to 10^6 is worth 7697× against the absolute grid and 2.08× against the relative one, so the relative grid leaves that attacker 8.2× worse off at a cap of 10^2 and $3.0 \cdot 10^4$× worse off at 10^6 (Section S11). Scale-invariance is thus worth nothing against a budget-limited boundary attacker and a great deal against a norm-limited one, which is the opposite of a defense one can recommend unconditionally.

The principle is known rather than new — adaptive threshold placement turns quantized measurements from a precision floor into an exponentially converging code [3, 14]. The contribution here is the negative one: it applies to significant-digit output quantization, so scale-invariant units are not by themselves a security property.

5.3 The floor that does survive: a discrete request format

Proposition 2 stops where the attacker's own resolution stops, which invites the question the rest of this paper deferred: what is $\inf_{\mathcal{Q}} F$? Once the request format is discrete — as every real one is, since a number must be serialized before it is sent — the infimum is not merely boundable but *exactly* computable, and it is positive.

Theorem 7 (Uniform floor under a discrete request format). *Let the secret be a scalar $\theta \neq 0$, let the oracle serve $Q_p(s\theta)$ rounding to p significant digits with ties away from zero, and let the attacker send any scale s expressible in d significant digits at any exponent, adaptively chosen, with no restriction beyond the format. Put*

$$T = \Big\{ \tfrac{N}{2M} 10^k \;:\; N \text{ odd},\ 2 \cdot 10^{p-1} < N < 2 \cdot 10^p,\ 10^{d-1} \le M < 10^d,\ k \in \mathbb{Z} \Big\},$$

and $T_\theta = \operatorname{sign}(\theta)\, T$, the thresholds on the secret's side of the origin (T is positive). Then (i) some *admissible query set distinguishes θ from θ' if and only if the signs differ, or a point of T_θ lies between them, counting one that meets the outer of the two. Write $u = |\theta|$, let a be the greatest point of T at or below u and b the least above it, and put $I = [a, b)$; $a = u$ exactly when $u \in T$. The secrets no admissible query set separates from θ are exactly $\operatorname{sign}(\theta)\, I$: every transcript's consistent set contains it, and the scales realizing a and b attain it. So $(b-a)/(a+b)$ is $\inf_{\mathcal{Q}}$, the minimax relative error on I, not a bound on it — an information radius, not an executable attack, since those scales depend on θ.* (ii) *$b - a \ge u\, 10^{-(p+d+1)}/4$, so every estimator has worst-case relative error at least $\frac{1}{8}\, 10^{-(p+d+1)}$, uniformly over every adaptive strategy at every budget.* (iii) *That constant is loose: T and the ratio in (i) are invariant under scaling by ten, so one decade gives the infimum exactly. For every $p \ge d$ it is at most $1/\big(2(10^d-1)(2\cdot 10^p-1)-1\big) \approx \frac{1}{4} 10^{-(p+d)}$ — some* twenty times *(ii) — witnessed by an explicit pair; over $p + d \le 6$ it equals that in twelve of the fifteen cells and is larger in the other three, each of which has $p < d$ (Section S13).*

Proof (sketch; full proof in Section S1). A transcript reveals only which side of each *achievable threshold* the secret lies on, and the achievable thresholds are exactly $\{\beta/s\}$ for β a cell boundary and s an admissible scale — which is the set T, since Q_p's boundaries are the odd multiples $N \cdot 10^k/2$. So the consistent set is the interval I of the statement, and the minimax relative error there is $(b-a)/(a+b)$: part (i). For (ii), two thresholds $\frac{N}{2M} 10^k$ and $\frac{N'}{2M'} 10^{k'}$ differ by $\frac{|NM' 10^k - N' M 10^{k'}|}{2MM'}$; the numerator is a nonzero integer multiple of a power of ten because N, N' are odd, so it is at least $10^{\min(k,k')}$, and $MM' < 10^{2d}$ with the exponents confined by $N, N' \sim 10^p$ bounds the gap below by $\frac{1}{4}\theta\, 10^{-(p+d+1)}$. Dividing by $a + b < 2b$ gives the bound. Both parts are for $\theta > 0$; the general case is exact reflection, since Q_p is odd, which is what T_θ records.

Remark 3 (The quantifier, stated where we can keep it). The secret above is a *scalar*, and for a scalar the theorem is unconditional: s ranges over the whole

request format, so no attacker is excluded and the bound really is over every strategy. An attacker recovering a matrix Δ coordinate by coordinate meets this floor at each entry. It does not follow that the matrix problem inherits it. A query x that is not a multiple of a basis vector constrains Δ through a *slab* $\{\langle x, \cdot\rangle \in \text{cell}\}$, and an intersection of slabs in several directions can be smaller than the box the per-coordinate intervals give. We have not settled whether combining directions beats coordinate probing, so we claim the floor for the scalar problem and for coordinate-probing attackers on the matrix problem, and not for every attacker on the matrix problem. Theorem 6 does not draw this distinction; here it is drawn explicitly.

Three things follow. First, within the scope Remark 3 fixes, this is the uniform statement Theorem 6 does not make: the quantifier sits outside $\mathcal{Q}$, so on that problem it is a defense guarantee and not a transcript certificate. Second, it is attained — an attacker that searches the boundary and the scale *together*, rather than scaling one fixed boundary as Proposition 2 does, tracks it (Section S13). Third, p and d enter *multiplicatively*. Serving more digits does not merely fail to help the defender; it hands the attacker a finer ruler, because a quantizer's boundaries are exactly the graduations its own precision provides. The two controls are one control, and neither alone fixes the floor.

Remark 4 (The request format is a modeling assumption, and it carries the theorem). "d decimal significant digits" describes a server that *canonicalizes* what it parses, not one that merely reads JSON: there the reachable scales are the binary64 grid, and the attacker aiming a threshold at θ misses only by that format's rounding unit, a displacement set by the request path with p playing no part. Worse, every cell boundary offers another scale to try; Section S13 measures the closest landing still falling as more are tried, below what binary64 itself resolves — a measurement out of resolution rather than a vanishing infimum, which over a finite format is attained but lies beneath any level a defender could rely on. We therefore state Theorem 7 for what it is: an exact result about a deliberately discretized request format, not an evaluated theorem about IEEE-754 parsing, and the two are not related by a constant — which is why we present it as a boundary of recoverability, not a recommendation.

What this leaves of the defense. Relative quantization defeats the amplification lever, and that claim survives intact: it is the attack an adversary can mount without fine control of its own queries, and against it the separation is real and large. What does not survive is the universal reading. A defender who quantizes relatively has closed one lever and should assume the other open unless the interface also bounds request precision — a defense on the query side rather than the answer side, and the one our measurements make binding. The change remains cheap, reformatting scores rather than touching the model, and is already the format used for probabilities, where normalization fixes the scale; the vulnerability is specific to APIs returning *unnormalized* scores. It is not an argument for more aggressive quantization (Section 8.5).

What limits the attacker. Corollary 1(1) assumes the amplified queries stay reachable and asymptotically homogeneous; both can fail, and that failure is the defender's other protection (Section S19). Under a top-k oracle amplification additionally drives the argmax into one cell, so most classes stop being returned; the attacker partly repairs this from the *public* base model at no query cost.

6 Hard-Label, Ranking, and Partial-Index Oracles

Theorem 8 (Boundary recovery and scale limits). *For binary affine classifiers, d affinely independent boundary points determine a functionally equivalent hard-label classifier up to sign and positive scale. For multiclass affine classifiers, or for public-feature heads, oriented recovery of every relevant pairwise boundary determines a functionally equivalent top-label classifier by pairwise voting, and complete index-only rankings reveal the same pairwise boundaries through rank-order flips. Partial top-k index-only oracles give the same guarantee only for class pairs co-visible in the returned set on both sides of the boundary.*

Hard-label and ranking transcripts do not identify logit scale or adapter magnitude. Moreover, for any $k < m$, there exist two distinct rank-one adapters for a known-base affine classifier that induce identical top-k index-only transcripts on a bounded query domain.

Proof. Deferred to Section S1. The non-identifiability construction hides a class below the top-k cut and varies its adapter row freely underneath; we run it as an executable certificate in Table 26.

7 Endpoint Oracles and the Downstream Quotient

Theorem 9 (Endpoint quotient and observability dichotomy). *Let a deployed classifier expose only endpoint logits $y(x) = Ch(x) + \beta$. Endpoint logits identify hidden effects only through the quotient map induced by C: two hidden perturbations $\delta_1(x), \delta_2(x)$ are endpoint-indistinguishable exactly when $\delta_1(x) - \delta_2(x) \in \ker C$ for every queried x.*

More generally, fix a query set Q and a linear space $\mathcal{S}$ of hidden adapter effects $\delta : Q \to \mathbb{R}^h$. Endpoint logits identify $\delta \in \mathcal{S}$ exactly iff $\mathcal{L}_Q : \delta \mapsto (C\delta(x))_{x \in Q}$ is injective on $\mathcal{S}$. Hence withholding hidden states and intermediate scores while returning only endpoint logits is utility-preserving for standard prediction but hides every hidden adapter component whose effect lies in $\ker C$.

Moreover, if $\ker C \neq \{0\}$, then for any nonzero $u \in \ker C$ and any nonzero linear form $v^\top x$, the rank-one hidden adapter effect $\delta(x) = u\, v^\top x$ is nonzero but endpoint-invisible for all inputs. This last part exhibits an effect *at the final hidden representation and does not assert that a perturbation of a given upstream module realizes it; realizability is architecture-specific, and Section 8.3 settles it empirically for one trained LoRA module rather than in general, as local conditioning with a measured radius (Section S6) rather than as exact indistinguishability.*

Proof. Deferred to Section S1. Every transcript depends on δ only through $(C\delta(x))_{x\in Q}$, and so the observable object is $\delta(x)$ modulo $\ker C$.

Theorem 10 (Jacobian observability). *Let $T_Q(\theta) = (y_\theta(x))_{x\in Q}$ be the endpoint-logit transcript of a hidden adapter with parameter vector θ, differentiable at θ_0 with Jacobian $J_Q(\theta_0)$. For every $v \in \ker J_Q(\theta_0)$, $T_Q(\theta_0 + \epsilon v) = T_Q(\theta_0) + o(\epsilon)$. Hence $\operatorname{rank} J_Q(\theta_0)$ upper-bounds the number of locally observable adapter degrees of freedom, and if it is less than $\dim(\theta)$, endpoint queries leave at least $\dim(\theta) - \operatorname{rank} J_Q(\theta_0)$ locally unobservable tangent directions. If the adapter-to-endpoint map is affine on a neighborhood along a subspace V, any $v \in V \cap \ker J_Q(\theta_0)$ gives exact transcript equality there.*

Corollary 3 counts what that leaves, the LoRA factorization's own gauge symmetry costing r^2 directions before any endpoint restriction. The positive endpoint cases all reduce to Theorem 1 once a hidden state is recovered, and are collected with it in Section S7: Proposition 3 transfers a hidden LoRA block on one ReLU cell; Corollary 2 inverts the endpoint exactly when $\operatorname{rank}(C) = h$, so any attack assuming final-hidden-state access applies unchanged, and only the rank-deficient case yields the quotient; Proposition 4 characterizes when a *subset* of output coordinates suffices, chosen by rank-revealing algebra on the *public* head at zero oracle cost — which is what reduces full-vocabulary probing to $h+1$ rows — and Corollary 4 shows a documented logit-bias interface [39, 40] instantiates it, including a scoreless variant.

8 Experiments

8.1 Methodology

We state the protocol first, since the threat model and recovery target change between settings.

Targets. Six families, from synthetic known-base affine targets through real PEFT LoRA checkpoints on GLUE/SST-2 and vision endpoint heads on CIFAR-10, each specified in Section S17; oracles are as defined in Section 3.

Query accounting. We report oracle calls, not equations: repetitions, forced rows, binary-search depth and probes that return no equation are all charged in full. Stopping is not adaptive — each attack spends its stated budget and then reports which class rows it solved, so failures are visible rather than hidden by an early exit — and rank-adaptive variants never receive the true rank. Metrics follow Definition 4.

Baselines and controlled validation. Every theorem hypothesis is first checked on controlled synthetic targets with exact ground truth, recovering at $5.69\cdot10^{-16}$ to $2.73\cdot10^{-12}$ in the positive cases and yielding executable certificates rather than failed attacks in the negative ones; prior LoRA extraction [53] reports functional mimicry, so a hard-label surrogate stands in as a functional baseline (Section S17).

Real-data public-feature heads. Against an adapted head over a public standardized-PCA/ReLU map on `digits` the whole oracle family reproduces the boundary row by row: exact recovery at 9 queries, and the scoreless row leaves the target unidentified in 80% of seeds while a functional surrogate still reaches 0.915 agreement — a restriction that removes the equations rather than perturbing them. Section S16 and Table 17 give the detail.

8.2 Observable transformer endpoint heads

Theorem 1 scales to real transformer endpoints where the adapted object is the final head over frozen public features. On 20 Newsgroups over frozen `bert-base-uncased`, 900 final-logit queries recover the low-rank head update to $6.52 \cdot 10^{-10}$ at 1.0000 label agreement, the augmented rank of 769 of 769 reported rather than assumed (Table 19) — the observable adapted head, not an arbitrary hidden one. It also scales to production-size vocabulary heads, where full column rank makes the endpoint invertible: `distilgpt2` (50,257 rows) recovers a rank-4 adapter to $3.09 \cdot 10^{-5}$ from 28 prompts, a scoreless index-bias variant to $1.69 \cdot 10^{-3}$, and BERT MLM to $7.32 \cdot 10^{-6}$ from 5 scored observations per row beyond the first 4, 176.5× fewer scalars than the 900-query full-logit transcript that recovers it to $7.64 \cdot 10^{-8}$. Rank-revealing QR on the *public* head selects the 769 rows at no oracle cost. Section S15 gives the detail.

8.3 Hidden adapters behind a rank-deficient endpoint

Moving the adapter *upstream* of the final head, where Theorem 9 predicts endpoint logits see it only through $\ker C$, we test both sides of that prediction on one real trained checkpoint. This is the paper's sharpest attack/impossibility contrast, and both halves use the *same* rank-4 query-projection LoRA module on `bert-base-uncased`/SST-2 (10 seeds, 0.8756 test accuracy). Under module-output access the update BA is recovered at $8.37 \cdot 10^{-15}$ relative error with 769 queries. Behind the two-class classifier head, 99.74% of hidden directions lie in $\ker C$, and 16 endpoint examples constrain only 32 of 6128 non-gauge update directions (0.52%). The certificate is constructive, not merely dimensional, and over ten seeds a step sweep makes the single example a stable regularity: at $t = 10^{-3}$, stepping along a *Jacobian*-null unit direction in factor space moves the merged update by $1.20 \cdot 10^{-3}$ and endpoint logits by $2.79 \cdot 10^{-9}$, flipping no labels, and over four decades the drift scales with exponent 1.996 against the update shift's 1.026 (float64). The direction is null to first order only, so what survives is curvature, $L = 2.22 \cdot 10^{-2}$. The drift is not zero, so an *exact*-logit oracle distinguishes the two in principle: severe ill-conditioning, not exact non-identifiability, the exact statement remaining Theorem 9's quotient. What the sweep buys is a radius *predicted*, not proved: a quantizer hides steps below $\sqrt{2\gamma/L}$, for γ the smaller of Theorem 6's two slacks — smaller because a null perturbation's sign is not ours to choose, and a slack not the cell width because rounding separates values closer than a cell that straddle an edge. On a 10^{-4} grid the slack gives $t = 5.88 \cdot 10^{-3}$, a merged-update movement of $7.03 \cdot 10^{-3}$

— 5.88× the certificate in either unit — at a precision Section 5's quantizers already supply. Section S6 gives the sweep, and Table 3 every row.

8.4 Adaptive attackers against score degradation

Holding the defense fixed and giving the attacker successive levers at $A = 64$ and $R_{\text{rep}} = 64$ reproduces Corollary 1 closely (Table 18). Repetition is worth 1.00× against deterministic rounding — unchanged to three significant figures, because rounding is a function of the query — and 8.6/7.2× against additive noise (predicted 8.0×); amplification is worth 70.8/62.8× (predicted 64×); the intercept probe cuts bias error 3.2× while leaving head error untouched, the two being governed by different columns of the design. End to end, adaptation improves recovery by 81× to 780× across the four defenses, at label agreement never below 0.9176. The top-k oracle is the honest limit (Table 29): amplification is actively *harmful* there, driving the argmax into one cell, so 40% of seeds leave a class row unsolved and coverage-aware selection cuts that only to 30%.

Where the lever exists. Amplification needs three structural conditions, and on real transformer and vision models each fails somewhere: LayerNorm, pixel-domain bounds and ReLU inactivity remove one apiece (Section S20). Which regime a deployment is in therefore decides what the units are worth, and it is a property of the feature map a defender can measure without knowing the adapter.

8.5 Absolute versus relative quantization at matched utility

Testing Theorem 5 fairly needs a defender-then-attacker game: the defender picks a quantizer and pays a utility cost, then the attacker best-responds over six levers — amplification, D-optimal selection, repetition, the intercept probe, rank projection, cell-width weighting. On `digits` (Table 24) amplification buys 949–1088× against an absolute grid and 1.2–1.3× against relative quantization, and at *identical* utility the attacker's best error is 456×, 388× and 430× larger under relative units. The residual 1.2–1.3× is expected: Theorem 5 gives exact invariance only where A is a power of the quantization base. Utilities coincide exactly here only because 78% of this target's logits lie in $[0.1, 1)$, where one significant digit *is* one decimal place; the trained adapter below does not rely on it. Table 30 extends this across the oracle family: *removing* the linear signal holds utility while driving the observable quotient to 0.0000 and 0.0026.

Top-1 survival is too coarse to settle a choice between quantizers alone. Re-matching on the served *distribution* (total variation or KL) selects the *same* relative partner everywhere and leaves all three gains unchanged, and proper scoring rules against true labels cannot arbitrate it (Section S10).

On a trained adapter. Everything above attacks an adapter nobody trained, so we rerun the whole game against the *same* rank-4 LoRA target as Section 8.3, retrained over 10 seeds (768 × 768, `bert-base-uncased`/SST-2, 0.8756 mean

accuracy), through $\mathcal{O}_{\text{module}}$, where Section S19 establishes all three conditions. The attacker chooses its own design — Theorem 4's $\|X^+\|_2$ lever — and utility is measured on the hidden states the deployed module really sees. Table 15 (Section S23) gives every setting.

Takeaway. The claim survives contact with a trained adapter, and the prediction is met almost exactly: amplification buys 1015/1026/1034× against the absolute grid — the predicted $A = 1024$ — and 1.9× against every relative setting. Pairing each absolute setting with a relative one that costs the service *less*, relative quantization leaves the attacker 178/179/182× worse off (95% paired bootstrap over 10 seeds: $[168, 187]$, $[171, 187]$, $[172, 193]$). Bootstrap coverage at $n = 10$ is not to be trusted, so also, distribution-free: relative is worse for the attacker on 10/10 seeds in all three pairings, exact two-sided sign test $p = 0.00195$ — the smallest 10 seeds can produce, so the design and not the effect limits it. The exact-arithmetic control sits at $1.8 \cdot 10^{-14}$ throughout, so no cell is a broken attack reported as a defense holding. The attacker here is the *weighted* one; unweighted it sees 252–256× on identical transcripts, so we report the smaller number. All of this is against amplification: under budget-limited boundary seeking (Section 5.2) the same target separates the families by 1.00×.

Where the breach generalizes. Exposure to amplification and to averaging is decided by two *independent* properties of a perturbation, so the four combinations form a taxonomy whose *absolute* row holds the deployed defenses [12, 42, 48]. All eight predicted outcomes hold, spending an absolute budget optimally rather than by rounding does not help the defender, and only changing the units recovers protection against *these* levers (Section S21) — not against Section 5.2's.

9 Limitations

We attack *observable* adapted heads, not arbitrary hidden ones, and the reachability hypothesis after Theorem 1 is verified rather than proved. The order-of-magnitude figures hold where all three conditions of Section S19 do; on a bounded-input $\mathcal{O}_{\text{end}}$ deployment the gain is an intrinsic 1.40× that one seed in ten reverses, and we surveyed no production API exposing the full lever. The endpoint impossibility results are local, finite-query certificates; the quantitative one measures conditioning, not indistinguishability (Section S6). Theorem 6 certifies a chosen query set, and $\inf_{\mathcal{Q}} F$ is zero while the attacker's scale is continuous (Section 5.2); Theorem 7 computes that infimum exactly, but for a *scalar* secret and only under a deliberately discretized request format, having no useful analogue under binary64 parsing (Remark 4). The noisy server we measure is Gaussian (Section S12). Sections S24 and S25 cover compute and disclosure.

10 Conclusion and Defensive Implications

Recoverability follows what an interface emits: score-bearing oracles make observable heads exactly recoverable, while hard labels leave a scale class, index-

only outputs a boundary arrangement, and a rank-deficient endpoint a quotient. Defenses resting on non-identifiability cost no utility and survive every adaptation (Theorems 8, 9) — the sturdiest finding here.

Every defense that merely *perturbs* scores fared worse the harder we attacked it. Units matter against amplification, but boundary seeking erases the distinction: quantizing bounds only an attacker that cannot finely control its queries. What bounded recovery was the *request*: its norm and its resolution, and coarsening that resolution only looks like a defense. Since $p + d \geq 2$, no scalar secret — and no matrix one (Remark 3) — can claim a floor above $2.93 \cdot 10^{-3}$: an error at which the adapter is already recovered, bought at 61 accuracy points on a pixel API (Section S14). The floor is nonincreasing in d, since a d-digit scale is also a $(d+1)$-digit one, and it is a ceiling on what can be guaranteed: individual secrets sit at larger per-secret floors (Table 10). So we make no rounding-based recommendation: treat the norm cap as load-bearing, relative units as a cheap improvement, and reserve *security* for interfaces that never emit the signal.

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Supporting Material

This material supports verification of the results. It is clearly labeled and placed after the references, as required. The main body is self-contained and intelligible without it.

S1 Deferred Proofs

The following proofs are omitted from the main body for length. Each is referenced from the statement it establishes.

Proof (of Theorem 2). The spectral gap separates the true row-space dimension from numerical or perturbation directions, so the right singular vectors above threshold span the row space of Δ. Proposition 1 then applies with $r = \rho$. If perturbations create singular values above threshold, the same computation recovers the numerical rank selected by the threshold, which is why the experiments report both the estimate and the spectral gap.

Proof (of Proposition 1). Once the row space is known, every adapter row has the form $\delta_c = \alpha_c B$. A returned score for class c gives $y_{i,c} - \langle h_{0,c}, \phi(x_i)\rangle = \langle \alpha_c, B\phi(x_i)\rangle + b_c$, so the unknowns are only the $r + 1$ scalars α_c and b_c. Full rank of the projected augmented design gives a nonsingular system, reducing the per-class requirement from $h + 1$ to $r + 1$.

Proof (of Theorem 3). Let $S \subseteq \mathbb{R}^h$ be the s-dimensional span of all adapter rows fully identified by the transcript. Since $s < r$, choose two distinct $(r - s)$-dimensional subspaces T_1, T_2 with $S \oplus T_1 \neq S \oplus T_2$; this is the step that uses $r < h$, since at $r = h$ both sums exhaust $\mathbb{R}^h$ and no two completions differ. Construct adapters whose observed rows equal the transcript-identified rows and whose unconstrained rows complete the row space using vectors from T_1 or T_2 respectively. The transcript contains no equations involving these complementary directions, so both adapters give identical returned scores on every query in the transcript, while their row spaces differ.

It remains to check that the *returned index sets* also agree, since a top-k transcript reveals them and a completion that reordered the argmax would be detectable. Scale each completing row by $t > 0$. This changes neither the rank nor the completed row space — $\operatorname{span}(tT_i) = \operatorname{span} T_i$ — and it does not touch the identified rows, so it does not touch the returned scores; it perturbs only the scores of the *unreturned* classes, continuously and proportionally to t. Let $\mu > 0$ be the smallest margin over the finitely many queries by which a returned score exceeds an unreturned one, which the hypothesis supplies. Choosing t small enough that every unreturned score moves by less than μ in either adapter leaves every returned index set unchanged. Both adapters then induce identical transcripts in the full sense.

Proof (of Theorem 5). For (1), asymptotic homogeneity gives $Z(A) = A(Z_\infty + o(1))$, and the head is linear in the features, so the exact score block satisfies $Y(A) = A(Y_\infty + o(1))$ with $Y_\infty = Z_\infty \Theta^\top$. The intercept is why the exact clause carries $b = 0$. With a bias the served block is $Y(A) = AZ_\infty\Theta^\top + b$, which is not $AY(1)$ for any $A \neq 1$, so $Q_\varepsilon(Y(A)) = A\,Q_\varepsilon(Y(1))$ fails and the mantissas are not carried to themselves — the same obstruction Theorem 6 states as $b_c = 0$. The asymptotic band is untouched, since $b/A \to 0$ is absorbed by the $o(1)$; only the exact clause needs the condition, and the gap is visible at once: on an exactly homogeneous synthetic linear head ($m = 6$ classes, $h = 4$ features, $n = 12$ queries, served at $p = 4$ significant digits, 20 seeds) the least-squares error across $A = 10^0, \ldots, 10^4$ is constant to floating-point resolution when $b = 0$, and moves in every seed when it is not, by a median $1.47\times$ and up to $2.12\times$. The check is recomputed by the test suite rather than read from

a stored run, since it is a two-second deterministic calculation. Relative quantization commutes with scaling by a power of the base, $Q_\varepsilon(cY) = c\,Q_\varepsilon(Y)$, so the perturbation obeys $E(A) = A(E_\infty + o(1))$; off that subsequence the two sides differ by at most one quantization step, which is again proportional to the score and so carries the same factor A. The head block of the design has pseudoinverse $Z(A)^+ = A^{-1}(Z_\infty^+ + o(1))$, using that Z_∞ has full column rank so the pseudoinverse is continuous there. Hence the recovered perturbation is $Z(A)^+E(A) = Z_\infty^+E_\infty + o(1)$ along that subsequence, a limit independent of A; when φ is exactly homogeneous the $o(1)$ terms vanish and the invariance is exact. Off the subsequence the one-step discrepancy in $E(A)$ does not vanish as A grows — it is proportional to the score, so it scales with A exactly as the signal does — and this is why the statement claims a band rather than a limit: what survives is $\|Z(A)^+E(A)\| \le \|Z_\infty^+E_\infty\| + \varepsilon\|Z_\infty^+Y_\infty\| + o(1)$, with no accompanying convergence. Under an absolute grid, $\|E\|_F \le \tau\sqrt{nm}/2$ irrespective of A, so the same computation yields $Z(A)^+E = A^{-1}(Z_\infty^+ + o(1))E$, which decays like $1/A$. Part (2) is immediate from determinism, and its closing claim — that neither lever moves the least-squares error, which stays $O(\varepsilon)$ for that estimator — follows by combining it with (1) and Theorem 4.

Proof (of Corollary 1). For (1), write the augmented design as $X = [Z\ \mathbf{1}]$. Scaling queries so that $Z \mapsto AZ$ leaves the constant column fixed. On the subspace spanned by the non-constant columns the singular values scale by A, so the corresponding block of X^+ scales by $1/A$, while a fixed absolute grid leaves $\|E\|_F$ unchanged. Theorem 4 then bounds the linear-head error by $O(1/A)$. The component of X^+ along the constant column is unchanged, so the intercept error is not improved. For (2), averaging R independent draws of mean-zero noise with standard deviation σ yields an error matrix with entrywise standard deviation $\sigma/\sqrt{R}$, hence $\mathbb{E}\|E\|_F$ scales by $1/\sqrt{R}$; deterministic rounding is a function of the query alone, so repeated queries return identical values and the averaged E equals the single-query E. For (3), let $\widetilde{\Delta} = \Delta + N$ with Δ of rank at most $R_{\max}$. The best rank-$R_{\max}$ approximation $\Pi(\widetilde{\Delta})$ satisfies $\|\Pi(\widetilde{\Delta}) - \Delta\|_F \le 2\|N\|_F$ by optimality of the truncated SVD and the triangle inequality, and discards the components of N orthogonal to the manifold. What that bound establishes is a constant factor, never a change of rate; it does not establish that projection strictly reduces the norm, and item (3) claims only the factor.

Proof (of Theorem 6). Rank first. For a one-sided direction, $\Delta + tUM^\top = U(SV^\top + tM^\top)$ and $\Delta + tNV^\top = (US + tN)V^\top$, each a product with an inner dimension of r, so both have rank at most r for *every* t — not merely to first order. (A two-sided combination $UM^\top + NV^\top$ is tangent to the manifold but leaves it, reaching rank $2r$, which is why the supremum is taken over one-sided directions only.) The bound is stated against an algorithm given the true rank, so the alternative must have rank exactly r, not merely at most r. It does, off a finite set: Δ has rank r, so some $r \times r$ minor of it is nonzero, and that minor of $\Delta + t\Xi$ is a polynomial in t of degree at most r which does not vanish at $t = 0$. A nonzero polynomial has at most r roots, so all but finitely

many $t \in [0, T(\Xi))$ give rank exactly r. Since F is a supremum over $t \to T(\Xi)$, deleting those finitely many values leaves it unchanged, and knowing r buys the algorithm nothing.

Indistinguishability next. Q is constant on each cell, so $Q(y') = Q(y)$ whenever y' lies in y's cell: whenever the displacement $y' - y$ is below $\gamma^{\uparrow}$ if it is positive, and above $-\gamma^{\downarrow}$ if it is negative. Perturbing the adapter by $t\Xi$ changes the score of query j on class c by exactly $t\, z_j^{\top} \Xi_c$, whose sign fixes which boundary of the cell it approaches; by the choice of t it stays short of that one, the opposite boundary being irrelevant to a displacement moving away from it. Every returned value is therefore unchanged, and the two adapters are transcript-equal. Definition 3 then places them in the same equivalence class, so no algorithm separates them; any estimator $\widehat{\Theta}$ has $\max\{\|\widehat{\Theta} - \Delta\|_F, \|\widehat{\Theta} - \Delta - t\Xi\|_F\} \geq t\|\Xi\|_F/2$ by the triangle inequality — and this is tight *for absolute loss*, since an estimator returning the midpoint achieves it. Taking the supremum over admissible t and Ξ gives exactly $F(\mathcal{Q})$ as defined, with the $\frac{1}{2}$ carried in the definition rather than dropped.

The reported metric is relative (Definition 4), and under it the midpoint is *not* optimal, because the two hypotheses are normalized by different norms. Write $D = t\|\Xi\|_F$, $n_1 = \|\Delta\|_F$ and $n_2 = \|\Delta + t\Xi\|_F$. An estimator at distance x from the first along the segment joining them has relative risk $\max\{x/n_1, (D-x)/n_2\}$, least where the two arguments agree, at $x = Dn_1/(n_1 + n_2)$; the exact two-point relative minimax is therefore

$$\frac{D}{n_1 + n_2}, \qquad \text{not} \qquad \frac{D}{2n_1}.$$

The scalar pair $\{1, 2\}$ makes the gap concrete: its relative minimax is $1/3$, not $1/2$. The two agree only when $n_2 = n_1$, and $D/(2n_1)$ *exceeds* the truth whenever $n_2 > n_1$, so the $\frac{1}{2}$ is not tight here and not conservative either. It is off by a controlled amount. Fix one admissible pair (t, Ξ) and write $F_{t,\Xi} = t\|\Xi\|_F/(2n_1)$, so that $D = 2F_{t,\Xi}n_1$; then $n_2 \leq n_1 + D$ gives $n_1 + n_2 \leq n_1\big(2 + 2F_{t,\Xi}\big)$ and

$$\frac{D}{n_1 + n_2} \;\geq\; \frac{F_{t,\Xi}}{1 + F_{t,\Xi}}.$$

The map $x \mapsto x/(1 + x)$ is increasing and continuous on $[0, \infty]$, so taking the supremum over admissible pairs passes it through to $F(\mathcal{Q}) = \sup_{t,\Xi} F_{t,\Xi}$ whether or not that supremum is attained, and sends $F = \infty$ to 1. Hence $F(\mathcal{Q})/(1+F(\mathcal{Q}))$ is the relative floor Theorem 6 states, and $F(\mathcal{Q})$ — which is what the tables report, and what $\frac{1}{2}$ makes tight for absolute loss — exceeds it by the factor $1 + F(\mathcal{Q})$. Every floor this paper reports is at most 10^{-2}, so that factor is at most 1.01: it moves one printed entry, the $9.90 \cdot 10^{-3}$ of Table 21, to $9.80 \cdot 10^{-3}$, and no other, the certificate sweeps near $2.5 \cdot 10^{-5}$ being corrected below their third significant digit. No comparison in the paper is closer than an order of magnitude, so none turns on it.

Since the argument uses only that the transcript is a function of the returned cells, it is indifferent to how $\mathcal{Q}$ was chosen, including adaptively.

Scaling last. Write $z_j(A)$ for the amplified design. For an absolute grid the numerator is capped, $\gamma_{jc}^{\uparrow}, \gamma_{jc}^{\downarrow} \leq \tau$, while asymptotic homogeneity gives

$|z_j(A)^\top \Xi_c| = A|z_{j,\infty}^\top \Xi_c|(1 + o(1))$. The minimum over j is what makes this bite: Z_∞ has full column rank, so every $\Xi_c \neq 0$ has some row with $|z_{j,\infty}^\top \Xi_c| \geq \sigma_{\min}(Z_\infty)\|\Xi_c\|$, whence $T(\Xi) \leq \tau(1 + o(1))/(A\sigma_{\min}\|\Xi_c\|)$ and $F = O(1/A)$; the floor vanishes, which is the direction the security claim needs. (A matching lower bound would additionally require the slacks to stay bounded away from zero, which holds generically but not for every Δ, so we claim only the upper bound.) For p significant digits the cell containing $y \in [10^k, 10^{k+1})$ has width 10^{k-p+1}, so the cell width is proportional to $|y|$ up to one decade. Suppose A is a power of the quantization base. If the score is exactly homogeneous and $b_c = 0$ then $y_{jc}(A) = Ay_{jc}(1)$ with A a power of ten, so the mantissas of the scores are literally unchanged, hence so is every slack ratio at that score; the factor A cancels between numerator and denominator and $F(A\mathcal{Q}) = F(\mathcal{Q})$ exactly.

Under the asymptotic homogeneity we actually assume this last step fails, and it is worth being exact about where. We have $y_{jc}(A) = A\big(z_{j,\infty}^\top \Delta_c + o(1)\big) + b_c$, so $y_{jc}(A)/A \to z_{j,\infty}^\top \Delta_c$; dividing by a power of ten does not move a mantissa, so the mantissa of $y_{jc}(A)$ converges to that of $z_{j,\infty}^\top \Delta_c$ — the limiting block's, not the $A = 1$ block's, and the two differ by $\Theta(1)$ whenever $Z_\infty \neq Z(1)$. The same holds for a nonzero intercept, whose contribution to the mantissa is $O(1/A)$ and so is lost in the limit rather than preserved from $A = 1$. Hence each slack ratio at $y_{jc}(A)$ converges to the relative slack Z_∞ induces, and $F(A\mathcal{Q}) \to F_\infty$, the floor evaluated at Z_∞. Provided no limiting score lies on a cell boundary — which would send its slack, and with it F_∞, to zero — $F_\infty > 0$, and since the subsequence is unbounded the limsup over all A is at least F_∞. What the argument does *not* give is equality there — off the subsequence F may fall on either side of F_∞, since a mantissa carried off its limiting position can sit nearer a boundary or further from one, so we claim no uniform bound and no strict inequality — nor $F_\infty = F(\mathcal{Q})$: the limit is a floor at a different transcript geometry, and Section S9 measures the gap. The security claim needs only that the limit is positive while the absolute grid's is zero, and that survives.

Off the subsequence we stop there, and the reason is the same one already recorded for the absolute case above. The mantissas do move within a fixed decade, but that constrains the cell *width*, not the distance to the cell's edge: as A varies continuously some $y_{jc}(A)$ crosses a boundary, and on either side of that crossing one of its two slacks is arbitrarily small. That alone does not send T to zero — the perturbation may take the sign that moves the score away, which is why the floor is not fragile at a typical amplification (Section S9) — but neither does anything hold the two directions apart: scores tight on opposite sides occur, and the attacker of Section 5.2 arranges them deliberately. Boundedness of the mantissa therefore gives no positive lower bound on F, and we do not assert one.

Proof (of Proposition 2). The search runs in the scale variable, which is the attacker's, rather than in the secret, which would require the bias to convert between the two.

One boundary. Fix $\beta = \beta_k$ and suppose $\theta > 0$; the case $\theta < 0$ is the reflection, and $\theta = 0$ is read off the first query. The exact score $y(s) = s\theta + b_c$ is strictly increasing in s, so $y(s) \geq \beta$ exactly when $s \geq s_k = (\beta - b_c)/\theta$. Because β is a

boundary between two cells of Q and Q is constant on cells, $Q(y(s))$ takes values in distinct cells on the two sides of s_k, so a single query at s reveals the sign of $s - s_k$. Note what this does *not* use: the attacker never has to place y at β, only to compare, so it needs no equation involving b_c. Given any bracket $[\ell, u] \ni s_k$, querying its midpoint and keeping the indicated half gives width $2^{-n}(u-\ell)$ after n responses. An initial bracket is supplied by two unamplified queries: $Q(y(0))$ and $Q(y(1))$ pin b_c and $\theta + b_c$ to the served precision, hence s_k to it as well, and the bracket is widened by doubling until the two endpoints report opposite sides.

Two boundaries. Running this for $\beta_1 \neq \beta_2$ yields s_1, s_2 with $\beta_k = s_k\theta + b_c$, two linear equations in the two unknowns whose determinant is $s_1 - s_2 \neq 0$ precisely because $\beta_1 \neq \beta_2$. Solving gives the displayed θ and b_c.

Halting. The condition is the attacker's own precision, not the oracle's; ε enters only through the initial bracket. If s is representable to d significant digits then the realized scale is $s(1+\eta)$ with $|\eta| \leq 5 \cdot 10^{-d}$, so once $u - \ell$ falls below $|s_k|\,|\eta|$ the reported side no longer tracks $s - s_k$ and the bracket stops contracting. Each s_k is therefore known to relative accuracy $\Theta(10^{-d})$. The quotient $(\beta_1-\beta_2)/(s_1-s_2)$ does not inherit that accuracy for free: differentiating, a relative perturbation η in each s_k moves θ by a relative $\kappa|\eta|$ at worst, with $\kappa = (|s_1| + |s_2|)/|s_1 - s_2|$ the condition number the statement names — which is why the statement carries κ rather than merely requiring $\beta_1 \neq \beta_2$, and why κ cannot be dropped: as $s_2 \to s_1$ it diverges and the two searches stop determining θ at all. The experiment takes β_1, β_2 a decade apart, where $\kappa = O(1)$ and the quotient does inherit the accuracy, with no dependence on ε. Interval cells are what make the bisection legitimate: monotone y crosses each boundary once, so a response is a threshold test on s; cells that were not intervals would return the same label on both sides and carry no bit. Finally $s_k = (\beta_k - b_c)/\theta$ is a finite constant, so the queries required are bounded in norm.

Proof (of Theorem 7). Two reductions first, since the argument below is written for a positive secret probed by a positive scale and neither restriction is free.

Signs of the secret. Significant-digit rounding is an odd function, $Q_p(-y) = -Q_p(y)$: the exponent it extracts depends on $|y|$ and the rounding of the mantissa is symmetric about zero. Hence the transcript produced by the secret $-\theta$ under any query set is the entrywise negation of the transcript produced by θ under the same query set, so the two induce the same partition of secrets into indistinguishable classes up to reflection. It therefore suffices to prove (i) for $\theta > 0$ with threshold set T; the case $\theta < 0$ follows by reflecting, which replaces T by $-T$, and the two cases are exactly the statement's T_θ. Secrets of opposite sign are separated by any single query, since the served values then differ in sign. This is why the statement is about T_θ and not T: T contains no negative number, so for two negative secrets it would vacuously certify that no query separates them, which is false.

Signs of the scale. Allowing $s < 0$ adds no thresholds. The boundaries of Q_p are symmetric about the origin, so for $s < 0$ the pullback $\{\beta/s\}$ of the negative

boundaries is the set of positive numbers $\{|\beta|/|s|\}$, which is what $|s|$ already contributes. We may thus take $s > 0$ throughout.

Ties. Resolving ties away from zero makes the cells of Q_p half-open, $[\beta, \beta')$ between consecutive boundaries, so every real number lies in exactly one cell and each boundary belongs to the cell above it. The convention is not cosmetic, and the theorem is false without one that sends every threshold the same way: under ties-to-even a boundary is assigned to whichever neighbour has an even last digit, so the *same* threshold can be resolved in opposite directions by two admissible scales. With $p = d = 1$ and $\theta = 3/2$ the scale 1 serves $Q_1(3/2) = 2$ and the scale 3 serves $Q_1(9/2) = 4$; the first response confines the secret to $[3/2, 5/2]$ and the second to $[7/6, 3/2]$, whose intersection is the single point $3/2$. Two queries would then determine θ exactly and the floor below would be zero. Away-from-zero removes this: every threshold goes to the cell above it, whatever its neighbours' last digits are. And it keeps Q_p odd, so the reflection above is untouched.

For (i), fix an admissible scale $s > 0$ and let $\theta > 0$. The served value $Q_p(s\theta)$ identifies the cell of Q_p containing $s\theta$, that is, the consecutive boundaries $[\beta, \beta')$ it lies in; dividing by s, it identifies which half-open interval delimited by $\{\beta/s : \beta \text{ a boundary}\}$ contains θ. A response therefore carries exactly the position of θ relative to those points, and no more. Take $\theta < \theta'$, which is no restriction. If some $t \in T$ satisfies $\theta < t \leq \theta'$, the scale realizing t puts θ below that cell edge and θ' at or above it, so the two served values differ. If no such t exists, then for every admissible s the two lie in the same half-open cell, so every served value agrees and the transcripts are identical — however many queries are asked and however they are chosen, since the argument is per-query. The half-open comparison is what makes this exact at the endpoints: a threshold falling on θ' separates the two, and one falling on θ does not. The consistent set is thus $[a, b)$, with a the greatest point of T at or below θ — so $a = \theta$ exactly when $\theta \in T$ — and b the least above it. The left endpoint belongs to it whether or not $a = \theta$: it is a threshold falling on the *smaller* of the two secrets, and by the sentence above such a threshold does not separate them. The interval has positive width, because T is locally finite — on a bounded interval away from the origin only finitely many (N, M, k) give a point $\frac{N}{2M}10^k$ there — so half the width is the infimum, rather than a lower bound on it, in absolute loss. The reported loss is relative (Definition 4), $|\widehat{\theta} - \theta|/|\theta|$, and there the minimax value is $(b - a)/(a + b)$: the estimator $\widehat{\theta} = 2ab/(a + b)$ equalizes $(\widehat{\theta} - a)/a$ and $(b - \widehat{\theta})/b$, any other estimator is worse at one endpoint, and the common value is $(b - a)/(a + b)$. Being a property of the interval it does not depend on where θ sits inside it; the interval being half-open, the worst case is attained at the left endpoint and approached as the secret tends to b. The infimum over query sets is attained too, and cheaply: querying the scale that realizes a returns the cell of $s\theta$, whose two edges pull back to a and to some further point of T; the same holds at b. Since a and b are the nearest such points, intersecting the two responses gives exactly the consistent set described above, closed at a and open at b, so two queries reach the floor and no larger budget improves on it. The

boundaries of Q_p are the midpoints between consecutive p-digit values, so within the decade $[10^e, 10^{e+1})$ they are $(m+\frac{1}{2})10^{e-p+1}$ for integers $m \in [10^{p-1}, 10^p)$; writing $N = 2m+1$ and absorbing the decade of s into k gives the stated form of T.

For (ii), take $\theta > 0$ by the same reflection, so that it suffices to bound the distance between adjacent points of T that straddle θ. If either neighbor lies outside $[\theta/2, 2\theta]$ the gap already exceeds $\theta/2$, which is far larger than claimed, so assume both lie inside it. Let $t = \frac{N}{2M}10^k$ and $t' = \frac{N'}{2M'}10^{k'}$ be two such distinct points and set $\kappa = \min(k, k')$. Then

$$|t - t'| = \frac{10^\kappa \left|N10^{k-\kappa}M' - N'10^{k'-\kappa}M\right|}{2MM'} \;\geq\; \frac{10^\kappa}{2MM'},$$

the numerator being a nonzero integer. Since $M, M' < 10^d$ the denominator is below $2 \cdot 10^{2d}$. For the numerator, $10^k = 2Mt/N$ with $t \geq \theta/2$, $M \geq 10^{d-1}$ and $N < 2 \cdot 10^p$, so $10^k \geq \theta 10^{d-p-1}/2$, and the same holds for k', hence for κ. Combining, $|t - t'| \geq \theta\, 10^{-(p+d+1)}/4$. Every gap of T near θ is therefore at least this wide. The bound holds at every θ the gap straddles, hence in the limit $\theta \to b$, giving $b - a \geq b\, 10^{-(p+d+1)}/4$; and $a < b$ gives $a + b < 2b$. So the minimax relative error of (i) obeys $(b-a)/(a+b) > (b-a)/(2b) \geq \frac{1}{8}\, 10^{-(p+d+1)}$, which is the claim. The bound is uniform in the query set because T does not depend on it.

For (iii), decade-invariance is immediate: multiplying $t = \frac{N}{2M}10^k$ by ten is the map $k \mapsto k+1$, which permutes T, and the ratio in (i) is scale-free, so the infimum over one decade is the infimum over all of T. One decade holds finitely many points, so that infimum is *computed* rather than estimated; Section S13 tabulates it for every $p + d \leq 6$.

The closed form is witnessed by an explicit pair whenever $p \geq d$. The witness uses the two extreme request mantissas, which are themselves a decade apart: $M = 10^d - 1$ for the lower point and $M = 10^{d-1}$ for the upper. Put

$$a = \frac{\left(2 \cdot 10^p - 2 \cdot 10^{p-d} - 1\right) \cdot 10}{2(10^d - 1)}, \qquad b = \frac{2 \cdot 10^p - 1}{2 \cdot 10^{d-1}}.$$

Both lie in T: a takes $N = 2 \cdot 10^p - 2 \cdot 10^{p-d} - 1$ with $M = 10^d - 1$ and $k = 1$, and b takes $N' = 2 \cdot 10^p - 1$ with $M = 10^{d-1}$ and $k = 0$. Both N and N' are odd and lie inside $(2 \cdot 10^{p-1}, 2 \cdot 10^p)$, with N an integer exactly because $p \geq d$ — which is where that hypothesis is spent — and both mantissas carry d digits. The two points share the decade $[10^{p-d}, 10^{p-d+1})$, and

$$b - a = \frac{N'(10^d - 1) - N \cdot 10^d}{2 \cdot 10^{d-1}(10^d - 1)} = \frac{1}{2 \cdot 10^{d-1}(10^d - 1)},$$

the numerator collapsing to 1. Dividing by $a + b$, which the same arithmetic puts at $\big(2(10^d - 1)(2 \cdot 10^p - 1) - 1\big)/\big(2 \cdot 10^{d-1}(10^d - 1)\big)$,

$$\frac{b-a}{a+b} = \frac{1}{2(10^d - 1)(2 \cdot 10^p - 1) - 1}$$

exactly; setting $d = p$ recovers the diagonal pair. So for every $p \geq d$ the infimum is at most the closed form, and enumeration shows the two agree in every enumerated cell with $p \geq d$. The witnessing pair is what licenses the two cells outside the enumerated range that this paper uses, $(p, d) = (7, 7)$ and $(7, 1)$ — and it licenses one direction only. Section S13 uses the closed form at both as an upper bound on the infimum, which is what the pair proves; we do not claim the floor equals it at either.

We do *not* claim the closed form bounds the infimum below for every (p, d); the witness above runs the other way, and only for $p \geq d$. Writing two normalized thresholds as $N10^{j}/(2M)$ and $N'10^{j'}/(2M')$, the elementary argument gives $(b - a)/(a + b) \geq 1/(2MN' - 1)$ only when the larger threshold carries the smaller decade shift, and loses a factor of ten otherwise — which is the regime the exceptional cells sit in. Enumeration finds no cell where the closed form exceeds the infimum, and three where it falls strictly below it: $(1, 2)$, $(1, 5)$ and $(2, 4)$.

Proof (of Theorem 8). For a binary affine classifier, hard labels reveal only the sign of $g(x) = (w_1 - w_0)^\top x + (b_1 - b_0)$, and boundary points satisfy $g(x) = 0$. With d affinely independent boundary points the augmented rows $(x_i, 1)$ span a d-dimensional subspace of $\mathbb{R}^{d+1}$, whose one-dimensional nullspace is the separating hyperplane up to scale; a single labeled validation point fixes the sign. For multiclass heads each class pair c, d has boundary $(h_c - h_d)^\top \phi(x) + (b_c - b_d) = 0$; hard-label queries or complete rankings locate such boundaries by bisection between inputs with opposite labels or opposite pairwise rank order, independent boundary feature vectors recover each boundary up to scale, and observed labels or rankings orient it. Pairwise voting over the recovered comparisons returns the same top class away from ties. For partial top-k this argument applies only when both classes are returned and their relative order is observed on both sides of the boundary. Scale is unidentifiable because g and αg induce identical labels for every $\alpha > 0$. For the final claim, fix $k < m$. The statement asserts existence, so the base is ours to choose: take a known head whose logit for one class $c^\star$ lies below the k-th largest by a margin $\mu > 0$ at every point of the bounded domain $\{x : \|\phi(x)\|_2 \leq \rho\}$. For $u \in \mathbb{R}^h$ with $0 < \|u\|_2 \leq \mu/(2\rho)$ put $\Delta_u = e_{c^\star} u^\top$, a nonzero rank-one update touching row $c^\star$ alone. Cauchy–Schwarz gives $|u^\top \phi(x)| \leq \mu/2 < \mu$ on the domain, so $c^\star$ stays strictly below the k-th largest logit throughout, while the remaining $m - 1$ rows — hence their scores and their mutual order — are untouched. Every top-k index-only response therefore coincides with the base model's. Any two admissible $u_1 \neq u_2$ give distinct rank-one adapters with identical index-only transcripts, which is the claim; Table 26 runs it as an executable certificate.

Proof (of Theorem 9). Endpoint equality for two hidden perturbations holds exactly when $C(\delta_1(x) - \delta_2(x)) = 0$, so the observable object is $C\delta(x)$, equivalently the class of $\delta(x)$ modulo $\ker C$. For the finite-query statement, transcripts depend on δ only through $(C\delta(x))_{x \in Q}$; if $\mathcal{L}_Q$ is injective, equal transcripts imply $\delta_1 = \delta_2$ in $\mathcal{S}$, and if a nonzero $\eta \in \mathcal{S}$ maps to zero then $\mathcal{L}_Q(\delta) = \mathcal{L}_Q(\delta + \eta)$ for all δ. For the rank-one family, $C\delta(x) = Cu\, v^\top x = 0$ for all x while $\delta \neq 0$.

Proof (of Theorem 10). Differentiability gives $T_Q(\theta_0+\epsilon v) = T_Q(\theta_0)+\epsilon J_Q(\theta_0)v+o(\epsilon)$. If $v \in \ker J_Q(\theta_0)$ the first-order term vanishes, so no finite-query endpoint transcript distinguishes θ_0 from $\theta_0+\epsilon v$ to first order, and the observable tangent space has dimension at most $\operatorname{rank} J_Q(\theta_0)$. The nullity bound follows from rank-nullity. On a neighborhood where T_Q is affine along V the Taylor expansion has no higher-order remainder along V, so $J_Q(\theta_0)v = 0$ implies exact transcript equality.

Proof (of Corollary 3). The transformation $(B, A) \mapsto (BR, R^{-1}A)$ leaves BA unchanged and has r^2 parameters on the full-rank factor stratum. The manifold of rank-r matrices in $\mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ has dimension $r(d_{\text{in}} + d_{\text{out}} - r)$, obtained by subtracting this gauge dimension from the $r(d_{\text{in}} + d_{\text{out}})$ factor parameters. A transcript with q queries and m logits has Jacobian rank at most qm. Applying Theorem 10 to the quotient space gives the bound.

Proof (of Proposition 3). Since C has full column rank, $C^\top$ has a right inverse, and for every queried point $f(x) - \beta = z(x)C^\top$ with $z(x) = \sigma((A_0 + \Delta)x + a)$, so $z(x) = (f(x)-\beta)(C^\top)^+$. Inside an all-active ReLU cell $z(x) = (A_0+\Delta)x+a$. Querying 0 recovers a, and querying ϵe_j recovers the j-th column of $A_0+\Delta$ from $(z(\epsilon e_j) - z(0))/\epsilon$. Subtracting public A_0 gives Δ. If the cell is not all-active, the same argument recovers only rows active on that cell.

Proof (of Corollary 2). Write $A = C^\top \in \mathbb{R}^{h \times m}$, so the endpoint returns $y(x) - \beta = h(x)A$ with β and A public. The map $v \mapsto vA$ on row vectors $v \in \mathbb{R}^{1\times h}$ has kernel $\{v : Cv^\top = 0\}$, which is $\ker C$.

If $\operatorname{rank}(C) = h$ then $\ker C = \{0\}$, and since A has h rows its range is all of $\mathbb{R}^h$, so $AA^+ = I_h$ and $(y(x)-\beta)A^+ = h(x)AA^+ = h(x)$. The inversion uses only public quantities, so it costs no extra oracle calls: any attack that assumes final-hidden-state access runs unchanged against the endpoint, one call per query.

If $\operatorname{rank}(C) < h$ then $\ker C \neq \{0\}$ and $h(x)$ is determined only modulo $\ker C$, which is the quotient Theorem 9 identifies. So the condition is not merely sufficient: full rank is exactly what separates inversion from the quotient.

Proof (of Proposition 4). The exact-coordinate transcript is $y_S(x) - \beta_S = h(x)C_S^\top$, which identifies all $h(x) \in \mathbb{R}^h$ iff C_S has full column rank; if the rank condition fails, a nonzero vector in $\ker C_S$ gives distinct hidden states with the same transcript. For shifted logits the transcript is $h(x)C_S^\top + \beta_S - \alpha(x)\mathbf{1}_S$ with one unknown scalar per query, a linear system in $(h(x), \alpha(x))$, so exact recovery for arbitrary $h(x)$ is equivalent to full column rank of $[C_S, -\mathbf{1}_S]$. Rank-revealing row selection uses only C, so it costs no oracle queries.

Proof (of Corollary 4). The odds identity follows by writing the biased probability of row i as $e^B p_i/(e^B p_i + 1 - p_i)$. Each forced row thus gives the corresponding original log-probability coordinate, and log-probabilities are logits up to a per-query additive shift. For the scoreless variant, row i becomes a winner exactly when the added bias reaches $\max_j y_j(x) - y_i(x)$, and finite binary search gives the stated tolerance. The rank condition is then the shifted-coordinate condition of Proposition 4.

S2 Deferred Figures and Algorithms

Referenced from the main body; deferred here for length.

Algorithm 1 Observable-head recovery from a scored top-k oracle (Theorem 1, Proposition 1)

Require: public H_0, ϕ; oracle $\mathcal{O}^{\text{score}}_{\text{top-k}}$; query pool $\mathcal{P}$; rank bound R; threshold τ
Ensure: recovered $\widehat{\Delta}, \widehat{b}$, or UNIDENTIFIED for unsolved rows

initialize per-class observation lists $L_c \leftarrow \emptyset$
for all $x \in \mathcal{P}$ **do**
  $(I, S) \leftarrow \mathcal{O}^{\text{score}}_{\text{top-k}}(x)$ ▷ 1 oracle call
  for all $(c, s) \in (I, S)$ **do**
    append $((\phi(x), 1), s)$ to L_c
  end for
  ▷ classes outside the returned set yield no equation: the failed probes
end for
for all classes c **do**
  if $|L_c| < h + 1$ **or** $\operatorname{rank}(L_c) < h + 1$ **then**
    mark c UNIDENTIFIED
  else
    solve the $(h + 1)$-dimensional least-squares system for $(\hat{h}_c, \hat{b}_c)$
  end if
end for
$M \leftarrow$ matrix of solved rows minus their public base rows
$\hat{\rho} \leftarrow \#\{\sigma_i(M) > \tau\}$
if $\hat{\rho} < R$ **then** ▷ row space not certifiable (Thm. 3)
  return with every remaining row UNIDENTIFIED
end if
$B \leftarrow$ top-R right singular vectors of M
for all classes c still UNIDENTIFIED **do**
  if L_c has projected augmented rank $R + 1$ on B **then**
    solve the reduced $(R + 1)$-dimensional system for $(\alpha_c, \hat{b}_c)$
  end if
end for
return $\widehat{\Delta} = \widehat{H} - H_0, \widehat{b}$

Why the row-space step is sound, and when it is refused. Every solved row of M is a row of Δ, so $\operatorname{row}(M) \subseteq \operatorname{row}(\Delta)$, a space of dimension at most R. If $\operatorname{rank} M = R$, the inclusion is an equality: the solved rows already span the adapter's row space, B is a basis for it, and Proposition 1 applies exactly. If $\operatorname{rank} M < R$, the transcript cannot establish that the solved rows suffice: as in the proof of Theorem 3, an unsolved row may carry a direction outside $\operatorname{row}(M)$ that its fewer than $h + 1$ observations leave unconstrained, and two adapters of rank at most R then share the transcript but not the row space. Algorithm 1 therefore takes the reduced step only in the first case, and otherwise reports the

remaining rows UNIDENTIFIED rather than guessing. The price is that a loose bound $R > \operatorname{rank} \Delta$ is never certified; an attacker may still run the reduced step on $\hat{\rho}$ directions and test each reduced solve for residual consistency on more than $\hat{\rho}+1$ observations, but that test can only reject a wrong basis, never certify a right one, and we report no result that relies on it. Under noisy scores, τ separates the spectrum as in Theorem 2, set above the error bound $\|X^{+}\|_2\|E\|_F$ of Theorem 4; our implementation runs the certified branch with R the declared rank.

Algorithm 2 Adaptive endpoint inversion via a forcible-row interface (Proposition 4, Corollary 4)

Require: public output head $C \in \mathbb{R}^{m\times h}$; oracle $\mathcal{O}_{\text{bias}}$; prompts $\mathcal{Q}$; bias magnitude B; tolerance ϵ
Ensure: recovered hidden states $\hat{h}(x)$ for $x \in \mathcal{Q}$

$S \leftarrow$ rank-revealing QR on C, selecting h rows (exact) or $h+1$ rows (shifted)
▷ selection uses only the public head: 0 oracle calls
for all $x \in \mathcal{Q}$ **do**
  for all $i \in S$ **do**
    if scored log-probability variant **then**
      query with bias B on row i; invert the odds identity ▷ 1 call
    else
      binary-search the bias making row i tie the current winner
      ▷ $\lceil \log_2(2B/\epsilon) \rceil$ calls
    end if
  end for
  solve the $|S|$-row linear system for $\hat{h}(x)$, and the shift $\alpha(x)$ if shifted
end for

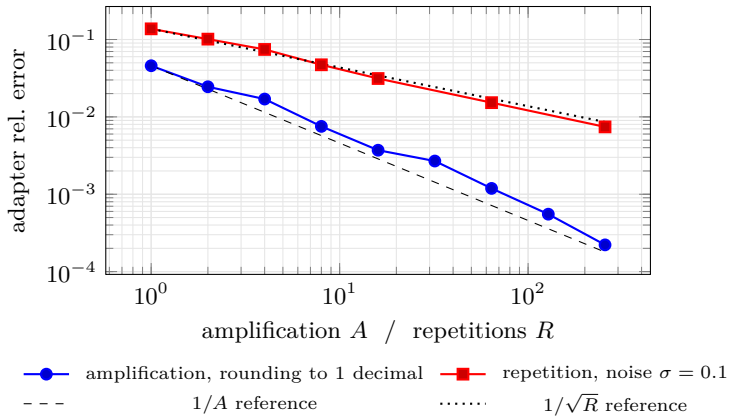


Fig. 1. Scaling laws of Corollary 1 on the raw-logit oracle, `digits`, 10 seeds. Amplifying queries 256× against a fixed absolute rounding grid reduces adapter error 206.7×, tracking the predicted $1/A$; averaging 256 repetitions against additive noise reduces it 18.5×, tracking the predicted $1/\sqrt{R}$ ($\sqrt{256} = 16$). The two levers are not interchangeable: repetition does nothing against rounding, and amplification does nothing against the intercept.

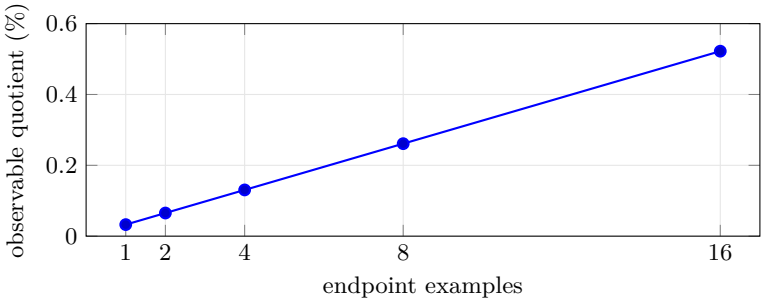


Fig. 2. Finite-query endpoint observability (Theorem 10) for the `bert-base-uncased`/SST-2 hidden LoRA module, after quotienting out LoRA factor gauge. The fraction observable *to first order* grows linearly in returned endpoint rows; even 16 examples expose only 0.5222% of the 6128-dimensional non-gauge update quotient. Being a Jacobian rank, this bounds the local tangent space, not nonlinear identifiability.

S3 Full PEFT Certificate

Table 3. Real PEFT LoRA checkpoint. **Dataset:** GLUE/SST-2. **Model:** `bert-base-uncased` with a trained query-projection LoRA module, rank 4, 10 seeds. **Oracles:** $\mathcal{O}_{\text{module}}$ for the recovery row; $\mathcal{O}_{\text{end}}$ for the certificate rows.

metric	mean	std / note
train / test accuracy	0.9341 / 0.8756	0.0066 / 0.0115
Module-output access (Thm. 1)		
module delta rel. err.	$8.37 \cdot 10^{-15}$	$7.18 \cdot 10^{-16}$
rank-projected rel. err.	$1.19 \cdot 10^{-15}$	$1.46 \cdot 10^{-16}$
module oracle queries		769 hidden dimension plus one
Endpoint-only access (Thm. 9, Cor. 3)		
endpoint classifier nullity	766	of 768
classifier-endpoint blind fraction	0.9974	quotient kernel
LoRA factor gauge nullity	16	r^2, quotiented out
LoRA update quotient dimension	6128	non-gauge update space
endpoint LoRA Jacobian rank	32	16 examples
non-gauge endpoint nullity	6096	local lower bound
quotient observable fraction	0.0052	local tangent space
real-test nullspace drift	$2.51 \cdot 10^{-15}$	$1.07 \cdot 10^{-15}$
real-test label flip rate	0.0000	utility preserved
real-test hidden shift norm	0.7995	0.0153
local non-gauge update shift	$1.20 \cdot 10^{-3}$	$3.86 \cdot 10^{-5}$
local endpoint max-logit drift, float32	$3.25 \cdot 10^{-6}$	float32 noise, see below
local endpoint max-logit drift, float64	$2.79 \cdot 10^{-9}$	Sec. S6

S4 Reported Metrics

Completing Section 3, which glosses these in the body.

Definition 4 (Reported metrics). *Let $\widehat{\Delta}$ be a recovered adapter and Δ the true one.*

- Relative error *is the relative Frobenius error $\|\widehat{\Delta} - \Delta\|_F / \|\Delta\|_F$. Bias error is reported separately as an absolute Euclidean norm, because the levers of Section 5 affect the two differently.*
- Test *or* label agreement *is the fraction of held-out inputs on which the recovered head and the true head predict the same top-1 label.*
- Observable quotient *is the fraction of relevant recovery directions the transcript constrains,* $\operatorname{rank} J_Q / \dim$ *of the non-gauge update space;* blind fraction *is* $\dim \ker C/h$*;* non-gauge endpoint nullity *is that dimension minus* $\operatorname{rank} J_Q$*, a lower bound on locally unobservable directions once LoRA gauge is removed.*

– Utility *of a deployed service is held-out label agreement unless a table says otherwise; because label agreement is blind to everything but the argmax, Section S19 also reports relative logit error and softmax KL, and Section S10 re-runs the defensive comparison of Section 8.5 under total variation, KL and two proper scoring rules.* Leakage *is the observable quotient. All are fractions, not composite scores.*

S5 Related Work: The Three Surveyed Strands

Completing Section 2, which keeps in the body only the two comparisons our claims rest on. Nothing here is required to follow the results; it is the positioning of this paper against the three literatures it sits between.

Cryptanalytic neural extraction. Carlini, Jagielski and Mironov [6] argued that neural-network extraction is naturally cryptanalytic and recovered ReLU networks at critical points; Canales-Martínez et al. [4] improved this to polynomial time, and later work optimized sign extraction [17], widened the activation family [10,36], and moved to hard labels [5,9,26]. We are complementary: we isolate a known-base low-rank secret and characterize the identifiability adapter structure adds.

Model stealing and equation solving. Prediction-API stealing predates that line. Tramèr et al. [48] demonstrated equation-solving extraction and studied rounding as a countermeasure; Section 1.1 states the difference. Related work covers substitute training [28, 41], production embedding projections recovered up to symmetry [7], softmax-bottleneck and logit leakage [8, 15, 16, 18], prompt inversion [37], low-rank structure in language-model distributions [19, 33], and output-geometry fingerprinting [55]; bounded top-logprob lists and `logit_bias` are documented production interfaces [39, 40]. Lowd and Meek [34] and Angluin [2] give the exact-learning lens; our hard-label and ranking attacks relate to learning halfspaces from signs, which matters for lower bounds because signs and rankings erase positive rescalings.

Low-rank adaptation, verification, and side channels. Adapters insert small trainable modules into frozen networks [23, 24]; their factors have asymmetric roles [56] and leak training-set information through spectra and norms [44]. StolenLoRA [53] and PHLoRA [49] attack LoRA deployments by substitute training and by decomposing full checkpoints, gray-box rather than oracle settings; a parallel line [22,25,32,43,45,46] asks whether adapters are valid, private, benign, or physically extractable. LoREnc [1] defends the deployment rather than the transcript, by truncating the dominant low-rank components of the base and compensating only authorized adapters; that attacks our hypothesis rather than our attack, since every positive result here needs the served base to be the *public* one, and a defender who can afford to serve a base nobody else holds has left the threat model this paper is about. We instead ask which API

outputs mathematically identify the adapter, and transformer-identifiability work motivates our quotient treatment [11, 20, 21, 51].

What exact parameters add over functional mimicry. Prior LoRA extraction targets functional mimicry [53], so it is fair to ask what exact parameters add. Factors leak fine-tuning-set information through their spectra [44], a channel a substitute trained on other data does not have; an adapter is a small detachable artifact, so recovery yields a redistributable asset and a license check; and gradient attacks and backdoor detection [46] need the weights themselves.

The full comparison to equation-solving extraction. Completing Section 1.1, which keeps only differences (i) and (iv) in the body.

Our observable-head recovery result (Section 4) is closest to the equation-solving attacks of Tramèr et al. [48], who recovered multiclass parameters from roughly $d + 1$ confidence-bearing queries and studied score rounding as a countermeasure. We differ in four ways. *(i)* Classical equation solving recovers a full matrix; here the base is public and the secret is a rank-r update, so once the row space is identified each further class row costs $r+1$ rather than $h+1$ observations (Proposition 1), an acceleration that Theorem 3 shows cannot be certified earlier. *(ii)* We characterize the maximal recoverable object across an oracle family (Definition 2) rather than one confidence API. *(iii)* Endpoint logits reveal exactly the adapter's image under the final head, so restriction yields a quotient rather than failure. *(iv)* They evaluate rounding against a *fixed* attacker, treating the rounded value as a fixed loss of precision; we let the attacker amplify (Section 5) and also choose where the thresholds fall relative to its own uncertainty (Section 5.2), which is what decides whether the defense holds.

Adaptive output-perturbation defenses. Two lines defend the score channel itself rather than the query stream, and both are worth separating from what we claim. ModelGuard [47] solves for a prediction perturbation that maximizes an information-theoretic objective subject to a utility constraint, and is explicitly evaluated against *adaptive* extraction attacks — the evaluation most of the earlier line omits. It is complementary to our result rather than a competitor to it: our taxonomy is about the *units* a perturbation is measured in, not its size or its optimality, and whether an optimized perturbation lands in our absolute row or our relative one depends on whether the solution it returns scales with the score magnitude, which its formulation does not require either way. What Theorem 6 contributes is that an absolute-scaled perturbation has a certificate that vanishes under amplification whatever objective produced it, and Corollary 1 that the loss is realized. We do not run it as a baseline, and we do not claim to beat it.

Caliber [50] is nearer to our own statement: it adds i.i.d. Gaussian noise to the logits and derives a closed-form minimax lower bound on the number of repeated queries needed to recover the clean logits at a fixed input. That is a per-input recovery-cost bound of the same species as Theorem 6's per-transcript certificate, and the contrast is instructive. Because its perturbation is *stochastic*, repetition

averages it away and the bound is a query *count*: the secret is recoverable, and the defense prices it. Our quantizers are deterministic, so repetition returns the identical value and buys nothing at any budget (Corollary 1(2)); no count of queries closes the gap, and the binding constraint moves to the request grid instead (Theorem 7). A stochastic defense sets a price and a deterministic one sets a floor, and the two are not interchangeable.

S6 The Curvature of the Null Direction

Section 8.3 exhibits one perturbation that the endpoint barely sees. One perturbation is an anecdote; the question is whether it obeys a law, and if so what radius that law predicts. This section sweeps the step size and answers both. The command is `run-null-curvature`; the measurement is over the same ten trained adapters as Table 3, 16 SST-2 validation examples, all medians over seeds.

Why float64, and what the float32 probe was measuring. The measurement must be run in double precision, and this is not a detail. A `bert-base-uncased` forward pass in float32 disagrees with the same pass in float64 by up to $4.08 \cdot 10^{-6}$ on these logits, whose own scale is 3.04. Any endpoint drift the float32 pipeline reports at or below that number is its own arithmetic. The $3.25 \cdot 10^{-6}$ reported by the float32 probe in Table 3 sits below it, and is therefore an upper bound pinned by rounding rather than a measurement of the perturbation; the float64 sweep puts the true drift at the same step at $2.79 \cdot 10^{-9}$, some three orders smaller. We report both rather than quietly replacing one with the other, because the correction runs in the direction that favors our claim and a reader is entitled to see that we noticed. The sweep below additionally probes at $t = 10^{-12}$ to locate its own arithmetic floor, $6.00 \cdot 10^{-15}$, and fits only over steps clearing that floor by $10\times$; all ten steps qualify.

The law. The fitted exponents are 1.996 for the endpoint drift and 1.026 for the update shift, against 2 and 1 predicted. The first-order term is dead — that is what "null" means, and the slope is the test of it — so the endpoint sees the direction only through the second-order term, with curvature $L = 2\,\mathrm{drift}/t^2$. At the smallest step the median seed gives $5.58 \cdot 10^{-3}$, but a median is not a bound and the radius below is a claim about every adapter we measured, so we take the *envelope*: the largest L any seed exhibits at any fitted step, which is $2.22 \cdot 10^{-2}$ at seed 3, a factor 3.99 above the median. With that constant the quadratic bound is satisfied at every measured point, which is checked rather than assumed; a radius written in the median would not hold for seed 3. The last row of the table is worth stating plainly: a perturbation whose merged update exceeds the update's own norm by $4.6\times$ still moves a logit by $2.3 \cdot 10^{-2}$, on logits of scale 3.

The radius, and why it is written in the cell slack. Under the quadratic law, an endpoint cannot separate two adapters differing by a null perturbation of

Table 4. Endpoint drift and update movement along a *Jacobian*-null direction, as a function of step size t, a displacement along a unit direction in LoRA factor-parameter space; the update-shift column converts it to merged-update movement. **Model:** `bert-base-uncased`/SST-2, rank-4 query-projection LoRA, 10 seeds, median over seeds, float64. The two columns scale with different powers of t, which is the content of the section: the endpoint sees the direction only through curvature.

step t	endpoint max-logit drift	update shift	shift / $\|BA\|$
10^{-4}	$2.791 \cdot 10^{-11}$	$1.196 \cdot 10^{-4}$	$9.529 \cdot 10^{-5}$
$3 \cdot 10^{-4}$	$2.512 \cdot 10^{-10}$	$3.588 \cdot 10^{-4}$	$2.859 \cdot 10^{-4}$
10^{-3}	$2.791 \cdot 10^{-9}$	$1.196 \cdot 10^{-3}$	$9.529 \cdot 10^{-4}$
$3 \cdot 10^{-3}$	$2.512 \cdot 10^{-8}$	$3.588 \cdot 10^{-3}$	$2.859 \cdot 10^{-3}$
10^{-2}	$2.791 \cdot 10^{-7}$	$1.196 \cdot 10^{-2}$	$9.529 \cdot 10^{-3}$
$3 \cdot 10^{-2}$	$2.511 \cdot 10^{-6}$	$3.588 \cdot 10^{-2}$	$2.859 \cdot 10^{-2}$
10^{-1}	$2.789 \cdot 10^{-5}$	$1.197 \cdot 10^{-1}$	$9.542 \cdot 10^{-2}$
$3 \cdot 10^{-1}$	$2.506 \cdot 10^{-4}$	$3.615 \cdot 10^{-1}$	$2.885 \cdot 10^{-1}$
10^{0}	$2.758 \cdot 10^{-3}$	$1.303 \cdot 10^{0}$	$1.036 \cdot 10^{0}$
$3 \cdot 10^{0}$	$2.340 \cdot 10^{-2}$	$5.771 \cdot 10^{0}$	$4.586 \cdot 10^{0}$
log-log slope	1.996	1.026	

norm below $\sqrt{2\gamma/L}$, where γ is the smaller of the served logit's two distances to its quantization cell's boundaries. The quantity is γ, not the cell width ε, and the distinction is not cosmetic: rounding is deterministic, so two values closer together than ε still serialize differently when they straddle a boundary, and what a drift must exceed to be seen is the slack, not the width. These are the two slacks Theorem 6 is stated in, minimized over the two directions because a null perturbation's sign is not ours to choose; writing $\sqrt{2\varepsilon/L}$ instead would be inconsistent with the paper's own floor, and would overstate the radius.

By how much is a measurement, not an estimate, because γ is a property of the served logits rather than of the grid. Table 5 reports both. The minimum slack over all ten seeds' logits runs two to four orders of magnitude below the cell width — at $\varepsilon = 10^{-6}$ some logit sits $1.21 \cdot 10^{-10}$ from an edge — and since the radius goes as $\sqrt{\gamma}$, the ε form overstates it by 16 to 91×. The radii the γ form gives are smaller, which is the direction that costs us: it makes the defensive claim weaker than the ε form would support. Every radius still falls inside the range of t actually swept, so none is an extrapolation; we mark the column rather than leaving the reader to check, because the quadratic bound must eventually fail as the endpoint map saturates, and a radius quoted past the data would be an assertion rather than a measurement.

What this does and does not establish. It does not prove that a perturbation of the studied LoRA module realizes an arbitrary hidden effect; Theorem 9 exhibits an effect at the final hidden representation, and realizability from an upstream module is architecture-specific. What the sweep establishes is the weaker, measured, and defensible statement: for this architecture and this trained module

Table 5. Invisibility radius implied by the curvature of Table 4. The radius is set by the measured cell slack γ, not the cell width ε; the ε column is retained only to show the size of that gap. The radius is no better than the quadratic law it comes from, so the last column records whether it lies inside the swept range.

cell width ε	min slack γ	radius $\sqrt{2\gamma/L}$	$\sqrt{2\varepsilon/L}$	update shift	in swept t?
10^{-2}	$1.221 \cdot 10^{-5}$	$3.313 \cdot 10^{-2}$	$9.482 \cdot 10^{-1}$	$3.963 \cdot 10^{-2}$	yes
10^{-3}	$7.845 \cdot 10^{-7}$	$8.398 \cdot 10^{-3}$	$2.998 \cdot 10^{-1}$	$1.004 \cdot 10^{-2}$	yes
10^{-4}	$3.845 \cdot 10^{-7}$	$5.880 \cdot 10^{-3}$	$9.482 \cdot 10^{-2}$	$7.032 \cdot 10^{-3}$	yes
10^{-6}	$1.215 \cdot 10^{-10}$	$1.045 \cdot 10^{-4}$	$9.482 \cdot 10^{-3}$	$1.250 \cdot 10^{-4}$	yes

there is a *Jacobian*-null direction whose endpoint signature is second-order, with a measured constant, and therefore a computable norm below which a finite-precision endpoint returned identical transcripts at every step we sampled. That is local conditioning with a radius, not exact indistinguishability, and it is what we claim.

One further limit is worth stating, because it separates a measurement from a theorem. The quadratic law is *fitted* at the ten sampled steps of Table 4 and checked at each of them; it is not verified on the continuum between them. A radius read off it is therefore a prediction supported at the sampled points, not a certificate over an interval. Turning it into one would need a bound on the Hessian — an operator-norm bound over a neighborhood, or interval arithmetic along the segment — which we do not compute. We use the radius only to say that the drift a quantizer must exceed is far above what this direction produces at the steps we measured, and every comparison in the paper is made at a measured step.

S7 Deferred Statements

The positive endpoint cases are collected next; all reduce to Theorem 1 once a hidden state is recovered.

Proposition 3 (Hidden LoRA block transfer on one ReLU cell). *Consider $f(x) = C\,\sigma((A_0 + \Delta)x + a) + \beta$ with A_0, a, C, β public, Δ an unknown rank-r adapter, σ ReLU, and $C \in \mathbb{R}^{m\times h}$ of full column rank. With raw-logit access and queries $0, \epsilon e_1, \ldots, \epsilon e_d$ inside an all-active ReLU cell of the adapted hidden layer, the attacker recovers $A_0 + \Delta$, a, and Δ exactly with $d+1$ queries.*

Proof. Deferred to Section S1.

Corollary 2 (Endpoint inversion under a full-rank output head). *If an endpoint returns exact logits $y(x) = h(x)C^\top + \beta$ with public $C \in \mathbb{R}^{m\times h}$, public β, and $\operatorname{rank}(C) = h$, then every queried final hidden state is exactly recoverable as $h(x) = (y(x) - \beta)(C^\top)^+$. Consequently any attack assuming final-hidden-state access also applies to such an endpoint oracle. If $\operatorname{rank}(C) < h$, this inversion fails and the endpoint exposes only the quotient of Theorem 9.*

Corollary 3 (LoRA quotient dimension bound). *For a rank-r LoRA update $\Delta = BA$ with B full column rank and A full row rank, the factorization has the gauge symmetry $(B, A) \sim (BR, R^{-1}A)$, $R \in \mathrm{GL}_r$, so r^2 factor directions are unidentifiable before any endpoint restriction. The rank-r update quotient has dimension $r(d_{\mathrm{in}} + d_{\mathrm{out}} - r)$. If q endpoint examples expose m logits each, the transcript has at most qm scalar equations, so finite-query endpoint access leaves at least $r(d_{\mathrm{in}} + d_{\mathrm{out}} - r) - qm$ locally unobservable update degrees of freedom when positive. The sharper empirical certificate replaces qm by the measured Jacobian rank.*

Two instantiation lemmas in the supporting material make this operational. Proposition 4 characterizes when a *subset* $S \subseteq [m]$ of output coordinates suffices — $\mathrm{rank}(C_S) = h$ for exact logits, $\mathrm{rank}([C_S, -\mathbf{1}_S]) = h+1$ for shifted ones such as normalized log-probabilities — with S chosen by rank-revealing linear algebra on the *public* head at zero oracle cost, which is what reduces full-vocabulary probing to $h+1$ rows; Corollary 4 shows a documented logit-bias interface [39,40] instantiates shifted coordinate access, including a scoreless variant.

These instantiate the endpoint theory used by the language-model experiments.

Proposition 4 (Coordinate-subset endpoint inversion). *For logits $y(x) = h(x)C^\top + \beta$ with C, β public, exact coordinate logits on $S \subseteq [m]$ identify arbitrary $h(x) \in \mathbb{R}^h$ iff $\mathrm{rank}(C_S) = h$. Shifted coordinate logits, including normalized log-probabilities or scoreless top-1 bias thresholds with unknown per-query additive shift, identify $h(x)$ iff $\mathrm{rank}([C_S, -\mathbf{1}_S]) = h + 1$. Thus h exact or $h + 1$ shifted coordinates are necessary and sufficient absent further priors, and S can be selected before querying by rank-revealing linear algebra on the public head, at zero oracle cost.*

Proof. Deferred to Section S1.

Corollary 4 (Forcible-row log-probability and top-1 interfaces). *Under the assumptions of Proposition 4, a restricted log-probability interface instantiates shifted-coordinate access when the API supports per-row logit biases large enough to force each requested row of S into the returned set and returns log-probabilities normalized over the full vocabulary with known temperature. If adding bias $B > 0$ to row i returns biased log-probability ℓ'_i, the original probability satisfies*

$$\frac{p_i}{1 - p_i} = e^{-B} \frac{e^{\ell'_i}}{1 - e^{\ell'_i}}.$$

A scoreless temperature-zero variant is possible when the API accepts per-row bias and returns only the winning index: binary search on the bias making row i tie the current winner recovers $y_i(x) - \max_j y_j(x)$ within the search tolerance, provided the required bias lies in the allowed range and tie-breaking is consistent. This abstracts documented API surfaces [39, 40]; the theorem still requires the stronger conditions stated above.

Proof. Deferred to Section S1.

S8 Two Limits of the Security Floor

Computing the floor. $F(\mathcal{Q})$ is evaluated by taking $T(\Xi)$ in closed form at each candidate direction and then hill-climbing one-sided directions, which merely exhibits an admissible alternative and so moves the bound, not the truth. The directional slack of Theorem 6 is what makes T worth computing: charging every score its distance to the *nearest* boundary instead would discard the far side of every cell at once. A direction no query moves is reported as 0 rather than ∞, which can only lower the reported floor. Over 10 seeds on `digits` (Table 21) the absolute floor falls from $9.90 \cdot 10^{-3}$ to $1.08 \cdot 10^{-5}$ as A runs 1 to 1024, slope -0.970 against the predicted -1; the relative floor stays within $[2.53, 4.55] \cdot 10^{-5}$, slope $+0.091$ against 0. Our best attacker tracks each (-1.005, $+0.040$), and the certificate is sound — never above a measured error — in all 120 configurations, though loose by 3–9$\times$ because one score near a cell boundary pins the whole polytope.

Two limits. First, we prove $F = O(1/A)$ for the absolute family, an *upper* bound — the direction the insecurity claim needs — but no matching lower bound, which would also require the slacks to stay away from zero, true generically but not for every Δ. Second, F is evaluated at the query set the attacker chose, so "the relative floor does not move" is a statement about $F(A\mathcal{Q})$, not about $\inf_{\mathcal{Q}} F$. The second limit is removed for a scalar secret under a discrete request format by Theorem 7, which computes $\inf_{\mathcal{Q}}$ exactly; it remains open in the generality stated here, and Remark 3 marks the boundary between the two.

S9 The Relative Floor Between Powers of the Base

Theorem 6(ii) is exact on one subsequence and deliberately silent off it, and the silence is worth defending: $F(A\mathcal{Q})$ is bounded *above* by a positive constant for all A, but not below. The tempting symmetry fails because bounded mantissas constrain the cell *width* and not the distance from a score to its cell's edge. This section measures what actually happens, on a fixed design ($n = 24$ queries, hidden width 6, $m = 3$ classes, rank 2, seed 0) with F evaluated in closed form along one fixed one-sided direction, so that every amplification is compared on the same perturbation rather than on whichever direction happened to win a search.

On the subsequence, scaling exactly. At $A = 10^k$ for $k = 0, \ldots, 6$ the relative floor is invariant to $1.9 \cdot 10^{-12}$, i.e. to floating-point noise, while the absolute floor falls from $1.54 \cdot 10^{-5}$ to $2.83 \cdot 10^{-12}$ across the same six decades. That contrast is the whole content of parts (i) and (ii), and it is exact rather than approximate.

On the subsequence, under asymptotic *homogeneity.* The sweep just described scales the design exactly, $z = Az_0$, which is the hypothesis under which part (ii) is an equality. The hypothesis Theorem 5 actually assumes is weaker, and

the difference is not cosmetic. Repeating the sweep with $z = Az_0 - \mu$ — what a standardizer does, same design, same fixed direction, same closed form, so the hypothesis is the only moving part — the floor is $2.50 \cdot 10^{-5}$ at $A = 1$, wanders as high as $4.02 \cdot 10^{-5}$ while the $o(1)$ still bites, and settles by $A = 10^8$ at $2.0565 \cdot 10^{-5}$. It converges, and not to where it started: F_∞ is 0.823 times $F(\mathcal{Q})$. That the limit is the right one is checkable rather than asserted — F_∞ agrees with the exactly homogeneous floor, which is the floor at Z_∞, to a relative $1.8 \cdot 10^{-4}$ — so the drift is the hypothesis and not the arithmetic. This is why part (ii) claims $F(10^k\mathcal{Q}) \to F_\infty$ and not $F(\mathcal{Q})$. What the security argument needs is that the limit is positive while the absolute grid's is zero, and that is exactly what the two sweeps show.

Off it. Sweeping A across one decade, the relative floor ranges over $[1.6{\cdot}10^{-2},\ 4.8]$ times its subsequence value: the upper end puts the lim sup strictly above F_∞ — which is why part (ii) claims only $\geq$ — and the lower end is the question. A single sweep cannot answer it — the smallest value it reports is equally consistent with a positive floor the grid never got below — so we refine on nested grids, each doubling the last, along which the running minimum can only fall. It falls from $2.7 \cdot 10^{-2}$ to $9.1 \cdot 10^{-4}$ over a 256-fold refinement, log-log slope -0.70, with no sign of a barrier. We find no evidence of a positive lower bound and assume nothing that would supply one; a finite refinement cannot establish that none exists, and we do not claim it.

Why the decay is slow, and why that is the interesting part. One might expect F to collapse wherever a served score meets a cell boundary, but it does not, and the reason is the directional slack Theorem 6 is stated in. A score at its cell's edge is immovable in one direction and a full cell width from the other, and the attacker takes whichever sign gives the longer chord; since a cell's two slacks sum to its width, no single score can be tight both ways. F collapses only when *two* scores sit near opposite edges at once — a codimension-two coincidence in a one-parameter family, which is exactly why a 256-fold refinement buys only a 30-fold decrease. So the honest reading of part (ii) is sharper than either a uniform bound or none at all: off the subsequence the floor is unprotected, and it is nonetheless not fragile at a typical amplification. An operator cannot rely on the second half, which is why the theorem states only the first.

Theorem 5(1) is exposed in the same place. One would like to say the least-squares error *converges* to a limit independent of A. The claim that matters — that amplification buys nothing — is untouched, and it holds: over $A = 1$ to 10^6 on a standardizer-style map (200 queries, width 8, 4 classes, 4 significant digits) the error stays inside $[1.18, 1.99] \cdot 10^{-4}$, against the $1/A$ decay an absolute grid would give over the same range. But convergence is more than the proof delivers, and more than is true: across the single decade $A \in [10^3, 10^4]$ the error ranges over $[8.26 \cdot 10^{-5},\ 3.58 \cdot 10^{-4}]$, a factor of 4.3. It oscillates in a band rather than settling into a limit, exactly as the one-step discrepancy in the proof allows, so the band is what we state. Nothing downstream used the limit — the attacker

Table 6. Service cost of each quantizer measured five ways, `digits`, 10 seeds, median over seeds. The two fidelity metrics separate the fine configurations by three and six orders; the two proper scoring rules separate them by under 0.2% and are not monotone in precision — 2 decimals scores *better* than 3 — because at this precision both are dominated by the classifier's own error, not by the quantizer's.

quantizer	top-1	TV	KL	Brier	NLL
absolute, 0 dec.	0.8218	$1.27 \cdot 10^{-1}$	$4.29 \cdot 10^{-2}$	0.3887	0.7144
absolute, 1 dec.	0.9574	$9.50 \cdot 10^{-3}$	$2.76 \cdot 10^{-4}$	0.5179	0.8806
absolute, 2 dec.	0.9947	$9.62 \cdot 10^{-4}$	$2.77 \cdot 10^{-6}$	0.5170	0.8794
absolute, 3 dec.	1.0000	$9.24 \cdot 10^{-5}$	$2.70 \cdot 10^{-8}$	0.5170	0.8795
relative, 1 sig.	0.9574	$8.71 \cdot 10^{-3}$	$2.30 \cdot 10^{-4}$	0.5176	0.8802
relative, 2 sig.	0.9947	$8.77 \cdot 10^{-4}$	$2.29 \cdot 10^{-6}$	0.5170	0.8794
relative, 3 sig.	1.0000	$8.52 \cdot 10^{-5}$	$2.23 \cdot 10^{-8}$	0.5170	0.8795
relative, 4 sig.	1.0000	$8.76 \cdot 10^{-6}$	$2.29 \cdot 10^{-10}$	0.5170	0.8795

comparisons in Section 8.5 compare errors at fixed A — so this narrows a claim without weakening a result.

S10 Utility Beyond the Top-1 Label

The matched-utility comparison of Section 8.5 pays its service cost in top-1 label survival. That single metric is a weak basis for a recommendation, so we recompute the same pairing under four further notions of utility, all measured on the oracle itself and independently of any attack: total variation and KL between the served and the exact predictive distribution, and Brier score and negative log-likelihood against the *true* held-out labels. Table 6 gives all five per configuration.

Two things follow, and they point in opposite directions.

The recommendation does not depend on the metric that produced it. Rematching each absolute setting to its nearest relative one under TV, and again under KL, reproduces the top-1 pairing exactly — 1 decimal with 1 significant digit, 2 with 2, 3 with 3 — so the security gains of Table 24 (456×, 388×, 430×) are unchanged. The relative grid is therefore not winning by hiding its cost where top-1 cannot see it: it is genuinely cheaper at equal fidelity, which is what Theorem 5 predicts, the served relative error being what both the quantizer and these metrics are expressed in.

A proper scoring rule cannot be used to match quantizers here at all, and we report the consequence rather than suppressing it. Because Brier and NLL vary by less than 0.2% across the fine configurations, selecting the "closest" partner under them is choosing between near-ties, and one cell duly inverts: matched on Brier, 1 decimal pairs with 4 significant digits and the gain reads 0.5× rather than 456×. That is an artifact of a saturated metric — the same table shows its non-monotonicity in precision, a property no meaningful cost measure has — and not evidence against the defense. The general lesson is that matched-utility

claims about output quantization should be stated in fidelity to the unquantized model; scored against ground truth, a well-tuned quantizer is invisible next to the model's own error, which is a statement about the model and not about the defense. Our recommendation is accordingly stated against TV and KL, where the metric can actually tell the two families apart.

S11 Budget-Limited and Floor-Limited Boundary Seeking

Table 2's identical columns invite a reading the data does not support: that boundary seeking erases the difference between the two quantizer families *as a matter of what the families permit*. It does not. The attacker of Proposition 2 bisects a fixed boundary and is cut off by its step allowance, not by the grid. Its query count settles the question arithmetically: with $h = 8$ coordinates, $m = 10$ classes and an allowance of 60 steps it issues $8 + 80 \cdot 60 = 4808$ queries in every cell of the sweep, which is the allowance exhausted exactly. A defense measured against an attacker that never reached its floor has not been measured.

The stronger attacker searches the boundary and the scale together. It runs as a second stage on the intercept $\hat{b}_c$ that Proposition 2 has already returned, and is charged for those queries too, so it inherits no knowledge the threat model withholds. To split a bracket at mid it needs some admissible scale s for which $\hat{b}_c + s \cdot \text{mid}$ lands on a cell boundary; for each s it snaps to the nearest real boundary b, obtains the threshold $(b - \hat{b}_c)/s$ it can actually test, discards thresholds that do not separate two cells, and takes the survivor nearest mid. An error e in $\hat{b}_c$ displaces the threshold it actually probes by only e/s, which the large scales this stage depends on drive below the floor it is chasing — so the estimate costs it nothing except where the request format is too coarse for stage one to have pinned b_c in the first place, which is visible in the table as the 4-digit column. Its reachable set is therefore the whole set $\{(b - b_c)/s\}$ rather than one scaled copy of a single boundary, and it halts when that set has no member inside the bracket — a property of the grid and the norm cap, not of the allowance. Table 7 reports it. It halts after 24.5–37.8 steps of the same 60, so every cell below is floor-limited.

Three readings follow. The norm cap is a real defensive lever, but only against an absolute grid: four decades of extra reach buy the attacker 7697× there and 2.08× against a relative grid, which is scale-invariance behaving exactly as Theorem 5 says it should — a relative cell is a fixed fraction of the value, so a larger scale reaches no finer a boundary. The separation the units buy therefore depends entirely on the cap: 8.2× at 10^2, 399× at 10^4, $3.0 \cdot 10^4$× at 10^6. Neither half of that range supports an unconditional recommendation, which is why Section 5.2's conclusion stands unchanged.

Third, these errors are 740–4459× *worse* than the same search run with the deployed bias handed to it, which the threat model does not permit. The gap is the honest price of recovering b_c first: at $d = 4$ the estimate $\hat{b}_c$ is itself good to only about 10^{-4}, and the residual displaces every probed threshold by e/s. The

Table 7. Joint (boundary, scale) search against both families, 10 seeds, median relative error of Δ, 4 significant digits of request precision. cap is the largest query scale the attacker may send. The search runs as a second stage on the intercept recovered by Proposition 2, and is charged for that stage's queries as well, so nothing here is given the deployed bias. Steps are the mean bisection steps of the second stage per coordinate–class pair out of an allowance of 60; the fixed-boundary attacker of Table 2 uses 60.0 in every cell.

cap	absolute (3 dec.)	steps	relative (4 s.d.)	steps
10^2	$3.89 \cdot 10^{-6}$	24.5	$3.18 \cdot 10^{-5}$	24.6
10^4	$5.66 \cdot 10^{-8}$	31.2	$2.25 \cdot 10^{-5}$	25.5
10^6	$5.05 \cdot 10^{-10}$	37.8	$1.53 \cdot 10^{-5}$	26.1

step counts are unchanged — 24.5–37.8 against the same allowance of 60, against 24.6–38.0 with the bias given — so the attacker still halts on the grid rather than on its budget, which is the claim this table exists to support; only the level at which it halts moves. The price falls away as the request format sharpens, exactly as e/s predicts: a factor of 740–4459 at $d = 4$, 70–200 at $d = 6$, 1.4–2.7 at $d = 8$, and 0.98–1.01 at $d = 12$, where the stage-one estimate has become finer than anything the second stage chases and the control is no longer distinguishable from the attacker (`results/adaptive_boundary_knownbias_digits/`).

We report $d = 4$ because it is the only request precision in the sweep at which the attacker's candidate scales are enumerated *exhaustively*: the 9000 four-digit mantissas fit under the per-exponent cap at every norm cap in the table. At $d \geq 6$ the candidate set is sampled, and sampling can only withhold scales the attacker could really send, so those cells bound the floor from above rather than measuring it. They are consistent with the table — 4133× to 4756× for the cap lever against the absolute grid, 1.17× to 1.24× against the relative one, and 7254× to 7604× of separation at 10^6 — but only the $d = 4$ row is a measurement of a floor, and it is the one the body quotes.

S12 Boundary Seeking Against a Noisy Server

Every boundary-seeking measurement elsewhere assumes the quantizer is the only thing between the attacker and the true logit. That is the strongest assumption these results rest on: the search works by bisecting on *which cell* a score fell in, so randomizing that answer near the cell wall removes what the bisection converges to. This section measures what a stochastic server costs the attacker. The server draws $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ fresh per query and adds it *before* rounding; the attacker is not told σ and does not model it, and may average R independent answers to one query at a cost of R queries. Table 8 reports the sweep.

Three regularities. First, once noise binds, the recovered error is proportional to it: over four decades of σ and both families the ratio err/σ stays in $[0.100, 0.127]$,

Table 8. Boundary seeking (unknown bias) against a server that perturbs the score before quantizing it. **Dataset:** synthetic low-rank head, 10 classes, $h = 8$, rank 2, 10 seeds, median over seeds; entries are relative error in Δ. Rows are the significant digits the attacker's own request carries, ∞ being unbounded request precision.

request	noiseless	$\sigma = 10^{-6}$		$\sigma = 10^{-3}$	
digits	($\sigma = 0$)	$R = 1$	$R = 64$	$R = 1$	$R = 64$
absolute grid, 3 decimals					
4	$1.88 \cdot 10^{-4}$	$1.88 \cdot 10^{-4}$	$1.89 \cdot 10^{-4}$	$2.54 \cdot 10^{-4}$	$2.04 \cdot 10^{-4}$
6	$2.01 \cdot 10^{-6}$	$1.97 \cdot 10^{-6}$	$2.03 \cdot 10^{-6}$	$1.01 \cdot 10^{-4}$	$1.69 \cdot 10^{-5}$
8	$1.81 \cdot 10^{-8}$	$1.12 \cdot 10^{-7}$	$6.34 \cdot 10^{-8}$	$1.05 \cdot 10^{-4}$	$1.37 \cdot 10^{-5}$
12	$1.76 \cdot 10^{-12}$	$1.21 \cdot 10^{-7}$	$4.67 \cdot 10^{-8}$	$1.07 \cdot 10^{-4}$	$1.31 \cdot 10^{-5}$
16	$4.79 \cdot 10^{-16}$	$1.16 \cdot 10^{-7}$	$5.56 \cdot 10^{-8}$	$1.23 \cdot 10^{-4}$	$1.48 \cdot 10^{-5}$
∞	$5.21 \cdot 10^{-16}$	$1.01 \cdot 10^{-7}$	$5.54 \cdot 10^{-8}$	$1.21 \cdot 10^{-4}$	$1.49 \cdot 10^{-5}$
relative grid, 4 significant digits					
4	$1.94 \cdot 10^{-4}$	$1.94 \cdot 10^{-4}$	$1.94 \cdot 10^{-4}$	$2.18 \cdot 10^{-4}$	$3.07 \cdot 10^{-4}$
6	$1.81 \cdot 10^{-6}$	$1.82 \cdot 10^{-6}$	$1.87 \cdot 10^{-6}$	$1.13 \cdot 10^{-4}$	$2.61 \cdot 10^{-4}$
8	$1.75 \cdot 10^{-8}$	$1.11 \cdot 10^{-7}$	$5.63 \cdot 10^{-8}$	$1.11 \cdot 10^{-4}$	$2.47 \cdot 10^{-4}$
12	$1.88 \cdot 10^{-12}$	$1.11 \cdot 10^{-7}$	$4.85 \cdot 10^{-8}$	$1.05 \cdot 10^{-4}$	$2.62 \cdot 10^{-4}$
16	$4.50 \cdot 10^{-16}$	$1.03 \cdot 10^{-7}$	$4.55 \cdot 10^{-8}$	$1.12 \cdot 10^{-4}$	$2.72 \cdot 10^{-4}$
∞	$4.26 \cdot 10^{-16}$	$1.03 \cdot 10^{-7}$	$4.88 \cdot 10^{-8}$	$1.26 \cdot 10^{-4}$	$2.55 \cdot 10^{-4}$

so the floor noise imposes is about $\sigma/9$ and is otherwise structureless. Second, averaging does *not* reliably buy the $\sqrt{R}$ that Corollary 1 prices a stochastic defense at. Over the 34 noise-bound cells $R = 64$ improves the error by a median of only 2.14× (range 0.40–8.28) while costing 64.0× the queries (median 8506.5 → 544,736). For *this* attacker the predicted rate appears in exactly one place: where σ matches the quantizer cell. On the absolute grid, whose cell is 10^{-3}, $\sigma = 10^{-3}$ gives a median 8.11× (range 5.97–8.28) against the predicted $\sqrt{64} = 8$. Away from that match the rate fails in both directions. Below it almost every draw rounds to the same representative, so the mean returns that representative and repetition is inert. Above it this estimator is actively *harmed*: on the relative grid, whose cell near the probed scores is about 10^{-4}, $\sigma = 10^{-3}$ spans some ten cells and $R = 64$ is a median 0.43× — an error 2.3× *worse* than a single query. The next paragraph is about how much of that is the estimator's fault.

Why the averaging attacker degrades, and what does not follow. Boundary seeking does not read a value; it reads *which cell* a score fell in. The mean of R quantized answers is not a cell representative — it lies off the grid, between representatives — so the attacker's test at the wall is answering a question it was not asked, and the finer the grid is relative to σ, the further off it lands.

What does *not* follow is that repetition costs an attacker anything. An attacker handed R draws may discard $R - 1$ of them, so no optimal attacker is worse off for being given more; whatever degrades above is the estimator and not

Table 9. What the attacker does with $R = 64$ repeated answers, against the same stochastic server. **Dataset:** synthetic low-rank head, 10 classes, $h = 8$, rank 2, 10 seeds, median relative error in Δ. *single* is one draw; *first* takes R and discards $R - 1$; *mean* averages the draws and *mode* majority-votes them, both then testing cell membership; *mean-thr* averages but tests against the published boundary. Only the last changes the decision rule, and only it improves every cell — approaching the predicted $\sqrt{R} = 8$ without attaining it (median 6.5×), while *mode* recovers much of the rate but not everywhere.

grid	digits	σ	single	first	mean	mode	mean-thr
absolute	12	10^{-6}	$1.19 \cdot 10^{-7}$	$1.10 \cdot 10^{-7}$	$4.41 \cdot 10^{-8}$	$1.65 \cdot 10^{-8}$	$2.09 \cdot 10^{-8}$
absolute	12	10^{-3}	$1.13 \cdot 10^{-4}$	$1.21 \cdot 10^{-4}$	$1.53 \cdot 10^{-5}$	$3.96 \cdot 10^{-5}$	$1.37 \cdot 10^{-5}$
absolute	∞	10^{-6}	$1.09 \cdot 10^{-7}$	$1.09 \cdot 10^{-7}$	$4.72 \cdot 10^{-8}$	$1.69 \cdot 10^{-8}$	$1.70 \cdot 10^{-8}$
absolute	∞	10^{-3}	$1.17 \cdot 10^{-4}$	$1.05 \cdot 10^{-4}$	$1.34 \cdot 10^{-5}$	$2.96 \cdot 10^{-5}$	$1.40 \cdot 10^{-5}$
relative	12	10^{-6}	$1.00 \cdot 10^{-7}$	$1.19 \cdot 10^{-7}$	$4.75 \cdot 10^{-8}$	$1.54 \cdot 10^{-8}$	$1.78 \cdot 10^{-8}$
relative	12	10^{-3}	$1.04 \cdot 10^{-4}$	$9.93 \cdot 10^{-5}$	$2.44 \cdot 10^{-4}$	$1.01 \cdot 10^{-4}$	$1.57 \cdot 10^{-5}$
relative	∞	10^{-6}	$1.16 \cdot 10^{-7}$	$1.07 \cdot 10^{-7}$	$4.20 \cdot 10^{-8}$	$1.73 \cdot 10^{-8}$	$1.89 \cdot 10^{-8}$
relative	∞	10^{-3}	$1.04 \cdot 10^{-4}$	$1.12 \cdot 10^{-4}$	$2.51 \cdot 10^{-4}$	$1.13 \cdot 10^{-4}$	$1.48 \cdot 10^{-5}$

the information. Table 9 separates the two, holding the server fixed and varying only what the attacker does with the draws. Discarding all but one reproduces the single-query error, 0.95 and 1.08× of it, as it must. Majority vote on the returned representative — an on-grid answer, so the cell test is again being asked the question it was built for — removes the degradation and recovers much of the rate, a median 5.2×, though not uniformly: one cell still sits at 0.92×. What recovers it in *every* cell is repairing the *test* rather than the estimate. Comparing the average against the published boundary itself, instead of against the representative of the cell above it, beats the single-query attacker in all eight cells by 5.6–8.3×, median 6.5×, against the $\sqrt{64} = 8$ that Corollary 1(2) predicts. In the two cells where averaging loses 2.3–2.4×, the same draws under the matched test *gain* 6.6–7.0×.

The reading is therefore narrower than the sweep alone suggests, and sharper. Corollary 1(2)'s $\sqrt{R}$ survives contact with a quantizing stochastic server, for an attacker whose decision rule matches its estimator; averaging is the right reply to noise added to a value, and against noise added *before* a quantizer it is the right estimate paired with the wrong test. The pairing costs, not the repetition. This weakens rather than sharpens the contrast with Caliber [50] drawn in Section 2, which prices a stochastic defense in repeated queries: that pricing presumes the attacker can average, and against this server a well-matched one can. What the contrast still rests on is the deterministic case, where repetition returns the identical value at any budget — the statement Corollary 1(2) actually makes, and the one Theorem 6 turns into a floor.

The two floors compose, empirically, as a maximum. The third regularity is the one that matters for a defender. Request coarseness and output noise do not

add; recovery is floored by whichever is larger. Writing F_{req} for the noiseless error at a given request precision, the prediction $\max(F_{\mathrm{req}}, \sigma/9)$ matches every one of the 48 measured (digits, σ) combinations to within a factor of 0.90–1.35. The practical reading is blunt: noise below the floor the request format already imposes buys the defender *nothing*. At 4-digit requests, raising σ from 0 to 10^{-6} changes the error by 1.0× — not at all — and even $\sigma = 10^{-3}$ buys 1.4×. Only where the attacker's requests are near-exact does noise do the work, and even there repetition buys it back only when σ happens to match the cell.

This does not weaken Proposition 2, which is a statement about a deterministic quantizer and remains exact. It bounds its reach: the machine-precision recoveries in Table 2 are a property of a *deterministic* server, and a stochastic one floors them at $\sigma/9$. It also leaves the paper's ranking of the levers unchanged, since both quantizer families answer to noise identically — the request format, not the units, is again what sets the floor.

S13 The Floor Under a Discrete Request Format

Theorem 7 is unusual among the paper's bounds in that the quantity it bounds is exactly computable, so it can be checked rather than only believed. For a given θ the achievable-threshold set T can be enumerated in exact rational arithmetic — for each admissible M only the few N nearest θ can bound the gap — which yields $\inf_{\mathcal{Q}}$ itself. Table 10 reports it against the theorem's closed form and against the joint-search attacker, over 10 seeds.

Two conventions are what make that enumeration meaningful, and both are worth naming. First, Q_p breaks ties away from zero, so its cells are half-open and each achievable threshold bounds a gap on exactly one side. A rule that resolves ties inconsistently — `numpy`'s default, ties-to-even, is one — lets two admissible scales split a single threshold in opposite directions, collapsing the consistent set to a point — not at every $\theta \in T$, but at those a pair of scales realizes with opposite orientations: six of the sixty-eight thresholds in a decade at $p = d = 1$, among them $\theta = 3/2$, for which the proof of Theorem 7 gives the two-query witness. Under ties away from zero none of the sixty-eight collapses, which is the property the floor needs. Second, a point of T can be named several ways — $3/2 = \frac{3}{2\cdot 1} = \frac{9}{2\cdot 3} = \frac{15}{2\cdot 5}$ at $p = d = 1$ — and is one threshold however many scales realize it, so the enumeration collects T as a set. Neither convention changes a number reported here; both change whether the floor exists at all.

Relative error is measured against the true secret (Definition 4), which is what makes $(b - a)/(a + b)$ the minimax value on a gap $[a, b)$ and fixes the estimator at the harmonic midpoint $2ab/(a + b)$. An earlier convention divided the *absolute* radius by the least favorable secret in the gap, giving $(b - a)/(2b)$; that quantity is never above the minimax one, so a server quoting it never claims more privacy than it has, but it is not the minimax error and it is approached rather than attained. Over the enumerated cells the two differ by at most 0.29%, so nothing in this section's tables moves at the precision printed; we report the minimax value because that is what the theorem asserts.

The enumeration is short enough to state in full, so a reader may check Theorem 7(iii) without our code. The listing below is self-contained — exact rationals throughout, with no floating-point value entering a comparison — and runs in a few seconds:

```
from fractions import Fraction

def uniform_floor(p, d):        # minimax rel. error, worst theta
    ten, pts = Fraction(10), set()
    for M in range(10**(d-1), 10**d):              # request mantissas
        for N in range(2*10**(p-1)+1, 2*10**p, 2): # served odd midpoints
            t = Fraction(N, 2*M)
            while t >= 10: t /= ten              # fold to one decade:
            while t <  1:  t *= ten              # T is decade-invariant
            pts.add(t)
    o = sorted(pts); ext = o + [o[0]*ten]          # wrap the decade
    return min((b-a)/(a+b) for a, b in zip(ext, ext[1:]) if b > a)

closed = lambda p, d: Fraction(1, 2*(10**d - 1)*(2*10**p - 1) - 1)

cells = [(p, d) for p in range(1, 6) for d in range(1, 6) if p + d <= 6]
agree = [c for c in cells if uniform_floor(*c) == closed(*c)]
print(len(agree), "of", len(cells), "cells attain the closed form")
print("exceptions:", [c for c in cells if c not in agree])
```

It prints `12 of 15 cells attain the closed form`, then `exceptions: [(1, 2), (1, 5), (2, 4)]`. Two things are worth reading off it. The enumerated infimum is never *below* the closed form, in any of the fifteen, so the closed form is a lower bound on the minimax recovery error throughout and a server that advertises it is never advertising more privacy than it has. And the three cells where the bound is not attained are the ones where the pair of achievable thresholds straddling the worst θ is not the pair the closed form assumes; there the true floor is larger, by 11.2% at $(1, 2)$, 0.02% at $(1, 5)$ and 0.10% at $(2, 4)$, in the order printed. All three have $p < d$, which is precisely the regime the witness of part (iii) does not reach; every enumerated cell with $p \geq d$ attains the closed form, as that witness requires.

The last column is the head-to-head with adaptive one-bit compressed sensing [3, 14]. Those results assume the adversary can name the threshold τ of each measurement; ours may name only a query scale, and reaches thresholds only at the images of cell boundaries already fixed by the quantizer. Run at the same budget of 40 queries, the free-τ attacker is simple bisection and returns $9.09 \cdot 10^{-14}$ regardless of (p, d) — it never encounters a floor, because in that model there is none. Doubling the budget to 80 makes the comparison sharper still: the free-τ error collapses to *exact* recovery in 63 of the 90 runs and never exceeds $2.0 \cdot 10^{-16}$, below the double-precision epsilon of $2.2 \cdot 10^{-16}$; meanwhile all 90 entries of the *attacker* column are unchanged — not merely agreeing to twelve digits but bit for bit (`results/precision_floor_budget80_rerun/`). That is the operational content of the word floor. So the exponential-decay guarantee

Table 10. The floor under a discrete request format, enumerated exactly except where marked, 10 seeds, median over seeds. *certified* is the exact minimax relative error $(b-a)/(a+b)$ on the gap of T containing the sampled secret: the minimax error over every adaptive query set at any budget. *uniform* is the same quantity at the *worst* secret — the infimum of Theorem 7(iii), enumerated over one decade of T, and the number a server may advertise; † marks the three cells where $p+d>6$ makes the enumeration too large, so the closed form $1/(2(10^d-1)(2\cdot 10^p-1)-1)$ stands there as a proxy rather than an enumerated infimum; all three have $p\geq 3$, and the closed form is exact at every $p\geq 3$ cell where the enumeration is also computable. *attacker* is the joint boundary/scale search, and *free* τ the adaptive one-bit compressed-sensing model, which may place thresholds freely and so has no floor to hit; both are run at a matched budget of 40 queries. Regressing $\log_{10}$ of the certified floor on $p+d$ gives slope -1.028 against the predicted -1, and the floor is nearly symmetric in the two precisions — $2.33\cdot 10^{-6}$ at $(2,4)$ against $3.06\cdot 10^{-6}$ at $(4,2)$, the sampled secret rather than the split accounting for the residual factor, since the closed form of Theorem 7(iii) is symmetric under $p\leftrightarrow d$ to within 0.5% here — which is what "multiplicatively" means.

served p	request d	certified	uniform	attacker	free τ	cert. / unif.	att. / cert.
2	2	$1.83\cdot 10^{-4}$	$2.54\cdot 10^{-5}$	$8.67\cdot 10^{-5}$	$9.09\cdot 10^{-14}$	7.2	0.47
2	3	$1.98\cdot 10^{-5}$	$2.52\cdot 10^{-6}$	$1.06\cdot 10^{-5}$	$9.09\cdot 10^{-14}$	7.9	0.53
2	4	$2.33\cdot 10^{-6}$	$2.52\cdot 10^{-7}$	$5.86\cdot 10^{-7}$	$9.09\cdot 10^{-14}$	9.3	0.25
3	2	$2.72\cdot 10^{-5}$	$2.53\cdot 10^{-6}$	$2.08\cdot 10^{-5}$	$9.09\cdot 10^{-14}$	10.8	0.76
3	3	$1.92\cdot 10^{-6}$	$2.50\cdot 10^{-7}$	$1.11\cdot 10^{-6}$	$9.09\cdot 10^{-14}$	7.7	0.58
3	4	$1.61\cdot 10^{-7}$	$2.50\cdot 10^{-8\dagger}$	$9.61\cdot 10^{-8}$	$9.09\cdot 10^{-14}$	6.4	0.60
4	2	$3.06\cdot 10^{-6}$	$2.53\cdot 10^{-7}$	$7.72\cdot 10^{-7}$	$9.09\cdot 10^{-14}$	12.1	0.25
4	3	$2.24\cdot 10^{-7}$	$2.50\cdot 10^{-8\dagger}$	$1.35\cdot 10^{-7}$	$9.09\cdot 10^{-14}$	8.9	0.60
4	4	$1.51\cdot 10^{-8}$	$2.50\cdot 10^{-9\dagger}$	$4.74\cdot 10^{-9}$	$9.09\cdot 10^{-14}$	6.0	0.31

does not transfer to a quantized API, and the 10^5–10^9 gap between the two columns measures what the discreteness of the request format is worth to a defender.

Three readings of the table need stating. First, *certified* and *uniform* answer different questions and differ by 6–12× for that reason alone: the former is the floor at a secret drawn uniformly from $[1,10)$, the latter the floor at the worst secret. Only the second is available to a server, which does not choose its adapter's entries. Comparing a sampled-θ floor against the uniform closed form conflates the two and inflates the apparent looseness to 121–245×; the honest statement is that Theorem 7(ii)'s constant is loose by 20× against (iii), because it is written to fix the *exponent* and throws the constant away three times in the proof. Second, the attacker sits *below* the certified floor, at 0.25–0.76× it, which is not a violation: the floor is a minimax statement over the secrets a transcript cannot separate, while the attacker is measured at a secret drawn uniformly inside that gap, and averages about half of it. A ratio near 1 would mean the attacker had found the worst case, not that the bound had failed. Third, the attacker does sit above the *uniform* floor in every enumerated cell, and above the quoted proxy

in the three daggered ones, which is the comparison that could have failed and did not.

The statement is for homogeneous scalar probing, which is what isolates the arithmetic; the served score in the experiments of Section 5.2 carries a known base contribution as well, and the achievable set is then $\{(b - b_c)/s\}$ rather than $\{b/s\}$. That shift changes the constant but not the exponent, and the measured floors there follow the same law.

The sign of the secret. T consists of positive numbers, so the characterization in Theorem 7(i) is a statement about $T_\theta = \mathrm{sign}(\theta)T$ and not about T; read with T and a negative secret it would assert that no query separates two negative scalars, which is false. The reduction is by reflection, and it is exact rather than approximate: significant-digit rounding is an odd function, so the transcript at $-\theta$ is the entrywise negation of the transcript at θ. Allowing negative *scales* likewise adds nothing, since the pullback of the negative boundaries through a negative scale is the same positive set. Because an argument of that shape is easy to get wrong, we ran the experiment again with θ drawn from $[-10, -1]$ (`run-precision-floor --negative-secrets`). All 90 rows reproduce the positive run exactly — certified floor, closed form, attacker, and free-τ baseline agree to the last bit — which is what exact reflection predicts and what an approximate symmetry would not give. The enumeration itself runs on $|\theta|$, so the negative run exercises that reflection rather than a second code path.

What p is on a real server. Everywhere else in this paper p is something *we* chose when we built a quantizer, so whether Theorem 7 describes a deployment rests on an assumption about servers. We removed it. We ran vLLM 0.11.0's OpenAI-compatible server on `facebook/opt-125m` on our own hardware — Section S25 rules out probing anyone else's — and read the raw HTTP response bytes rather than a parsed float, since a parsed float cannot report how many digits were serialized. Across 96 served log-probabilities the wire carried 14–17 significant digits, and every one of them recovered the server's own `float32` exactly. The stock configuration applies no output rounding whatever. The digit count therefore overstates what leaks — it is an artifact of widening a `float32` before printing it, not a grid the server chose. If one models this server by Q_p at all, the parameter to use is the arithmetic's own 7 decimal digits — a modeling choice, not a rounding grid we measured, and the next subsection is why the distinction matters. Serving `float16` weights does not lower it: logits are reduced in `float32` regardless, and all 96 values again fit `float32` and no narrower width.

Two things follow. The output-precision defense is not something a deployment inherits by default; at $p \approx 7$ an operator who has not deliberately rounded is near the permissive end of Table 10, not the protective one. And with p fixed by the arithmetic, the only remaining control is the request precision d — the same conclusion Section S11 reaches from the attacker's side. The next section asks what that control is worth, and the answer is sobering.

The request format is an assumption, and it is doing the work. Theorem 7 assumes the attacker names its scale in d decimal significant digits. No JSON endpoint works that way: it accepts a decimal string, parses it to binary64, and multiplies. Those are different threshold sets, and the difference is not a constant. To place a threshold at θ the attacker wants $s = b/\theta$; if it can send that to within a relative u, the threshold lands within $O(u\theta)$ of θ, so the displacement is set by the rounding unit of the *request path* and not by p at all.

It is worse than a small floor. The attacker does not get one attempt: every cell boundary b offers another scale, each landing somewhere else inside that $O(u)$ window, and it keeps the closest. Measured by refinement — the same test Section S9 applies to Theorem 6 — the closest landing keeps falling as more boundaries are tried, from $4.09 \cdot 10^{-19}$ at 250 boundaries to $1.41 \cdot 10^{-20}$ at 8000, with no sign of settling; under binary32 parsing it falls from $2.61 \cdot 10^{-10}$ to $1.33 \cdot 10^{-11}$, tracking that format's coarser unit. Both sequences are still falling when the refinement stops, and the binary64 figures sit below that format's own rounding unit, so the experiment cannot separate a small positive limit from none: we claim only that the infimum Theorem 7 computes has no useful analogue here, not that it is zero — over a finite request format it is attained. The decimal grid is not a conservative model of JSON parsing; it is a different assumption, and it is the assumption carrying the theorem.

What the defense would cost if it were deployed. This is worth stating plainly rather than leaving in a bound. For $p \geq d$, Theorem 7(iii) gives $\approx \frac{1}{4} \cdot 10^{-(p+d)}$ as an upper bound on the floor, so the defender needs that quantity *large*, which means $p+d$ *small*. In the decimal model at $p = d = 7$ that bound is $2.5 \cdot 10^{-15}$: not protection against parameter recovery but the arithmetic noise of the pipeline. Only p there is anchored in a measurement — not a grid any server applies, as the measurement above found, but the seven decimal digits of the `float32` width we measured its logits to carry — while $d = 7$ is the model's own assumption, answering to nothing on the server at all, since its request path parses binary64. Because the constraint binds on the sum, coarsening the request does raise the floor, and provably rather than plausibly: part (ii) puts the floor at $(p, d) = (7, 1)$ above $1.2 \cdot 10^{-10}$, while the witness puts the floor at $(7, 7)$ below $2.5 \cdot 10^{-15}$; so the coarsening is worth at least $4.9 \cdot 10^{4}$, and $1.1 \cdot 10^{6}$ if the two witnesses are tight. It is bought from far too low a base: even at $d = 1$ the witness holds the floor below $2.8 \cdot 10^{-9}$. Over the fifteen enumerated cells a 10^{-4} floor needs $p + d \leq 3$, and a 10^{-6} floor needs $p + d \leq 5$. Three significant digits in total, across the answer and the request, to buy the weaker of the two. We take that to be the honest reading: the request-resolution floor is a genuine and exactly computable obstruction, and it is not a defense anyone can deploy at usable precision. It belongs in the paper as a boundary of what is recoverable, not as a recommendation. The command is `run-request-model`.

Why the decimal reading is safe in this direction, and only this one. Theorem 7 models both sides of the interface as decimal, and a stock server is neither: its logits are `float32` and its request path parses a decimal string to binary64.

Those threshold sets are not the decimal ones and, as the previous section shows, are not related to them by a constant. Substituting $p \approx 7$ into Table 10 therefore does not *certify* a deployment, and we do not use it that way. It is used only in the direction that survives the mismatch. The decimal model is *optimistic for the defender*, and the refinement measurement says by how much: against binary64 request parsing the attacker's closest landing reaches $1.41 \cdot 10^{-20}$ and is still falling, some five orders of magnitude below the $2.5 \cdot 10^{-15}$ the decimal model would concede it at $p = d \approx 7$. So the conclusion drawn here — that a server built this way, serving `float32` logits and parsing requests as binary64, has no usable request-resolution defense — holds *a fortiori* of the configuration we measured: its real interface gives the attacker more room than the theorem's, not less. One server is one measurement, and we have not surveyed others; what generalizes is the mechanism, not the number. The converse is what we do not claim. A positive floor for a binary interface would need a threshold analysis of binary64 parsing rather than of Q_p, and we leave it open; it is the natural next theorem; Section 9 already records the scope restriction that makes it one.

S14 What the Rounding Defense Costs a Legitimate User

The previous section reads Theorem 7 from the defender's side and finds the budget binds on $p+d$. That leaves a question: what does spending that budget do to the people the service is for? This section measures it, on three deployments, and it is built so the answer could have come back the other way — if coarsening were cheap, the defense would be viable and the paper would have to say so.

The obstruction is arithmetic before it is empirical. The closed-form benchmark of Theorem 7(iii), $1/\big(2(10^d-1)(2\cdot 10^p-1)-1\big) \approx \frac{1}{4}\cdot 10^{-(p+d)}$, fixes a budget $p+d \le \lfloor -\log_{10}(4F)\rfloor$ for a target floor F. It equals the floor in the twelve enumerated cells that attain it and lies below it — never above — in the other three, so reading budgets off it can only understate what a server may serve. It does not: substituting the enumerated infimum for the closed form leaves every row of Table 11 unchanged, the largest gap between them being 11.2% at $(1, 2)$. We read this off part (iii) rather than off the proven constant $\frac{1}{8}\cdot 10^{-(p+d+1)}$, which is some twenty times smaller and would make every entry below one digit more pessimistic than the truth. A server must serve at least one significant digit and accept at least one, so $p+d \ge 2$ in any deployment whatsoever. Table 11 is the consequence.

The largest floor *any* deployable server can claim is therefore $2.93 \cdot 10^{-3}$, at $p = d = 1$ — a server serving one significant digit and parsing one. An attacker held to a relative error of three parts in a thousand has recovered the adapter, and every setting a real deployment would choose is orders below that. So the floors that would protect something are not expensive — they do not exist, at any price, and no utility measurement can change that. Tightening the constant moves each row by a digit and moves the conclusion not at all.

Table 11. What each target floor costs in served and requested precision, at the closed-form floor of Theorem 7(iii), which is the exact infimum where enumeration attains it and below it elsewhere; no row changes if the enumerated infimum is used instead. A server serves at least one significant digit and accepts at least one, so $p+d \geq 2$ always, and every floor above $2.93 \cdot 10^{-3}$ is out of reach at any price.

target floor	required $p+d$	deployable?
10^{-2}	≤ 1	no
10^{-3}	≤ 2	only $p=d=1$
10^{-4}	≤ 3	yes
10^{-6}	≤ 5	yes

The scalar quantifier, and why it does not weaken this. Theorem 7 is proved for a scalar secret, and the object a deployment cares about is a matrix; Remark 3 keeps that distinction rather than eliding it. It costs this section nothing, because the gap runs the defender's way. The floor is an infimum over attacker strategies, and an attacker on the matrix problem may always probe coordinate by coordinate, meeting the scalar floor at each entry — so the matrix strategies are a superset and their infimum can only be smaller. Whatever combining directions turns out to be worth, the matrix floor is at most the scalar one: the numbers in Table 11 are a *ceiling* on what a matrix deployment could advertise, and a conclusion that the ceiling is already too low to protect anything holds a fortiori beneath it. The open matrix case could make this defense look worse than we report. It cannot make it look better.

What the one reachable setting costs. Each deployment is a public base model with an adapted head, the structure the recovery theory assumes. The served answer at a cell is $Q_p(f(Q_d(x)))$ against a baseline of $f(x)$; accuracy is on held-out true labels and total variation is between the served and unquantized softmax, since a user consuming *scores* loses more than the argmax reports (Section S10). Table 12 reports it.

Table 12. Held-out accuracy under request and answer canonicalization, against the unquantized baseline, at the only deployable floor ($p=d=1$) and at the next setting down. The last column is the total variation between the served and unquantized softmax at $p=d=1$, for a user who consumes scores rather than the argmax.

deployment	baseline	$p=d=1$ $F=2.93\cdot10^{-3}$	$p=d=2$ $F=2.54\cdot10^{-5}$	TV at $p=d=1$
vision, pixel domain	0.8270	0.2190	0.8230	0.755
vision, normalized domain	0.8270	0.2825	0.8090	0.671
tabular, raw domain	0.9736	0.6000	0.9722	0.308
tabular, standardized	0.9736	0.9750	0.9750	0.043
embedding interface	0.8840	0.8840	0.8820	0.015

On a pixel API the strongest floor a server can claim, $F = 2.93 \cdot 10^{-3}$ at $p = d = 1$, removes 61 accuracy points, and still removes 54 if the operator canonicalizes after its own normalization instead of on the wire. That cost is real and survives the choice. It is also the only expensive row: at the tightened floor of Theorem 7(iii) a 10^{-4} target is met at $p + d \leq 3$ rather than only at $p = d = 1$, and the cheapest such cell costs the pixel API 3.65 points instead of 61. That does not rescue the defense — it lowers the price of a floor at which the adapter is already recovered — but it is the honest accounting, and reading the cost off part (ii)'s constant instead would have overstated it by more than an order of magnitude.

The tabular row measures the quantizer's placement, not the defense. Canonicalizing raw features costs 37 accuracy points; canonicalizing the same requests after the operator's scaler costs nothing measurable ($0.9736 \to 0.9750$, inside the spread over ten splits). Both are discrete grids from the attacker's side — they differ by an invertible affine map — so the 37-point figure is an artifact of where an operator put its quantizer. We report both, because the first on its own would support our thesis for the wrong reason.

The embedding interface is the other caution. Request canonicalization is free there, which reads as good news for the defense until one notices that the secret at that interface is matrix-valued, and Theorem 7 is scalar: it is cheap exactly where it is unproven. At the text interface the request carries no numeric field at all, so there is no d, and the theorem is silent rather than favorable.

What this settles. Serving coarsely is survivable on its own — $p = 1$ with exact requests costs the vision endpoint 3.1 points and the SST-2 endpoint 0.2 — and a 10^{-6} floor is nearly free everywhere. Neither observation helps, because neither setting bounds an attacker anywhere near a useful error. The defense's problem is not its price. It is that the settings worth buying are not deployable, and the deployable ones are worth nothing. The command is `run-request-utility`; the run takes 880 seconds.

S15 Observable Transformer Endpoint Heads

Section 8.2 states both endpoint recoveries. This section records what stands behind them — the settings, the scope of the claim, and a fidelity measurement the body has no room for.

Classifier endpoint. The 20 Newsgroups recovery is over a 12-layer public transformer at ten seeds, and its augmented rank of 769 of 769 is *reported* rather than assumed — the reachability check of Section 4. That check is what confines the claim: what is recovered is the observable adapted head, not an arbitrary hidden transformer adapter. Table 19 gives every setting.

Fidelity beyond top-1 agreement. Label agreement is the weakest useful fidelity metric and it saturates here: across the exact, rounded, relatively quantized and noisy arms it reads 1.0000, 0.9987, 0.9957 and 0.9937 while the parameter error moves over twelve orders of magnitude, so a reader who trusts it alone would conclude the four attacks are interchangeable. They are not. Measuring the recovered model's *posterior* against the true one on the same held-out set separates them cleanly: mean KL runs $1.7 \cdot 10^{-18}$, $1.7 \cdot 10^{-2}$, $1.9 \cdot 10^{-1}$ and $6.8 \cdot 10^{-1}$ nats, and the worst-case total variation runs $1.6 \cdot 10^{-11}$, 0.25, 0.71 and 0.67. A relatively quantized oracle therefore leaves an attacker whose top-1 label agrees 99.6% of the time but whose posterior is nearly disjoint from the target's on the worst query, and whose calibration error differs by $3.7 \cdot 10^{-3}$.

The same two divergences priced on the *served* answer say what the defense costs its legitimate consumers, and this is where the comparison becomes a security statement rather than a trade: quantizing to 3 decimals moves the served posterior by $8.7 \cdot 10^{-9}$ nats and to 4 significant digits by $2.0 \cdot 10^{-7}$ nats, against the $1.7 \cdot 10^{-2}$ and $1.9 \cdot 10^{-1}$ the same quantizers cost the attacker — six orders of magnitude of asymmetry, on one scale, with top-1 utility at 1.0000 either way.

Full-vocabulary endpoints. What makes a production-size head invertible at all is full column rank, which is Corollary 2. The `distilgpt2` and MLM figures the body quotes are ten-seed means, and the scoreless index-bias variant reaches $1.69 \cdot 10^{-3}$ at *exact* top-1 agreement — so dropping the scores costs the attacker precision, not identifiability (Tables 27 and 28). The MLM row is where Proposition 1 is at its clearest, each row beyond the first four needing five scored observations rather than 769. The operational point is query cost: rank-revealing QR on the *public* head selects those 769 rows for free — out of the MLM head's 30,522, and out of `distilgpt2`'s 50,257 for that row.

S16 Real-Data Public-Feature Heads

Table 17 gives the oracle family row by row behind the public-feature result of Section 8.1. The target is an adapted head over a public standardized-PCA/ReLU map on `digits`, $h = 8$, rank 2, $m = 3$, over ten seeds, and the shape of the table is that the scored oracles reproduce the boundary — Theorem 1 at nine queries — while the index-only oracle leaves the target unidentified. Two things in it do not fit in the body. Dataset-manifold queries outperform random Gaussian probes, which is a reachability effect and not a property of the head: what the attacker can reach decides the conditioning. And the hidden adapter of Proposition 3 recovers at $2.84 \cdot 10^{-11}$ on these same features, which validates that proposition's local-cell assumption and nothing past it — a controlled check, not a deployment result.

S17 Remaining Experimental Protocol

Reported values. Every figure in the paper is a mean over the stated seeds with a population standard deviation, except where a table says it is a median; metrics themselves follow Definition 4.

Comparison to prior attacks. Prior LoRA extraction [53] reports functional mimicry, not comparable on parameter error, so Table 17 adds a hard-label surrogate as a functional baseline. Prior hard-label extraction [5,9] locates critical points to recover full networks; ours bisects for pairwise boundaries in a known-base head, which is why it recovers geometry but provably not scale.

Targets. Six families: (i) synthetic known-base low-rank affine targets; (ii) public-feature adapted heads on scikit-learn `digits` and `wine`, the public map being standardized PCA followed by ReLU and the adapted head a low-rank approximation to a shifted-label ridge head; (iii) observable endpoint transformer heads over frozen public features; (iv) controlled hidden LoRA-block targets on one stable ReLU cell; (v) real PEFT LoRA checkpoints trained on GLUE/SST-2; and (vi) vision endpoint heads over frozen `resnet18` and `vit_b_16` features on CIFAR-10. Oracles are as defined in Section 3.

Controlled validation. Every theorem hypothesis is first checked on controlled synthetic targets with exact ground truth (Table 26): positive cases recover at machine precision, $5.69 \cdot 10^{-16}$ to $2.73 \cdot 10^{-12}$, and both negative cases are executable certificates rather than failed attacks.

Completing Section 8.1. The design pool is the training input pool unless a table says random Gaussian, and scored top-k uses $k = 2$ with $m = 3$ on the public-feature targets. A defense is always applied to the oracle, never to the attack, and every table states which oracle it uses.

Query, training and evaluation data. An extraction attacker is entitled to query whatever the API accepts, so drawing the design from the training input pool is a modeling choice rather than leakage — but it does need saying what it cannot inflate. Our primary metric is relative error against the *true parameters*, which no overlap between the query set and the training set can flatter, since the target is the weight matrix and not a prediction. Where we do report a functional metric, it is computed on inputs the attacker never queried: on the transformer endpoints the query set is exactly the 900 training texts and agreement is measured on the 300 held-out ones.

How to read the intervals. Every interval we report is the 2.5th to 97.5th percentile of a bootstrap that resamples *seeds*, paired where the comparison is paired — a nominal 95% percentile interval. Its *coverage* is only approximate at ten seeds, where the resampling distribution is supported on a limited number of distinct means, so we read these as spread rather than as calibrated coverage. That is why the claims rest on separations of two to three orders of magnitude and on Theorem 6, which is a bound rather than a measurement, and never on one interval clearing another by a narrow margin. Where a comparison is paired we also report the exact sign test, which assumes nothing about the resampling distribution; at ten seeds its smallest attainable two-sided value is $p = 0.00195$,

so it can confirm a consistent direction but cannot express how large the effect is, and we quote it alongside the interval rather than instead of it. Two further conventions: the frontier point estimate is a minimum over A of a per-seed mean, which cannot itself be resampled, so the interval beside it resamples per-seed ratios and is centered within 1.5% of the quoted value rather than exactly on it; and intervals are omitted where a row records a failed attack. Section S18 restates these conventions per experiment, which is where they become checkable.

S18 Data Ledger: Training, Query, and Evaluation Sets

Section S17 states our data conventions in prose. This section states them per experiment, because a convention is only checkable against the particular runs it is supposed to govern. Every entry below was read from the experiment source and cross-checked against the configuration recorded in the tracked result file, not from our recollection of the design.

Two conventions decide how to read Table 13. First, our primary metric is relative error against the *true parameters*, and no overlap between an attacker's query set and a target's training set can flatter it: the estimand is the weight matrix, not a prediction. Overlap matters only for the functional metrics — label agreement and softmax divergence — and those are the columns the ledger exists to police. Second, an extraction attacker is entitled to query whatever the API accepts, so drawing a design from the training input pool is a modeling choice and not leakage; what would be leakage is *scoring* a recovered model on inputs it was fitted from.

The one entry that needs its own sentence. The † on `vision-endpoint` marks the only place where the evaluation inputs are not disjoint from the design, and we flag it rather than let the table imply otherwise. That panel's reported utility figures are *quantizer degradation*: the served output against the exact output on the same inputs, which is a property of the rounding and not a generalization claim. Nothing there is scored on unseen data because nothing there is a claim about unseen data; the attack column of that panel is relative parameter error, which the previous paragraph explains is immune to the question. The internal variable is named `held_out`, which is a misnomer; we note it here so that it does not mislead a reader of the code.

What the ledger confirms. Every experiment making a *generalization* claim computes its functional metric on inputs the attack never queried; the `vision-endpoint` quantizer-degradation diagnostic marked above is the sole exception, and it makes no such claim. Three further cases deserve naming because they are the ones where a reader might reasonably suspect otherwise: `endpoint-transformer` queries *exactly* its target's 900 training texts and scores agreement on the disjoint 300; `module-amplification` draws its design from isotropic Gaussians rather than from data at all, and measures utility on real CLS states; and the public feature maps in `realdata` — the standardizer and

Table 13. Per-experiment data ledger. "Target trained on" is what the victim adapter was fitted from; "attacker queries" is the design the attack submits; "functional metric on" is the set any agreement or divergence figure is computed over. Dashes mark experiments with constructed ground truth, which have no training data and report no functional metric.

experiment	target trained on	attacker queries	functional metric
`synthetic`, `lower-bound`	— constructed	per table	—
`security-floor`, `floor-margin`	— constructed	per table	—
`request-model`	— constructed	enumerated	—
`adaptive-boundary`	— constructed	adaptive	—
`precision-floor`	— constructed	enumerated	—
`realdata` (digits, wine)	65% stratified	from z_{train}	z_{test}
`defense-sweep`	65% stratified	from x_{train}	z_{test}
`quantization-frontier`	65% stratified	200 of 800	z_{test}
`endpoint-transformer`	900 texts	the same 900	300 held out
`mlm-output-head`	— public head	900 contexts	128 held out
`lm-endpoint-inversion`	— public head	28 prompts	12 held out
`lora-checkpoint` (SST-2)	1800 GLUE	769 probes	500 GLUE val.
`trained-adapter-defense`	1800 GLUE	900 probes	500 GLUE val.
`module-amplification`	1200 CLS	900 Gaussian	300 CLS
`vision-endpoint`	frozen features	corpus at A	service outputs†
`request-utility` (vis.)	5000 CIFAR tr.	—	2000 CIFAR test
`request-utility` (tabular)	60% stratified	—	40% held out
`request-utility` (text)	1800 GLUE	—	500 GLUE val.

the PCA — are fitted on the training split alone, so the held-out features are not constructed with knowledge of themselves.

S19 Where the Amplification Lever Exists

Corollary 1(1) needs three things, and a deployment breaks at least one unless it exposes a continuous-vector interface (Table 20; Section S20 gives the measurements).

(i) φ must not be scale-invariant. The ratio $\|\varphi(Ax)\|/(A\|\varphi(x)\|)$ separates three regimes: *exactly* homogeneous at 1.0000 for the identity map that $\mathcal{O}_{\text{module}}$ and embedding-input endpoints expose; *asymptotically* homogeneous for `digits` and a BatchNorm `resnet18`; and *scale-invariant*, decaying like $1/A$, for both LayerNorm models we test. So the 7.7–8.5× in endpoint-recovery error that relative units buy on the BERT text endpoint at matched utility (Table 14), with both attacks still succeeding except on a single six-digit seed, is caused by normalization discarding the scale, not by tokens being discrete: that surface is continuous and amplifiable. The same split runs through vision, with `resnet18` exposed and `vit_b_16` not, so the regime is readable from the architecture.

(ii) The amplified query must be reachable. Those ratios scale the normalized tensor. An API taking ordinary images does not allow that — pixels are bounded, so contrast can only be raised about mid-gray and then clipped, which drives the `resnet18` ratio to 0.0009, indistinguishable from the LayerNorm case.

Where all three conditions do hold, through $\mathcal{O}_{\text{module}}$ at $h = 768$, the feature map is the identity and Corollary 1(1) is exact rather than asymptotic: over 10 seeds amplification buys a mean $1026\times$ against the absolute grid, against a predicted $A = 1024$, a 0.2% miss. At 3 seeds the same mean sits 2% low, so the extra seeds buy accuracy on a quantity the theory pins rather than a number we were free to fit.

(iii) The amplified design must keep rank. Even in the tensor domain `resnet18` does not deliver the lever, and the diagnostic shows it is the *attack* breaking rather than the hypothesis: a feature coordinate stops activating, the design falls one short of the rank it needs, and recovery fails *with no quantizer present*. The same mechanism sets where the `digits` breach reproduces ($302\times$, $190\times$, $135\times$ at widths $8, 16, 32$) and where it does not (`wine`, $3.2\times$ and $0.1\times$, the per-class design rank-deficient and most seeds unable to solve a class). Presenting the breach as universal would be an overclaim; the lever is unavailable there, not ineffective.

Takeaway. Which regime a deployment is in decides what the units are worth. Where the lever is absent but the attack is healthy — the pixel-domain `resnet18` endpoint, exact-arithmetic recovery at 10^{-12} — the two families differ only intrinsically, and that difference is both small and unreliable. Over 10 seeds the advantage of relative units averages $1.40\times$ with a standard deviation of 0.39 and a range of $[0.93, 2.08]$: on one seed in ten the absolute grid is the better of the two. It is an effect rather than noise — 9/10 seeds favor relative units, exact two-sided sign test $p = 0.0215$, and the percentile bootstrap over seeds gives $[1.18, 1.64]$, excluding 1 — but it is not an effect a defender can count on in any single deployment. Pooling the seeds before taking the ratio, which is what a leak/cost slope comparison does (absolute 35.8, relative 58.7), reports $1.64\times$ instead; that value sits at the very top of the seed-level interval, and preferring the per-seed mean to it is the conservative reading, not a cosmetic one. Where all three conditions hold, by contrast, the lever is worth what the theory says and the gap widens by two to three orders of magnitude; the next two subsections measure that, first on `digits` and then on a trained adapter. The contrast is the finding. With no scale to exploit, the choice of quantizer family buys a small factor that sometimes vanishes entirely; wherever an interface hands the attacker a scale, it buys orders of magnitude. A defender who reads the second number as evidence for the first has bought almost nothing.

S20 Deferred Measurements for the Three Regime Conditions

These are the numbers behind Section S19; Table 20 gives every row.

(i) Homogeneity of φ. The asymptotically homogeneous maps settle at 1.73 (`digits`) and 0.144 (BatchNorm `resnet18`); the scale-invariant ones decay to 0.0009–0.0010 by $A = 1024$, `bert-base-uncased` through `inputs_embeds` and `vit_b_16` alike.

(ii) Reachability of the amplified query. The ratios above scale the normalized tensor, which a pixel API does not expose. Clipping contrast amplification to the valid image range drives the `resnet18` ratio to 0.0009 by $A = 1024$ — from 0.2262 at $A = 4$ — indistinguishable from the scale-invariant LayerNorm case, so the lever is unavailable on the bounded input domain even though it exists on the tensor one.

(iii) Rank of the amplified design. On `resnet18` in the tensor domain, by $A = 256$ a feature coordinate has stopped activating for all 2000 queries, the design falls to rank 512 of the 513 needed, and recovery fails at $8.36 \cdot 10^{-1}$ with no quantizer present. The homogeneity hypothesis itself holds throughout — the ratio settles at 0.1445 rather than decaying (Table 20) — so this is the attack breaking, not the theory. On `wine` the per-class design similarly loses rank (8.1 of 9 and 10.1 of 13), leaving 80% and 100% of seeds unable to solve a class, which is why the breach measures only 3.2× and 0.1× there.

What relative units buy at the endpoint. Section S19 reads the consequence of (i) off a real endpoint, and that comparison carries its own table because a single ratio hides how much it moves. Table 14 gives ten seeds at each of two precisions. Both quantizers hold label utility at exactly 1.0000 on every seed, so neither gap is bought with accuracy. Relative units leave the attacker worse off on 10/10 seeds at both precisions — exact two-sided sign test $p = 0.00195$, the smallest ten seeds can produce — so the *direction* is as firm as this design allows. The *size* is not: per-seed ratios run from 3.58 to 16.65 at eight digits and 3.70 to 22.68 at six, and the 8.53× and 7.70× we quote are ratios of the means, which is not the mean of the ratios (8.86 and 8.25). We therefore read this gap as a direction and one significant figure, not as a calibrated constant. One seed at six digits carries a relative error above 1, where the recovered update is further from the truth than zero is — a failed attack rather than a defeated defense — and it is kept rather than dropped.

S21 The Perturbation Taxonomy Experiment

Exposure to amplification and averaging is decided by two *independent* properties of a perturbation, so the four combinations form a taxonomy whose *absolute* row holds the deployed defenses — score rounding [48] and output perturbation [12, 42] — and whose relative row is our repair. Testing each cell against both levers predicts eight outcomes rather than one.

Takeaway. All eight predictions hold, with non-overlapping intervals wherever the theory predicts a separation. Amplification breaches absolute units whether or not the perturbation is random and does nothing in relative ones; averaging

Table 14. Absolute versus relative quantization at the observable BERT text endpoint, ten seeds at each precision. **Model:** `bert-base-uncased` on 20 Newsgroups (`sci.crypt`/`comp.graphics`), rank-2 endpoint head, 900 final-logit queries. Columns are relative recovery errors of the endpoint update under an absolute grid and under significant-digit quantization at the same digit count; both hold label utility at 1.0000 on every seed. The per-seed ratio is $\varepsilon_{\text{rel}}/\varepsilon_{\text{abs}}$; the mean row divides the two means, which is the number Section S19 quotes and is not the mean of the column above it.

	eight significant digits			six significant digits		
seed	absolute	relative	ratio	absolute	relative	ratio
0	$2.40 \cdot 10^{-4}$	$2.60 \cdot 10^{-3}$	10.81	$3.55 \cdot 10^{-2}$	$2.68 \cdot 10^{-1}$	7.55
1	$1.53 \cdot 10^{-3}$	$5.47 \cdot 10^{-3}$	3.58	$2.02 \cdot 10^{-1}$	$8.90 \cdot 10^{-1}$	4.41
2	$1.72 \cdot 10^{-3}$	$9.07 \cdot 10^{-3}$	5.27	$1.84 \cdot 10^{-1}$	$8.87 \cdot 10^{-1}$	4.83
3	$7.33 \cdot 10^{-4}$	$3.71 \cdot 10^{-3}$	5.06	$3.52 \cdot 10^{-2}$	$7.99 \cdot 10^{-1}$	22.68
4	$4.36 \cdot 10^{-4}$	$3.57 \cdot 10^{-3}$	8.20	$4.48 \cdot 10^{-2}$	$3.51 \cdot 10^{-1}$	7.84
5	$7.59 \cdot 10^{-4}$	$7.59 \cdot 10^{-3}$	9.99	$9.27 \cdot 10^{-2}$	$5.08 \cdot 10^{-1}$	5.48
6	$1.44 \cdot 10^{-3}$	$8.48 \cdot 10^{-3}$	5.90	$9.84 \cdot 10^{-2}$	$5.07 \cdot 10^{-1}$	5.15
7	$5.01 \cdot 10^{-4}$	$8.34 \cdot 10^{-3}$	16.64	$7.55 \cdot 10^{-2}$	$2.79 \cdot 10^{-1}$	3.70
8	$1.86 \cdot 10^{-3}$	$3.09 \cdot 10^{-2}$	16.65	$3.37 \cdot 10^{-1}$	$3.87 \cdot 10^{0}$	11.49
9	$6.08 \cdot 10^{-4}$	$3.97 \cdot 10^{-3}$	6.53	$8.83 \cdot 10^{-2}$	$8.25 \cdot 10^{-1}$	9.35
mean	$9.82 \cdot 10^{-4}$	$8.37 \cdot 10^{-3}$	8.53	$1.19 \cdot 10^{-1}$	$9.19 \cdot 10^{-1}$	7.70

breaches stochastic perturbations whether or not the units are relative, against a predicted $\sqrt{R} = 8.0$, and is worth 1.0× against both deterministic cells. A defense must fix both properties, and exactly one cell does.

The result is about units, not about grids. Spending the same absolute budget optimally rather than by rounding — displacing every returned vector by the *full* allowed norm in an unhelpful direction, deterministically in the query, the mechanism class utility-constrained output perturbation occupies [42] — does not help the defender: 256.3× against rounding's 301.8×, intervals overlapping, averaging again 1.0×. That is what Theorems 4 and 6 predict, since they see the perturbation only through $\|E\|_F$ (Section S22). Only changing the units recovers the protection.

So an absolute defense's safety is *conditional* on architecture and feature geometry, which a defender must audit per endpoint and re-audit whenever either changes — Section S19 gives the test. It sometimes holds, and we report the case against ourselves: on the BERT text endpoint, the most deployment-like interface here, rounding to three decimals defeats recovery outright (relative error $1.23 \cdot 10^{2}$ against $6.52 \cdot 10^{-10}$ on exact scores, Table 30), and no adaptation repairs it because LayerNorm has removed the lever. Relative quantization removes that dependence on feature geometry — but only against this lever, and Section 5.2 is the audit it does *not* escape.

Attacking our own recommendation. Every lever above was built against a perturbation whose size is independent of the score; relative quantization violates that on purpose, so testing it only against those levers would repeat the mistake of Section 5. Told that the error is proportional to the signal, one reweights: least squares with weights the reciprocal cell width, which is observable from the returned precision and free of query cost. It works, buying 2.6–4.6× (Table 23), and is the attacker reported throughout this paper. Two things matter more than its size. It does not compound with A, which is what Theorem 5 claims and a growing gain would have refuted; and against *every* absolute setting the same lever is worth 1.00× with a degenerate interval — not resistance but an identity, since a constant cell width makes the weights uniform, so an absolute defense does not even offer a weighting to exploit. Section S22 reports the one design adaptation that backfires.

S22 Deferred Discussion of the Taxonomy Experiment

Why the optimal absolute defense is still an absolute defense. The last row of Table 22 spends the defender's budget better than rounding does: instead of snapping each score to a grid, which wastes most of the allowed norm on scores that happen to sit near a cell center, it displaces every returned vector by the *full* allowed norm in the direction least useful to a least-squares solve, deterministically in the query so that repetition cannot average it away. At equal budget this dominates rounding pointwise, and it is the mechanism class that utility-constrained output-perturbation defenses [42] occupy. That it still changes nothing, as Section S21 reports, is forced rather than lucky: the two mechanisms are matched by construction in $\|E\|_F$, and that is the only channel through which the bounds see a perturbation at all — which is the point of stating the taxonomy in terms of units rather than mechanisms. We test the mechanism class rather than reimplement a published system, which would risk a strawman.

A design adaptation that backfires. Section S21 reports the estimator adaptation that helps the attacker against relative quantization. The natural *design* adaptation does not. Since a relative grid makes small scores individually more precise, an attacker may try to steer the design toward small-magnitude queries; wherever that constraint binds it is a net loss (0.67–0.93×). Rows of small norm cluster near a hyperplane, so the design matrix loses conditioning faster than the per-entry precision improves, and $\|X^+\|_2$ in Theorem 4 grows by more than $\|E\|_F$ shrinks. The two adaptations therefore pull in opposite directions, which is why we report the weighted attacker with an unconstrained design as the strongest one we found.

S23 Detailed Experimental Tables

These are the full settings behind results the main body summarizes. They are provided for verification; the main body reports the numbers that carry the argument.

Table 15. Absolute versus relative quantization against a *trained* rank-4 LoRA module, $h = 768$, `bert-base-uncased`/SST-2, 10 seeds (mean test accuracy 0.8756, s.d. 0.0121). "Leak" is the attacker's best relative error over $A \leq 1024$, minimized per seed for *both* families; each absolute row is paired with a relative setting that degrades the service *no more*.

absolute	utility	cost	leak	amp. gain	paired relative	cost	leak
2 dec.	0.9884	$3.67 \cdot 10^{-3}$	$2.37 \cdot 10^{-5}$	1015×	3 sig. digits	$1.11 \cdot 10^{-3}$	$4.21 \cdot 10^{-3}$
3 dec.	0.9986	$3.67 \cdot 10^{-4}$	$2.36 \cdot 10^{-6}$	1026×	4 sig. digits	$1.11 \cdot 10^{-4}$	$4.23 \cdot 10^{-4}$
4 dec.	0.9998	$3.67 \cdot 10^{-5}$	$2.33 \cdot 10^{-7}$	1034×	5 sig. digits	$1.11 \cdot 10^{-5}$	$4.24 \cdot 10^{-5}$

Table 16. Roadmap. Each principal or supporting result, the oracle it concerns, the maximal recoverable object, and the experiment that instantiates it. Principal results are marked $\star$. The three quantization rows are meant to be read in order: Theorem 6 certifies a floor at the transcript the attacker chose, Proposition 2 chooses a different transcript and collapses it, and Theorem 7 restores a floor only once the attacker's own requests are discretized.

Result	Oracle	Recoverable object	Experiment
Def. 3	any	transcript equivalence class	framework
Thm. 1*	$\mathcal{O}_{\text{raw}}$, $\mathcal{O}^{\text{score}}_{\text{top-k}}$	exact head, bias, and Δ	Tables 26, 17, 19
Prop. 1	$\mathcal{O}^{\text{score}}_{\text{top-k}}$	further rows at $r+1$ cost	Tables 26, 28
Thm. 2	$\mathcal{O}^{\text{score}}_{\text{top-k}}$	rank and row space from a bound	Tables 27, 28
Thm. 3*	$\mathcal{O}^{\text{score}}_{\text{top-k}}$	*not* the row space, if $s < r$	Table 26
Thm. 4*	degraded scores	error bound; factors into two terms	Sec. 5, Table 18
Cor. 1	degraded scores	$1/A$ and $1/\sqrt{R}$ scaling	Table 18, Fig. 1
Thm. 5*	relative quantization	non-decay of the error, not a limit	Tables 24, 25
Thm. 6*	any deterministic quantizer	floor at the attacker's *own* transcript; not uniform over query sets	Tables 21, 5
Prop. 2	any deterministic quantizer	*collapses* that floor: bracket halves per query	Tables 2, 7
Thm. 7*	quantizer + d-digit requests	exact $\inf_{\mathcal{Q}}$ for a scalar secret; matrix case open	Table 10
Thm. 8*	$\mathcal{O}_{\text{label}}$, $\mathcal{O}^{\text{idx}}_{\text{top-k}}$	boundaries up to positive scale	Tables 26, 17
Thm. 9*	$\mathcal{O}_{\text{end}}$	quotient modulo $\ker C$	Table 3
Thm. 10	$\mathcal{O}_{\text{end}}$	local observability certificate	Fig. 2
Cor. 3	$\mathcal{O}_{\text{end}}$	non-gauge nullity lower bound	Table 3
Prop. 3, Cor. 2, Prop. 4, Cor. 4	$\mathcal{O}_{\text{raw}}$, $\mathcal{O}_{\text{end}}$, $\mathcal{O}_{\text{logprob}}$, $\mathcal{O}_{\text{bias}}$	hidden state, then reduce to Thm. 1	Tables 27, 28

Table 17. Real-data public-feature adapted head. **Dataset:** scikit-learn `digits`, classes $\{0, 1, 2\}$, 10 seeds. **Model:** public standardized-PCA/ReLU feature map with $h = 8$, rank-2 adapted head over a shifted-label ridge head, $m = 3$ classes, $k = 2$. Mean $\pm$ population standard deviation.

oracle	test agreement	diagnostic
$\mathcal{O}_{\text{raw}}$	1.0000 ± 0.0000	9.0 queries
$\mathcal{O}^{\text{score}}_{\text{top-k}}$, random queries	0.9691 ± 0.0926	800.0 queries
$\mathcal{O}^{\text{score}}_{\text{top-k}}$, dataset queries	1.0000 ± 0.0000	800.0 queries
$\mathcal{O}^{\text{score}}_{\text{top-k}}$, rounded	0.9963 ± 0.0059	one decimal
$\mathcal{O}^{\text{score}}_{\text{top-k}}$, noisy	0.9936 ± 0.0052	$\sigma = 0.05$
$\mathcal{O}_{\text{label}}$ boundaries	0.9277 ± 0.1354	2.50 ± 0.81 of 3 pairs
complete ranking boundaries	1.0000 ± 0.0000	3.00 ± 0.00 of 3 pairs
$\mathcal{O}^{\text{idx}}_{\text{top-k}}$	unidentified	0.80 ± 0.40 failure rate
hard-label surrogate	0.9149 ± 0.0556	functional baseline

Table 18. Per-lever validation of Corollary 1 on the raw-logit oracle. **Dataset:** `digits`, classes $\{0, 1, 2\}$, 10 seeds, mean over seeds, $h = 8$, rank 2, declared bound $R \leq 4$, design 200. Each row adds one lever to the row above; the parenthesized factor is the improvement contributed by *that* lever alone. Amplification uses $A = 64$ and repetition $R_{\text{rep}} = 64$, so Corollary 1 predicts $64\times$ and $\sqrt{64} = 8.0\times$ respectively, and predicts that repetition does nothing against deterministic rounding. The full ablation over all four defenses, and the top-k oracle, are Table 29.

attacker	rounding, 1 dec.	noise, $\sigma = 0.1$	Cor. 1 predicts
non-adaptive	$4.58 \cdot 10^{-2}$	$1.17 \cdot 10^{-1}$	—
+ D-optimal design	$3.97 \cdot 10^{-2}$ (1.2×)	$8.18 \cdot 10^{-2}$ (1.4×)	better conditioning
+ repetition	$3.97 \cdot 10^{-2}$ (1.00×)	$9.53 \cdot 10^{-3}$ (8.6×)	$1\times$ / $\sqrt{R} = 8.0\times$
+ amplification	$5.61 \cdot 10^{-4}$ (70.8×)	$1.52 \cdot 10^{-4}$ (62.8×)	$A = 64\times$
+ intercept probe	$5.64 \cdot 10^{-4}$ (1.0×)	$1.50 \cdot 10^{-4}$ (1.0×)	head unchanged
overall adapter gain	81×	780×	
bias error, first → last	$1.29 \cdot 10^{-2} \to 6.74 \cdot 10^{-3}$	$2.65 \cdot 10^{-2} \to 2.55 \cdot 10^{-3}$	intercept lever only
label agreement, final	0.9968	0.9995	utility preserved

Table 19. Observable endpoint-head recovery. **Dataset:** 20 Newsgroups. **Model:** frozen public `bert-base-uncased`, 10 seeds. **Oracle:** $\mathcal{O}_{\text{end}}$ on text queries. Recovery solves for the low-rank classifier-head update over public transformer features, not for arbitrary hidden transformer adapters.

metric	mean	std / note
test accuracy	0.8493	0.0176
endpoint head delta rel. err.	$6.52 \cdot 10^{-10}$	$2.50 \cdot 10^{-10}$
held-out label agreement	1.0000	0.0000
endpoint final-logit queries	900	final logits only
augmented feature rank	769	full rank (of 769)

Table 20. What the amplification lever requires, and where each requirement fails. **Top:** $\|\varphi(Ax)\|/(A\|\varphi(x)\|)$; pinned at 1 makes Corollary 1(1) exact, a positive constant makes it asymptotic, and decay like $1/A$ means the map discarded the scale. The ratio is 1 at A=1 by definition, so that column is omitted. Pixel rows clip contrast amplification to the valid image range. **Bottom:** attacks at matched service cost, 10 seeds; "leak" is the attacker's best error over A, "exact" the same solve with no quantizer — the control that says whether a defense is being measured at all.

feature map	interface / domain	A=4	16	64	1024	regime
identity	$\mathcal{O}_{\text{module}}$ / embedding	1.0000	1.0000	1.0000	1.0000	exactly homog.
`digits` PCA/ReLU	$\mathcal{O}_{\text{raw}}$	1.4839	1.6697	1.7183	1.7337	asympt. homog.
`resnet18` (BatchNorm)	$\mathcal{O}_{\text{end}}$, tensor	0.2567	0.1672	0.1488	0.1445	asympt. homog.
`resnet18` (BatchNorm)	$\mathcal{O}_{\text{end}}$, *pixel*	0.2262	0.0612	0.0152	0.0009	scale-invariant
`vit_b_16` (LayerNorm)	$\mathcal{O}_{\text{end}}$, tensor	0.2700	0.0688	0.0168	0.0010	scale-invariant
BERT CLS (LayerNorm)	$\mathcal{O}_{\text{end}}$, `inputs_embeds`	0.2471	0.0616	0.0153	0.0009	scale-invariant
$\mathcal{O}_{\text{module}}$, *`bert-base-uncased`*, $h = 768$, random *head: all three requirements hold. This is the control for Table 15, which repeats it against a* trained *adapter; the two are different targets and their numbers should not be read as replicates.* absolute, 2/3/4 dec., 10 seeds: amp. gain 1043/1031/1004×, mean 1026× against the predicted A=1024; relative gain 2×; relative leaks 225–238× less.						
$\mathcal{O}_{\text{end}}$, *`resnet18`*, $h = 512$, *pixel domain: lever absent, attack healthy.* Leak / cost slope over 10 seeds: absolute 35.8, relative 58.7, so pooled the relative family leaks 1.64× less at equal cost; per seed the advantage is 1.40 ± 0.39, range $[0.93, 2.08]$, favoring relative on 9/10 seeds (sign test $p = 0.0215$); exact 10^{-12}.						
$\mathcal{O}_{\text{end}}$, *`resnet18`, tensor domain: the attack breaks, not the defense.* Rank / dead coords: $A \leq 64$: 513, 0; $A \geq 256$: 512 of 513 needed, 1; exact err. $8.36 \cdot 10^{-1}$.						

Table 21. Certified indistinguishability floor of Theorem 6, `digits`, 10 seeds, mean over seeds. The floor is the largest adapter perturbation that provably leaves every returned score inside its own quantization cell, so $F/(1+F)$ lower-bounds the relative error of *any* algorithm at that query scale; the column is F, which is that bound up to a factor below 1.01 here; "attacker" is the best error our implementation achieves. The certificate never exceeded a measured error in any of the 120 configurations. The slope, not the level, is what Theorem 6 predicts.

family	quantity	A=1	4	16	64	256	1024
absolute	floor	$9.90 \cdot 10^{-3}$	$2.17 \cdot 10^{-3}$	$5.62 \cdot 10^{-4}$	$1.75 \cdot 10^{-4}$	$4.26 \cdot 10^{-5}$	$1.08 \cdot 10^{-5}$
1 dec.	attacker	$3.97 \cdot 10^{-2}$	$8.97 \cdot 10^{-3}$	$2.33 \cdot 10^{-3}$	$5.61 \cdot 10^{-4}$	$1.38 \cdot 10^{-4}$	$3.78 \cdot 10^{-5}$
relative	floor	$2.53 \cdot 10^{-5}$	$2.80 \cdot 10^{-5}$	$3.15 \cdot 10^{-5}$	$4.22 \cdot 10^{-5}$	$4.55 \cdot 10^{-5}$	$4.32 \cdot 10^{-5}$
3 s.d.	attacker	$2.24 \cdot 10^{-4}$	$2.64 \cdot 10^{-4}$	$2.73 \cdot 10^{-4}$	$2.74 \cdot 10^{-4}$	$3.10 \cdot 10^{-4}$	$3.00 \cdot 10^{-4}$
Log–log slope against A		absolute: floor -0.970, attacker -1.005 (predicted -1)					
		relative: floor $+0.091$, attacker $+0.040$ (predicted 0)					

Tightening the certificate. Charging each score only the cell boundary its *signed* displacement travels toward, rather than the nearer of the two, raises the floor 2.2–3.4×; hill-climbing the admissible polytope instead of sampling it raises it again, to 5.9–11.3× the untightened floor, uniformly in A. Both phases only exhibit admissible perturbations, so they move the bound and not the truth, and the slopes are unchanged.

Table 22. Defense taxonomy, `digits`, 10 seeds, $A = 256$, $R = 64$. Cells give the factor by which *amplification* / *averaging* reduces the attacker's error, as a mean over seeds with 95% bootstrap intervals. Corollary 1 predicts amplification to breach the absolute row, averaging the stochastic column.

	deterministic	stochastic
absolute	1 decimal place 302× [251, 353] / 1.0×	additive noise 0.1 252× [205, 310] / 8.0× [7.0, 9.1]
relative	3 significant digits 0.9× [0.6, 1.2] / 1.0×	multiplicative noise 0.1% 0.8× [0.7, 1.0] / 8.7× [7.6, 9.8]
Strongest member of the absolute–deterministic cell		
adversarial	worst-case displacement, $\|E\| = 0.05$ per query: 256.3× [210.9, 306.4] / 1.0×	

Table 23. Attackers adapted to the *relative* error model, `digits`, 10 seeds, design 100. Entries are the factor by which the lever improves on ordinary least squares on the *same* seed, as a mean over seeds with 95% paired percentile bootstrap intervals. Theorem 5 claims scale-invariance, so what would refute it is a gain growing with A; a bounded constant does not. The absolute rows are an identity, not a measurement: one cell width makes the GLS weights uniform.

lever	family / setting	A=1	16	256	1024
GLS	absolute, 1 dec.	1.00 [1.00, 1.00]	1.00	1.00	1.00
	absolute, 2 dec.	1.00 [1.00, 1.00]	1.00	1.00	1.00
	relative, 3 sig.	2.86 [2.21, 3.62]	4.40 [3.57, 5.25]	4.08 [2.65, 5.89]	3.33 [2.58, 4.07]
	relative, 4 sig.	2.58 [2.08, 3.18]	4.63 [3.53, 5.70]	3.27 [2.65, 4.10]	3.03 [2.07, 4.01]
+ magnitude steering	absolute, 1 dec.	0.93 [0.85, 1.05]	0.75 [0.65, 0.87]	0.71 [0.59, 0.84]	0.81 [0.72, 0.92]
	absolute, 2 dec.	0.88 [0.78, 0.99]	0.67 [0.58, 0.77]	0.85 [0.74, 0.96]	0.87 [0.71, 1.04]
	relative, 3 sig.	2.98 [2.33, 3.71]	4.23 [3.31, 5.25]	3.85 [2.58, 5.47]	3.46 [2.61, 4.32]
	relative, 4 sig.	2.69 [1.89, 3.64]	5.38 [3.75, 7.15]	3.77 [2.92, 4.69]	3.49 [2.42, 4.65]

Read the magnitude-steering block against the GLS block directly above it: every interval overlaps its GLS counterpart on the relative family, so the design lever adds nothing there, while on the absolute family it costs the attacker up to a third of its accuracy. Under relative units a small-magnitude score does carry a proportionally precise equation, but rows with uniformly small scores cluster near a hyperplane, and the conditioning lost exceeds the precision gained. The probe pass is charged to the attacker's query count.

Table 24. Utility–security frontier, `digits`, 10 seeds, mean over seeds, attacker best-responding over $A \in \{1, \dots, 1024\}$. "Amp. gain" is what amplification alone buys, predicted to be 1× for the relative family. Paired rows have *identical* utility.

absolute grid	utility	attacker	amp. gain	relative	utility	attacker	amp. gain
1 decimal	0.9527	$3.78 \cdot 10^{-5}$	1088×	1 sig. digit	0.9527	$1.72 \cdot 10^{-2}$	1.2×
2 decimals	0.9952	$4.25 \cdot 10^{-6}$	949×	2 sig. digits	0.9952	$1.65 \cdot 10^{-3}$	1.3×
3 decimals	0.9989	$4.10 \cdot 10^{-7}$	982×	3 sig. digits	0.9989	$1.76 \cdot 10^{-4}$	1.3×

Table 25. Verification of the asymptotic-homogeneity hypothesis of Theorem 5 (Remark 2), 10 seeds, mean over seeds. "Ratio" is $\|\varphi(Ax)\|/(A\|\varphi(x)\|)$: it does not remain at 1, so exact homogeneity fails. "Defect" is $\|Z(A)/A - Z(4A)/4A\|_F / \|Z(4A)/4A\|_F$, the quantity the proof requires to vanish; it decays like $1/A$ on both datasets, so $Z(A)/A$ converges and the theorem applies asymptotically.

dataset	quantity	A=1	4	16	64	256	1024
`digits`	ratio	1.000	1.484	1.670	1.718	1.731	1.734
`digits`	defect	0.560	0.140	0.0342	0.0085	0.0021	0.0005
`wine`	ratio	1.000	6.911	8.587	9.008	9.113	9.139
`wine`	defect	0.910	0.197	0.0471	0.0116	0.0029	0.0007

Table 26. Controlled validation of the oracle boundary on synthetic targets. These check the theorem hypotheses before moving to real-data and transformer-scale settings. Means over three seeds unless noted.

setting	oracle / condition	outcome	result
Affine low-rank head, $d = 32$	$\mathcal{O}_{\text{raw}}$	$5.69 \cdot 10^{-16}$ rel. err.; 1.000 agreement	Thm. 1
Affine low-rank head, $d = 32$	$\mathcal{O}^{\text{score}}_{\text{top-k}}$, $k = 3$	all 8 rows recovered; 1.000 agreement	Thm. 1
Multiclass affine head	$\mathcal{O}_{\text{label}}$	3/3 pairwise boundaries; 1.000 agreement	Thm. 8
Public ReLU feature head	$\mathcal{O}_{\text{raw}}$ / $\mathcal{O}^{\text{score}}_{\text{top-k}}$	$9.89 \cdot 10^{-14}$ / $1.38 \cdot 10^{-14}$ rel. err.	Thm. 1
Low-rank-assisted top-k	row space observed first	6/6 classes using 34 observations	Prop. 1
Local hidden LoRA block	full-rank public downstream map	$2.73 \cdot 10^{-12}$ rel. err.; 1.000 agreement	Prop. 3
Public ReLU feature head	ranking / co-visible partial $k = 2$	all pairwise boundaries; 1.000 agreement	Thm. 8
Partial top-k lower bound	class never returned	indistinguishable rank-one adapters, 10/10 seeds	Thm. 8

Table 27. Full-vocabulary endpoint recovery. **Model:** `distilgpt2`, 10 seeds; public LM output head with 50,257 vocabulary rows and 768 hidden columns. **Target:** rank-4 low-rank final-representation update over a public 16-dimensional feature projection of the base hidden state. **Oracles:** $\mathcal{O}_{\text{end}}$, $\mathcal{O}_{\text{logprob}}$ with forced rows, and scoreless $\mathcal{O}_{\text{bias}}$.

metric	mean	std / note
Setting		
hidden dimension / vocabulary rows	768	50,257
adapter feature dimension / rank	16	4
public adapter-feature rank	16	full rank
output-head rank / condition number	768	$2.74 \cdot 10^{3}$
softmax-augmented rank	769	full column rank
Hidden-state recovery (Cor. 2, Prop. 4)		
full-logit hidden rel. err.	$1.77 \cdot 10^{-9}$	$1.91 \cdot 10^{-10}$
top-1/logit-bias hidden rel. err.	$9.34 \cdot 10^{-10}$	$6.70 \cdot 10^{-11}$
top-1/index-bias hidden rel. err.	$8.69 \cdot 10^{-8}$	$6.44 \cdot 10^{-9}$
Adapter recovery (Thm. 1)		
full-logit adapter rel. err.	$3.09 \cdot 10^{-5}$	$1.10 \cdot 10^{-5}$
top-1/logit-bias adapter rel. err.	$7.76 \cdot 10^{-6}$	$1.25 \cdot 10^{-6}$
top-1/index-bias adapter rel. err.	$1.69 \cdot 10^{-3}$	$3.62 \cdot 10^{-4}$
held-out top-1 agreement	1.0000	all three oracles
Unknown rank, given only $R \leq 8$ (Thm. 2)		
rank estimate, logits / log-prob	4.0000	plateau 10^{-4}–10^{-1}
rank estimate, index-only	4.0000	plateau $3 \cdot 10^{-3}$–10^{-1}
rank-projected err., logits / log-prob	$1.61 \cdot 10^{-5}$ / $4.07 \cdot 10^{-6}$	
rank-projected err., index-only	$7.86 \cdot 10^{-4}$	$2.73 \cdot 10^{-4}$
Query cost (Alg. 2)		
full-logit endpoint queries	28	prompts
logit-bias calls, all / adaptive	$1.41 \cdot 10^{6}$ / $2.15 \cdot 10^{4}$	forced rows
index-bias calls, all / adaptive	$3.38 \cdot 10^{7}$ / $5.17 \cdot 10^{5}$	incl. binary search
sampled exact / shifted rows for rank	768 / 795	random subset mean
adaptive exact / shifted rows	768 / 769	QR-selected
adaptive exact / shifted condition	$1.43 \cdot 10^{4}$ / $2.56 \cdot 10^{4}$	public head

Table 28. Full-vocabulary masked-LM output-head recovery. **Model:** `bert-base-uncased` MLM, 10 seeds, 30,522 vocabulary rows, 768 hidden columns. **Target:** synthetic rank-4 update to the vocabulary output head over the public transformed MLM feature. **Oracles:** $\mathcal{O}_{\text{end}}$ on masked contexts, and a selected-coordinate scored-token oracle. Contrast with the classifier-head quotient of Table 3: the same hidden dimension is recoverable when the endpoint exposes a full-rank vocabulary head.

metric	mean	std / note
output-head rank / min singular value	768	1.702
output-head condition number	98.67	fixed head
augmented feature rank	769	full rank
output-head adapter rel. err.	$7.64 \cdot 10^{-8}$	$1.22 \cdot 10^{-8}$
output bias rel. err.	$1.99 \cdot 10^{-10}$	$3.40 \cdot 10^{-11}$
held-out max logit error	$6.60 \cdot 10^{-10}$	$1.70 \cdot 10^{-10}$
held-out top-1 agreement	1.0000	0.0000
endpoint full-logit queries	900	masked contexts
coordinate oracle calls	30,522	scored-token rows
coordinate scalar observation reduction	$1.76 \cdot 10^{2}$	Prop. 1
coordinate adapter rel. err.	$7.32 \cdot 10^{-6}$	$6.70 \cdot 10^{-6}$
unknown-rank estimate	4.0000	0.0000, $\tau = 10^{-3}$
unknown-rank adapter rel. err.	$1.46 \cdot 10^{-6}$	$1.15 \cdot 10^{-6}$, $\tau = 10^{-3}$
adaptive exact / shifted rows	768 / 769	feature adapter
adaptive exact / shifted feature err.	$2.79 \cdot 10^{-9}$ / $8.01 \cdot 10^{-9}$	subset inversion

Table 29. Adaptive attacker versus score degradation. **Dataset:** `digits`, classes $\{0, 1, 2\}$, 10 seeds, mean over seeds, $h = 8$, rank 2, declared bound $R \leq 4$, design size 200, $A = 64$, $R_{\text{rep}} = 64$. Relative error is of the linear head; bias error is separate. "fail" is the fraction of seeds with at least one unsolved class row.

defense	attacker	calls	rel. err.	bias err.	agreement	fail
Panel A: raw-logit oracle $\mathcal{O}_{\text{raw}}$						
rounding, 1 decimal	non-adaptive	200	$4.58 \cdot 10^{-2}$	$1.29 \cdot 10^{-2}$	0.9926	0.00
	+ D-optimal design	200	$3.97 \cdot 10^{-2}$	$1.62 \cdot 10^{-2}$	0.9862	0.00
	+ repetition	12800	$3.97 \cdot 10^{-2}$	$1.62 \cdot 10^{-2}$	0.9862	0.00
	+ amplification	12800	$5.61 \cdot 10^{-4}$	$2.19 \cdot 10^{-2}$	0.9782	0.00
	+ intercept probe	16896	$5.64 \cdot 10^{-4}$	$6.74 \cdot 10^{-3}$	0.9968	0.00
rounding, 0 decimals	non-adaptive	200	$1.02 \cdot 10^{0}$	$3.95 \cdot 10^{-1}$	0.9080	0.00
	full adaptive	16896	$6.45 \cdot 10^{-3}$	$1.19 \cdot 10^{-1}$	0.9176	0.00
noise, $\sigma = 0.05$	non-adaptive	200	$6.37 \cdot 10^{-2}$	$1.41 \cdot 10^{-2}$	0.9910	0.00
	+ repetition	12800	$4.96 \cdot 10^{-3}$	$2.08 \cdot 10^{-3}$	0.9979	0.00
	full adaptive	16896	$8.40 \cdot 10^{-5}$	$1.42 \cdot 10^{-3}$	0.9979	0.00
noise, $\sigma = 0.1$	non-adaptive	200	$1.17 \cdot 10^{-1}$	$2.65 \cdot 10^{-2}$	0.9787	0.00
	full adaptive	16896	$1.50 \cdot 10^{-4}$	$2.55 \cdot 10^{-3}$	0.9995	0.00
Panel B: scored top-k oracle, $k = 2$						
exact scores	non-adaptive	200	$3.07 \cdot 10^{-15}$	$3.95 \cdot 10^{-16}$	1.0000	0.00
	+ amplification	12800	$2.39 \cdot 10^{-1}$	$2.31 \cdot 10^{-1}$	0.8681	0.40
	+ class coverage	12800	$1.84 \cdot 10^{-1}$	$1.73 \cdot 10^{-1}$	0.8995	0.30
rounding, 1 decimal	non-adaptive	200	$7.59 \cdot 10^{-2}$	$1.86 \cdot 10^{-2}$	0.9899	0.00
	+ amplification	12800	$2.43 \cdot 10^{-1}$	$2.67 \cdot 10^{-1}$	0.8287	0.40
	+ class coverage	12800	$1.88 \cdot 10^{-1}$	$2.02 \cdot 10^{-1}$	0.8697	0.30

Table 30. Utility-vs-leakage defense sweep. Public-feature rows use `digits` over 10 seeds; the BERT endpoint rows use the 10-seed `bert-base-uncased` experiment of Table 19 and the two hidden-LoRA rows the 10-seed SST-2 checkpoint of Table 3. "Utility" and "observable quotient" are as in Definition 4. Errors are those of the *non-adaptive* attacker; the last column reports what the adaptive attacker of Section 8.4 then does. A relative error above 1 means the recovered update is further from the truth than zero is, so the row records a *failed attack*, not a defeated defense, and is marked accordingly.

API restriction	rel. err.	utility	obs. quot.	adaptive attacker
digits scored top-k, rounded 1 dec.	$6.34 \cdot 10^{-2}$	0.9628	0.9667	defense defeated
digits scored top-k, noise $\sigma = 0.1$	$7.60 \cdot 10^{-2}$	0.9553	0.9667	defense defeated
digits top-k indices only	not identifiable	n/a[‡]	0.0000	no equations
BERT endpoint head, rounded logits	$1.23 \cdot 10^{2}$	0.9987	1.0000	defense holds[†]
BERT endpoint head, noisy logits	$3.41 \cdot 10^{3}$	0.9937	1.0000	defense holds[†]
BERT/SST-2 hidden LoRA, clf. endpoint	not identifiable	1.0000	0.0026	not identifiable
BERT/SST-2 hidden LoRA, module output	$8.37 \cdot 10^{-15}$	1.0000	1.0000	no defense

[†]The two BERT endpoint rows are the paper's most deployment-like setting and the absolute grid *wins* there: the exact-score attack succeeds ($6.52{\cdot}10^{-10}$, Table 19), so the quantizer and not the solver is what breaks recovery, and Table 20 shows why adaptation cannot repair it — LayerNorm makes this feature map scale-invariant, so the amplification lever is absent. This is Section S21's point in its strongest form: an absolute defense's safety is conditional on architecture, and here the condition happens to hold. [‡]Utility is Definition 4's agreement between the *recovered* and true heads, so it is undefined where no head is recovered: this row solves for nothing. The served labels are untouched — that is the row's point — but that is the API's own accuracy, not a recovery metric, and the two are not interchangeable.

S24 Compute, Hardware, and Reproduction Budget

Hardware. Every number in this paper was produced on a single Apple M1 laptop: 8 CPU cores, 8 GB of unified memory, macOS 26.6.1. No GPU or accelerator was used at any point, and none is required: the source contains no CUDA or MPS device selection, so a reproducer inherits the same CPU-only path regardless of what hardware is available. This is a deliberate property of the attack rather than an accident of our budget. The recovery step is a least-squares solve on a system the interface itself constructs, and the transformer and vision experiments spend their time in forward passes over a few thousand short inputs; neither needs training-scale compute. The defensive reading is the uncomfortable one — the attacks in this paper are within reach of a commodity laptop,

so an operator cannot treat adversary compute as the control that protects an adapter.

Software. Python 3.12.12 with numpy 2.4.4, scikit-learn 1.8.0, torch 2.11.0, torchvision 0.26.0, transformers 5.8.0, peft 0.19.1, datasets 4.8.5, sentencepiece 0.2.1, and pytest 9.0.3. The repository declares version ranges capped at the next major release of each dependency and ships a lock file pinning this exact set; `uv sync` uses the lock, so a reproducer gets these versions and not merely compatible ones. The caps are not a claim that every version in range reproduces the tracked numbers — transformers crossed from 4.x to 5.x during this work — but a guard against a lockless resolve silently jumping a major boundary.

Storage and downloads. The result files total under 10 MB. Model weights are not tracked: the transformer and vision experiments download public checkpoints and datasets on first use, which occupies roughly 4.3 GB of cache. That download is the dominant one-time cost of a cold reproduction and is not included in the wall-clock figures below, which were measured with the cache already warm.

Wall-clock. Table 31 times every one of the 32 reproduce commands behind this paper, each measured on the machine above with the model cache already populated and nothing else running. Running the table end to end takes 6.0 hours, and the distribution is extremely uneven: one command accounts for 54% of it, and the remaining thirty-one, together with the test suite, finish in 167 minutes combined. Two entries time a smaller configuration than the paper reports, and both are named here rather than left to a note in the data: the `adaptive-boundary` entry omits the joint (boundary, scale) search priced below, and `trained-adapter-defense` trains three seeds where Table 15 reports ten, so reproducing that table costs about ten times this entry's training. The 6.0 hours is therefore a floor on the cost of reproducing the paper, not the cost. One experiment is deliberately absent: the served-precision probe of Section S13 reads bytes off a self-hosted vLLM server rather than computing against the local model cache, so its wall-clock would measure that server and not this paper.

The command accounting for that 54% is `run-trained-adapter-defense`, at 3 h 13 m, and it is worth being precise about what it buys. It is not an attack. Almost all of its time goes into *training* the victim adapter — a BERT LoRA over 1800 examples for two epochs — which is the honest baseline a defender pays to have a model at all. The attacks themselves are cheap by comparison: the entire adaptive boundary-seeking experiment of Section 5.2 runs in 10 s, the quantization frontier in 37 s, and the full 120-configuration security-floor sweep in 12 s. The one attack that is not cheap is the joint (boundary, scale) search of Section S11, at 1 h 25 m — and what it spends the time on is enumerating the scales its *own* request format can express, which is the attacker paying for the defender's discretization rather than for the model. This asymmetry is the practical form of the paper's argument. The defender's cost is training; the attacker's cost is a least-squares solve, and no amount of the former buys protection from the latter.

Table 31. Wall-clock for all 32 reproduce commands and the test suite, in seconds, on the single M1 CPU described above with a warm model cache. Left: the core experiments, which need only `numpy` and `scikit-learn`. Right: the sixteen commands requiring the optional `vision` and `lora` extras. The first endpoint-transformer entry is the small DistilBERT smoke command; the two that follow are the BERT runs behind Table 19, and the two after those are the ten-seed quantizer comparisons behind Table 14.

core command	s	extras command	s
`pytest` (252 tests)	200	`vision-endpoint`	842
`synthetic`	< 1	`trained-adapter-defense`	11591
`ablation`	2	`endpoint-transformer` (smoke)	19
`realdata`	13	`endpoint-transformer` (BERT)	449
`defense-sweep`	2	`endpoint-transformer` (SST-2)	305
`adaptive-defense`	25	`endpoint-transformer` (BERT, 8 digits)	450
`quantization-frontier`	37	`endpoint-transformer` (BERT, 6 digits)	454
`defense-taxonomy`	26	`lm-endpoint-inversion`	218
`security-floor`	12	`mlm-output-head`	203
`adaptive-boundary`	10	`lora-checkpoint`	21
`lower-bound`	< 1	`lora-checkpoint` (Jacobian)	20
`floor-margin`	14	`lora-checkpoint` (stronger)	865
`precision-floor`	44	`module-amplification`	129
`noise-aggregators`	929	`null-curvature`	457
`request-model`	< 1	`request-utility`	880
`realdata` (wine)	3	`adaptive-boundary` (noise)	3376
`defense-sweep` (wine)	2		
subtotal	1319	subtotal	20279

Reproducibility record. The paper itself carries the record needed to audit these results and to rebuild them: the entry point and measured wall-clock for every experiment that runs on the machine described here (Table 31), the data each target is trained and evaluated on (Table 13), the machine and library versions above, and, in the text beside each result, the configuration wherever it departs from that entry point's defaults. We do not reproduce the option strings themselves or name a result file at every table; what the paper gives is the experiment, its inputs, its cost and its settings, not a transcript of a shell session. Each experiment writes a result file, and a test suite of 252 checks re-reads those files and asserts the figures printed here — so the numbers in this paper are machine-checked against the runs that produced them rather than transcribed by hand. The source code and result files are archived at `https://github.com/ImKKingshuk/IndoCrypt2026-LoRA-RBQL`.

Environmental cost. The total is small enough to state plainly: one laptop CPU, no cluster, no accelerator, and no hyperparameter search — the configurations are fixed in the reproduce commands and the runs are seed-deterministic. Development cost more than the 6.0 hours of one full pass, since some settings were

re-run at more seeds as the paper developed, but it remains bounded by a small multiple of it. We report this because a reader is entitled to know whether a result depends on compute they do not have, and here it does not.

S25 Responsible Disclosure, Dual Use, and Detectability

What was attacked. Every target in this paper is a model we trained ourselves or an openly licensed checkpoint evaluated locally: BERT and DistilGPT-2 from public weights, LoRA adapters we fine-tuned on SST-2, a low-rank classifier-head update we trained on 20 Newsgroups, a vision head on public images, and synthetic linear targets. We issued no queries to any third-party prediction API, paid or free, and we did not attempt to extract any deployed commercial adapter. There is therefore no affected vendor to notify, and nothing here was learned by probing a service whose terms we had not agreed to. The corresponding gap is stated in Section 9: we did not survey production APIs for one exposing the full amplification lever, so our claims about deployed interfaces are inferences from documented output formats, not measurements of them.

Dual use. The results cut both ways, and we think the defensive edge is the sharper one. The two impossibility results (Theorems 8, 9) tell a defender what to stop emitting; the boundary-seeking result (Section 5.2) tells a defender that a control they might have believed in — scale-invariant score quantization — does not survive an adaptive adversary, which is information they cannot act on unless someone publishes it. The offensive content is a least-squares solve against a linear system that the interface itself constructs, which is not a difficult idea and, in the equation-solving form, has been public since Tramèr et al. [48]. Our contribution to an attacker is mainly the observation that adaptivity beats precision, and that observation is already standard in quantized compressed sensing [3, 14].

Legality and scope. Nothing in this paper authorizes extraction from a service one does not own. Querying a third-party API to reconstruct its parameters will in general breach that service's terms of use, and may implicate trade-secret or computer-misuse law depending on jurisdiction; the concrete harm is to the fine-tuning investment, since an adapter is often the only proprietary part of an otherwise public stack. We present these attacks as measurements of what an interface leaks, for use by the party that operates it.

Detectability and cost. An operator is not defenseless while it decides what to do. The attacks are not cheap and not quiet. Recovering the 10×8 target of Section 5.2 takes 4808 queries, and their structure is conspicuous: the boundary attacker sends one-hot inputs whose single nonzero entry converges to a fixed scale, so the transcript is a long run of near-identical, near-degenerate probes differing in their last significant digits. Amplification is louder still — inputs at $1024\times$ the natural norm. At transformer scale the costs are larger: $2.15 \cdot 10^4$

calls behind a forcible-row endpoint and $5.17 \cdot 10^5$ behind a scoreless one for the rank-4 update of Section 8.3. Rate limiting and norm caps do *not* stop boundary seeking, for the reason Section 5.2 gives, but per-account query accounting and outlier detection on input geometry are orthogonal controls that these attacks do nothing to evade, and we did not try to make them evasive.

What we do not claim to leak. We recover adapter parameters, not training data. The spectra of recovered updates reflect the fine-tuning optimization as much as its data, and we make no membership-inference or data-reconstruction claim from them; treating a recovered spectrum as evidence about individual training examples would go well beyond anything we measured.

S26 Disclosure of AI Tool Usage

The authors did not use generative AI tools to draft, revise, or edit the manuscript, conduct the literature review, design the methodology, generate or review code, create or process datasets, or analyze results. All text, figures, tables, and code in this paper are solely the work of the authors.